

Advanced Data Analysis

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2021-11-20

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Chapter 1

Prerequisites

More books by the author can be found here:

- Data Analysis: the first book in the data analysis series, which covers data analysis basics and causal inference
- Marketing Research
- Communication Theory

This book's skeleton is based on several courses. Hence, I'd like to credit professors accordingly.

Course	Professor
Data Analysis 3	Christopher Wikle
Analyzing Unstructured Data	Kunpeng Zhang
Deep Learning	Yann LeCun

Compare to the last book, this book should differ as follows:

- It focuses on prediction models in machine learning (both supervised and unsupervised), where the first book is more about interpretation and causal inference

Part I

MACHINE LEARNING

Chapter 2

Introduction

Resources

- INTRODUCTION TO MACHINE LEARNING
- An introduction to Statistical Learning
- The Elements of Statistical Learning

Machine learning is also known as statistical learning

- two types of learning:
 - supervised learning: predicting or estimating output based on inputs
 - unsupervised learning: understanding the relationships among inputs without output

	Unsupervised	Supervised
Continuous	• Clustering & Dimension Reduction (SVD, PCA, K-means)	• Regression (linear, polynomial) • Tree-based methods
Categorical	• Association Analysis (Apriori, FP-growth) • Hidden Markov	• Classification (KNN, Trees, Logistic, Naive Bayes, SVM)

3 types of data

- Structured Data (tables form)
- Semi-structured Data (e.g., XML, JSON, HTML)
- Unstructured Data

- Texts
 - Images: Videos, photos,
 - Networks: Social Networks
- ML is concerned with the the development, the analysis, and the application of algorithm that allow computers to learn
- Learning:
 - A computer learns if it improves its performance at some tasks with experience (i.e., data)
 - Extracting a model of system from the sole observation (or the simulation)
 - A model = some relationships between the variables used to describe the system.
- Goals:
 - Prediction
 - Inference
- Learning if useful when:
 - Human expertise does not exist (navigating on Mars)
 - Humans are unable to explain their expertise (speech recognition)
 - Solution changes in time (routing on a computer network)
 - Solution needs to be adapted to particular cases (user biometrics)
- Applications in different domains:
 - Retail: Market basket analysis, customer relationship management (CRM)
 - Finance: Credit scoring, fraud detection.
 - Manufacturing: Optimization, troubleshooting
 - Medicine: medical diagnosis
 - Telecommunications: Quality of service optimization, routing
 - Bioinformatics: Motifs, alignment WEb mining, search engines.
- **Steps**
 - Data generation: data types, sample size
 - Preprocessing: normalization, missing values, feature selection/extraction
 - Machine Learning: Hypothesis, choice of learning paradigm/algorithm.
 - Hypothesis validation: cross-validation, model deployment.
- Types of machine learning:
 - Supervised: manually label variables
 - Unsupervised:
- Training and testing

- Training is the process of making the machine learn
- No free lunch:
 - * Training set and testing set come from the same distribution
 - * Need to make some assumptions.
- Factors affecting the performance:
 - Types of training provided
 - The form and extend of prior knowledge
 - Type of feedback provided
 - The learning algorithms used
- Two factors:
 - Modeling
 - Optimization
- Algorithms:
 - ML depends on algorithms.
 - The algorithms control the search to find and build the knowledge structures
 - The learning algorithms should extract useful information from training examples.
- Types of learning (algorithms):
 - Supervised learning ($\{X_n \in R^d, y_n \in R\}_{n=1}^N$)
 - * Prediction
 - * Classification (discrete labels), regression (real values)
 - Unsupervised learning ($\{X_n \in R^d\}_{n=1}^N$)
 - * Clustering
 - * Probability distribution estimation
 - * Finding association (in features)
 - * Dimension reduction
 - Semi-supervised learning
 - Reinforcement learning
 - * Decision making (robot, chess machine)

2.1 General Framework

- Y = output
- $X = (X_1, \dots, X_p)$ = p inputs

Under supervised learning, additive error model:

$$Y = f(x) + \epsilon$$

where

- $f(\cdot)$ = unknown fixed function
- ϵ = random error ($E(\epsilon) = 0; \epsilon \perp X$)

For prediction (blackbox)

$$\hat{Y} = \hat{f}(X)$$

where

- Reducible Error: $\hat{f}(\cdot)$ can be improved (in principle)
- Irreducible Error: ϵ because it's not a function of the inputs, which might contain unknown variables or uncontrolled variation

$$\begin{aligned} E(Y - \hat{Y})^2 &= E[f(X) + \epsilon - \hat{f}(X)]^2 \\ &= [f(X) - \hat{f}(X)]^2 + \text{var}(\epsilon) \\ &= \text{reducible} + \text{irreducible} \end{aligned}$$

For inference, $f(X) = E(Y|X)$, but the distribution of $Y|X$ is unknown; hence, must be estimated.

We try to find $\hat{f}$ such that $Y \approx \hat{f}(X) \forall (X, Y)$

Ways to estimate f :

- Parametric
 1. Make an assumption about the function form of $f(X)$
 2. Try to fit data to that specified form (i.e., estimate the parameters) using OLS in regression or MLE in general.
- Nonparametric

Tradeoff between prediction accuracy and model interpretability

Restrictive Model	Flexible Model
More inference	More prediction
More interpretable	

2.2 Assessing Model Accuracy - Model fit

2.2.1 Continuous output: Mean squared error (MSE)

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{f}(x_i))^2$$

Generalization Error is a measure of how accurately an algorithm or model is able to predict outcome values for test set.

Tradeoff between flexibility of the method and training MSE, but there is an U-shaped relationship between flexibility and test MSE. This is a manifestation of bias-variance tradeoff

$$E(MSE) = E(y_0 - \hat{f}(x_0))^2 = \text{var}(\hat{f}(x_0)) + [\text{bias}(\hat{f}(x_0))]^2 + \text{var}(\epsilon)$$

where 0 subscript means training data set

- $E(MSE) \geq \text{var}(\epsilon)$ because ϵ is irreducible model error and serves as the lower bound on the quality of potential model fit
- More flexible means more variance, but less bias.
- Less flexible (parametric) means less variance, but more bias.

2.2.2 Categorical output: training error rate

Training error rate is the proportion of mistakes in classification from applying $\hat{f}$ to the training set:

$$\frac{1}{n} \sum_{i=1}^n I(y_i \neq \hat{y}_i)$$

and **test error rate** is

$$\frac{1}{n} \sum_{i=1}^n I(y_0 \neq \hat{y}_0)$$

Bayes classifier is a simple classifier where it assigns each observation to the most likely class, given its predictor values (i.e., conditional probability that $Y = j | x_0$)

In other words, assigns x_i to class j where $P(Y = j | X = x_0)$ is largest

Bayes classifier should produce the lowest possible test error rate, known as Bayes error rate:

$$1 - E_x(\max_j [P(Y = j|X)])$$

- where it is greater than 0 if the classes overlap
- analogous to the irreducible error

In reality, we don't know $P(Y|X)$, hence, we try to approximate this conditional probability.

For example, KNN

$$P(Y = j|X = x_0) \approx \frac{1}{K} \sum_{i \in N_0} I(y_i = j)$$

2.2.3 Curse of dimensionality

Example: with a p -dimensional hypercube of unit volume, to capture a fraction (r) of the observations in a subcube, the length of a side of the sub-cube averages to $e_p(r) = r^{1/p}$

$$e_1(.01) = .01; e_1(.1) = .1; e_{10}(.01) = .63; e_{10}(.1) = .8$$

To cover 1% of the observation in 10 dimensional cub, we have to cover 60% of the entire sample space.

Hence, there is no “local” neighborhood in high-dimensions.

2.2.4 Reminder

Linear Regression

$$Y = f(X) + \epsilon$$

where $f(X) = \beta_0 + \sum_{j=1}^p X_j \beta_j$ and X_J can be:

- quantitative inputs
- transformations of quantitative inputs
- polynomial terms
- interactions
- Discretize (i.e., dummy) variable of quantitative inputs.

We assume additive independent errors.

Relationship between n and p	Result
$n \gg p$	low bias, low predictive variance, fit well with test data
$n \approx p$	substantial variability, overfit, fit poorly with test data
$n < p$	no unique solution, variance is infinite

And we fit to (i.e., find β) to minimize the residual sum of squares (RSS):

$$RSS(\beta) = \sum_{i=1}^n (y_i - f(x_i))^2 = \sum_{i=1}^n (y_i - \beta_0 - \sum_{j=1}^p x_{ij}\beta_j)^2 = (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^T (\mathbf{y} - \mathbf{X}\boldsymbol{\beta}) \equiv \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2$$

which is the square of the L^2 norm

differentiating with respect to β

$$\mathbf{X}^T (\mathbf{y} - \mathbf{X}\boldsymbol{\beta}) = 0$$

then

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$$

and

$$\text{var}(\hat{\boldsymbol{\beta}}) = (\mathbf{X}^T \mathbf{X})^{-1} \sigma^2 = (\mathbf{X}^T \mathbf{X})^{-1} \text{var}(\boldsymbol{\epsilon})$$

Prediction

$$\hat{\mathbf{y}} = \mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$$

where

$$\mathbf{H} \equiv \mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T$$

known as the hat matrix, which projects orthogonally the $\mathbf{y}$ vector onto the subspace of $\mathbf{X}$

From the Gauss-Markov theorem, OLS estimates are unbiased and have the smallest variance among all linear unbiased estimates.

For an estimator $\tilde{\boldsymbol{\beta}}$

$$MSE(\tilde{\boldsymbol{\beta}}) = E(\tilde{\boldsymbol{\beta}} - \boldsymbol{\beta})^2 = \text{var}(\tilde{\boldsymbol{\beta}}) + [E(\tilde{\boldsymbol{\beta}}) - \boldsymbol{\beta}]^2 = \text{variance} + \text{bias}^2$$

Since OLS is an unbiased estimator $\tilde{\beta} = \hat{\beta}$

We could get smaller MSE if we allow the estimator to be biased. Hence, OLS might not be the best variance-bias trade-off.

Inference

$$\hat{\sigma}^2 = \frac{1}{n-p-1} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

assuming that $\epsilon \sim N(0, \sigma^2)$ then

$$\hat{\beta} \sim N(\beta, (\mathbf{X}^T \mathbf{X})^{-1} \sigma^2)$$

$$(n-p-1)\hat{\sigma}^2 \sim \sigma^2 \chi_{n-p-1}^2$$

If p is large (i.e., a lot of covariates), you will likely find at least one variable to be significant by chance, thus, you could find a significant partial t-statistics. The F-test does not suffer from this issue because it accounts for the number of predictors.

Issues with linear regression

- Nonlinearity of the response -predictor relationship
- correlated errors
- Non-constant error variance
- high-leverage points
- Collinearity

Rule of thumb:

- parametric methods will tend to outperform non-parametric methods when the number of observations per predictor is small.

2.2.5 Model Assessment

The “generalization” performance of a learning method is how well it predicts independent test data, which is an objective way to choose between models (in terms of **model selection** and **model assessment**)

Suppose we have

- Y = output variable
- X = vector of inputs
- $\hat{f}(X)$ = prediction model estimated from a training set T

Consider errors between Y and $\hat{f}(X)$ through some loss function. For example, **squared-error loss**

$$L(Y, \hat{f}(X)) = (Y - \hat{f}(X))^2$$

or **absolute error loss**

$$L(Y, \hat{f}(X)) = |Y - \hat{f}(X)|$$

2.2.5.1 Types of Error

Test error (or generalization error) is the prediction error over an independent test sample, and is defined as

$$Err_T = E_{X^\circ, Y^\circ}[L(Y^\circ, \hat{f}(X^\circ))|T]$$

where X and Y are assumed to be drawn randomly from their joint population distribution and training set T is fixed

Thus, the test error refers to the error over an independent sample when $f(\cdot)$ is estimated with this specific training set

Expected prediction error (expected test error)

$$Err = E_T E_{X^\circ, Y^\circ}[L(Y^\circ, \hat{f}(X^\circ))|T] = E_T[Err_T]$$

which averages over possible training sets

Training error is the average loss over the specific training sample, given by

$$err = \frac{1}{n} \sum_{i=1}^n L(y_i, \hat{f}(x_i))$$

Even though the interest is in the test error, Err_T , which is almost impossible without a separate test data set. because within a single data set, we attempt to approximate such a validation step by adjusting the training error (e.g., AIC model selection) or by efficient re-use (cross-validation/bootstrap) of the training sample, but then we can only get good estimates of Err (expected prediction error), not Err_T

The bias-variance tradeoff when considering loss function is related to the performance of the training error as a measure of test error

2.2.5.2 Model Error/ Model Selection

Training error is more optimistic (less than) the true test error

To read more on quantifying this optimism (Hastie et al., 2001) Ch. 7.4 where the amount that $\bar{err}$ underestimates the true test error depends on how strongly y_i affects its own prediction.

In other words, the larger $cov(\hat{y}_i, y_i)$, the more optimistic the training error is.

The greater the effective number of parameters (less degrees of freedom) (more flexible model), the larger this co variance

(Hastie et al., 2001) Ch. 7.5 showed that most of model selection criterion (e.g., AIC, BIC C_p) try to estimate prediction error by estimating the optimism and adjusting the training error accordingly to obtain an unbiased estimate of prediction error.

Alternatively, cross-validation and bootstrap methods provide direct estimates of the expected prediction (test) error, Err

Note:

- A model with more effective number of parameters is more flexible

Let

$$\mathbf{y} = (y_1, \dots, y_n)'$$

then a linear fitting method can be written as $\hat{\mathbf{y}} = \mathbf{S}\mathbf{y}$

where $\mathbf{S} = n \times n$ matrix depending on the input vectors x_i , but not on y_i

For example, in linear regression $\mathbf{S} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$

Then the effective number of parameters (df) is

$$df(\hat{\mathbf{y}}) = df(\mathbf{S}) = trace(\mathbf{S})$$

which is the sum of the diagonal elements fo $\mathbf{S}$

If $\hat{\mathbf{y}}$ arises from an additive-error model, $Y = f(X) + \epsilon$, with $var(\epsilon) = \sigma_\epsilon^2$, then

$$df(\hat{\mathbf{y}}) = \frac{\sum_{i=1}^n cov(\hat{y}_i, y_i)}{\sigma_\epsilon^2}$$

2.2.5.3 Cross-validation

If you have a large data set, you could randomly divide it into a training set (to fit the model) and a validation (or holdout, or test) set (to evaluate the test error rate relative to your loss function of choice such as MSE). But it has 2 issues:

- It can be highly variable
- With fewer observations to fit the model, the statistical models perform worse, and validation set error rate may overestimate the test error rate

```
library(tidyverse)
library(ISLR2)
train <- sample(392, 120) # select random subset of 120 out of 392
Auto <- Auto %>% drop_na()
lm.fit <- lm(mpg ~ horsepower, data = Auto, subset = train)

attach(Auto)
mean((mpg - predict(lm.fit, Auto))[-train] ^ 2) # MSE in the validation set (not the training set)

## [1] 27.3254

# Quadratic regression
lm.fit2 <- lm(mpg ~ poly(horsepower, 2), data = Auto, subset = train)
mean((mpg - predict(lm.fit2, Auto))[-train] ^ 2) # MSE

## [1] 21.62483

# Cubic regression
lm.fit3 <- lm(mpg ~ poly(horsepower, 3), data = Auto, subset = train)
mean((mpg - predict(lm.fit3, Auto))[-train] ^ 2) # MSE

## [1] 21.58103
```

2.2.5.3.1 Leave-One-Out CV (LOOCV) A special case of K-fold CV when $K = n$

$$CV_{(n)} = \frac{1}{n} \sum_{i=1}^n MSE_i$$

where $MSE_i = (y_i - \hat{f}^{-i}(x_i))^2$ (alternative loss function could also be used)

- LOOCV has low bias as an estimate of the expected test error rate, but very high variance
- Individual MSE's in LOOCV are highly correlated because they pretty much have the same data set except one observation.
- LOOCV is expensive to implement

```

Auto <- Auto %>% drop_na()
library(tidyverse)
glm.fit <- glm (mpg ~ horsepower , data = Auto) # allows to be passed to cv.glm()
coef(glm.fit)

## (Intercept)  horsepower
## 39.9358610  -0.1578447

# exactly the same as glm without binomial link
lm.fit <- lm(mpg ~ horsepower , data = Auto)
coef(lm.fit)

## (Intercept)  horsepower
## 39.9358610  -0.1578447

library(boot)
glm.fit <- glm (mpg ~ horsepower , data = Auto)
cv.err <- cv.glm (Auto , glm.fit)
cv.err$delta # cross-validation estimate for the test error

## [1] 24.23151 24.23114

# the first delta is the standard CV estimate, the second delta is a bias-corrected ve

# automate to different polynomial
cv.error <- rep (0, 10)
for (i in 1:10) {
  glm.fit <- glm(mpg ~ poly(horsepower , i), data = Auto)
  cv.error[i] <- cv.glm(Auto , glm.fit)$delta[1] # cross-validation estimate for the
}
cv.error # a drop from lienar to quadratic fit

## [1] 24.23151 19.24821 19.33498 19.42443 19.03321 18.97864 18.83305 18.96115
## [9] 19.06863 19.49093

```

2.2.5.3.2 K-Fold Cross-Validation We randomly split the data set into K equal size parts (or folds). Each fold is held out and the model is trained on the remaining $K-1$ folds and then the loss function of choice is evaluated on the fold that was held out

Consider $k = 1, \dots, K$ folds

let $\hat{f}^{-k}(x)$ denote the fitted function computed with the k -th fold removed. We compute the loss based on the samples in the k -th fold.

For example, with squared-error loss, we compute MSE_k for the observations in the k -th fold, $k = 1, \dots, K$. The K -fold cross-validation estimate is

$$CV_{(K)} = \frac{1}{K} \sum_{k=1}^K MSE_k$$

Empirically, good choices for K are 5 or 10.

```
cv.error.10 <- rep(0, 10)
for (i in 1:10) {
  glm.fit <- glm(mpg ~ poly(horsepower, i), data = Auto)
  cv.error.10[i] <- cv.glm(Auto, glm.fit, K = 10)$delta[1] # K-fold cross validation
}
cv.error.10
```

```
## [1] 24.21946 19.39499 19.34473 19.40776 19.00064 19.24864 18.63274 18.94513
## [9] 19.10414 19.11054
```

2.2.5.3.3 Generalized Cross-Validation Recall PRESS statistics

$$\frac{1}{n} \sum_{i=1}^n [y_i - \hat{f}^{-i}(x_i)]^2 = \frac{1}{n} \sum_{i=1}^n \left[\frac{y_i - \hat{f}(x_i)}{1 - S_{ii}} \right]^2$$

where S_{ii} is the i-th diagonal element of $\mathbf{S}$ (or the i-th element of the “hat matrix” diagonal in linear regression). Hence, we only have to fit the model once.

Generalized cross-validation (GCV) is a similar approximation when the trace of $\mathbf{S}$ can be computed more easily than the individual elements S_{ii}

$$GCV(\hat{f}) = \frac{1}{n} \sum_{i=1}^n \left[\frac{y_i - \hat{f}(x_i)}{1 - \text{trace}(\mathbf{S})/n} \right]^2$$

2.2.5.3.4 CV and Classification Common loss function is 0-1 loss where the error rate is defined as

$$L(G, \hat{G}(X)) = I(G \neq \hat{G}(X))$$

where $\hat{G}(X)$ is the highest estimated conditional probability given the inputs.

Alternatively, we could also use deviances that factor in the estimated conditional probability directly. If $p_k(X) = P(G = k|X)$ and $\hat{G}(X)$ = the k that gives the maximum $\hat{p}_k(X)$

$$L(G, \hat{p}(X)) = -2 \sum_{k=1}^K I(G = k) \log \hat{p}_k(X) \equiv -2 \log \hat{p}_G(X)$$

which is -2 times the log-likelihood (i.e., the deviance).

Note:

- CV must be applied to the entire sequence of modeling steps (i.e., samples must be removed before any selection or filtering steps are applied so long as these steps involve the outputs
 - An exception: if there was an unsupervised step in the process associated with the inputs. Since it's unrelated to the outputs, we don't need to do CV on this component.

2.2.5.4 Bootstrapping

- To quantify the uncertainty associated with a given estimator
- Get new data sets by sampling from the observed data set many times with replacement
- Sample B different sets and calculate the quantity of interest about which you want to do inference $\hat{\alpha}^{*1}, \dots, \hat{\alpha}^{*B}$
- Then, you look at the distribution of samples and apply Monte Carlo procedures to do inference
- For example, you want to get the standard error

$$SE_B(\hat{\alpha}) = \sqrt{\frac{1}{B-1} \sum_{r=1}^B (\hat{\alpha}^{*r} - \frac{1}{B} \sum_{r'=1}^B \hat{\alpha}^{*r'})^2}$$

- In principle, you could evaluate model error, but it often overfits because the training and test samples contain some of the same observations, which can be improved by “leave-one-out” bootstrap procedures, but there are still large training set bias problems with the bootstrap.
- (Hastie et al., 2001) give an estimator that helps reduce this bias.
- In general, cross-validation is more effective for evaluating model performance, and bootstrapping is more effective when evaluating the properties of estimators (for inference).
- Rule of thumb for “big data” prediction problem, we can combine cross-validation and hold-out validation:
 - Use K-fold CV on the training data to choose the parameters that control flexibility of the model
 - Use the entire training data to fit the model selected by CV
 - Use a large hold-out sample to evaluate the test error from this fitted model (compare to the CV error).

```
summary(Portfolio)
```

```
##           X                Y
## Min.      :-2.43276   Min.    :-2.72528
## 1st Qu.: -0.88847   1st Qu.: -0.88572
## Median   :-0.26889   Median   :-0.22871
## Mean     :-0.07713   Mean     :-0.09694
## 3rd Qu.:  0.55809   3rd Qu.:  0.80671
## Max.     :  2.46034   Max.     :  2.56599
```

```
alpha.fn <- function(data, index) {
  X <- data$X[index]
  Y <- data$Y[index]
  (var(Y) - cov(X, Y)) / (var(X) + var(Y) - 2 * cov(X, Y)) # statistic of interest
}
```

```
alpha.fn(Portfolio, 1:100)
```

```
## [1] 0.5758321
```

```
alpha.fn(Portfolio, sample(100, 100, replace = T)) # randomly select 100 obs from 1:100 with repl
```

```
## [1] 0.6030941
```

```
# Another faster way
```

```
boot(Portfolio, alpha.fn, R = 100) # R = number of replicates (samples)
```

```
##
## ORDINARY NONPARAMETRIC BOOTSTRAP
##
## Call:
## boot(data = Portfolio, statistic = alpha.fn, R = 100)
##
##
## Bootstrap Statistics :
##      original      bias    std. error
## t1* 0.5758321  0.004565119  0.09480566
```

Another example: SE estimate for linear model

```
boot.fn <- function(data, index){
  coef(lm(mpg ~ horsepower, data = data, subset = index))
}
boot.fn(Auto, 1:392)
```

```
## (Intercept)  horsepower
##  39.9358610  -0.1578447
```

```

boot.fn(Auto, sample(392,392,replace =T))

## (Intercept)  horsepower
##  38.7599822  -0.1496408
boot(Auto, boot.fn,1000)

##
## ORDINARY NONPARAMETRIC BOOTSTRAP
##
##
## Call:
## boot(data = Auto, statistic = boot.fn, R = 1000)
##
##
## Bootstrap Statistics :
##      original      bias    std. error
## t1* 39.9358610  0.0539484426 0.876417119
## t2* -0.1578447 -0.0005380178 0.007604873
# there is a difference between the 2 SE estimates
summary(lm(mpg ~ horsepower, data = Auto))$coef

##              Estimate Std. Error  t value    Pr(>|t|)
## (Intercept) 39.9358610  0.717498656  55.65984 1.220362e-187
## horsepower  -0.1578447  0.006445501 -24.48914 7.031989e-81
# Quadratic example
# boot.fn <- function(data,index){
#   coef(
#       lm(mpg ~ horsepower + I(horsepower^2),
#         data = data, subset = index)
#   )
# }
# boot(Auto, boot.fn, 1000)
# summary(
#   lm(mpg ~ horsepower + I(horsepower^2), data = Auto)$coef
# )

```

The difference in SE estimates from bootstrapping and from a linear model comes from:

1. LM estimates σ^2 using the RSS, which relies heavily on the linear model being correct. In other words, for example, if we have non-linearity in our data, the residuals (e) will be inflated, and leads to $\hat{\sigma}^2$ being inflated as well.
2. LM assumes x_i fixed and variability comes from $\text{var}(\epsilon_i)$

2.3 Improving Prediction Accuracy and Interpretability

If we have

1. large number of predictors
2. highly dependent predictors
3. low interpretability

then, we can either

1. Subset Selection
2. Shrinkage Methods (also known as “*regularization*”): force (shrink) p predictors’ parameters towards zero to reduce variance. It can also be used for variable selection
3. Dimension Reduction: Project the p predictors in an M -dimensional subspace where $M < p$ (i.e., compute M different linear combination or projection of the p predictors and use them as new predictors). (e.g., principal component analysis)

2.3.1 Subset Selection

- Also known as Variable Selection (for more detail click the hyperlink)

2.3.1.1 Best Subset Selection

- Fit all p one by one, then all $p(p-1)/2$ two predictors models, etc. to have 2^p possible models. Then you find the best one using
 - C_p
 - AIC
 - BIC
 - Adjusted R^2
 - cross-validation error measure
- A possible algorithm is the leaps and bounds algorithm by Furnival and Wilson (1974) (up to 40 p)
- Usually, we do best-subset first then apply cross-validation

```
library(tidyverse)
library(ISLR2)
head(Hitters) # check dataset
```

```
##               AtBat Hits HmRun Runs RBI Walks Years CAtBat CHits CHmRun
## -Andy Allanson    293   66     1   30  29    14     1    293   66     1
```

```
## -Alan Ashby      315  81    7  24  38    39   14  3449  835    69
## -Alvin Davis     479 130   18  66  72    76    3  1624  457    63
## -Andre Dawson   496 141   20  65  78    37   11  5628 1575   225
## -Andres Galarra 321  87   10  39  42    30    2   396  101    12
## -Alfredo Griffin 594 169    4  74  51    35   11  4408 1133    19
##
##           CRuns CRBI CWalks League Division PutOuts Assists Errors
## -Andy Allanson    30   29    14    A         E    446    33    20
## -Alan Ashby      321 414   375    N         W    632    43    10
## -Alvin Davis     224 266   263    A         W    880    82    14
## -Andre Dawson    828 838   354    N         E    200    11     3
## -Andres Galarra  48   46    33    N         E    805    40     4
## -Alfredo Griffin 501 336   194    A         W    282   421    25
##
##           Salary NewLeague
## -Andy Allanson    NA         A
## -Alan Ashby      475.0        N
## -Alvin Davis     480.0        A
## -Andre Dawson    500.0        N
## -Andres Galarra  91.5         N
## -Alfredo Griffin 750.0        A
```

```
# check missing values
sum(is.na(Hitters$Salary)) # 59 missing data points
```

```
## [1] 59
```

```
# eliminate missing values
Hitters <- Hitters %>%
  drop_na(Salary)

# Best subset selection (by RSS)
library(leaps)
regfit.full <- regsubsets(Salary ~ ., Hitters)
# summary(regfit.full)
```

2.3.1.2 Stepwise Selection

- With very large numbers of p , stepwise selection can be less computational intensive
- Forward stepwise selection
 - can be used when $p > n$, but only models up to order $n - 1$
- Backward stepwise selection
 - can't be applied when $p > n$
- Hybrid stepwise

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- When applying cross-validation, to account for the variability, we apply the **one-standard-error rule**
 - Evaluates cross-validation criteria (MSE) for each model size
 - Select the smallest model where the estimated test error is within one SE of the lowest point on the curve (i.e. more parsimonious model)

2.3.2 Shrinkage Methods

- Constrain the Least Squares solution in a way that regularizes (shrinks) the coefficients towards 0.
- This leads to biased estimates, but reduce variance

2.3.2.1 Ridge regression

- can also deal with multicollinearity
- Adding a constant to the diagonal of the $\mathbf{X}^T \mathbf{X}$ matrix to get well-behaved (low variance) regression estimates even when $\mathbf{X}^T \mathbf{X}$ is nearly singular.

In LS regression, we minimize the residual sum of squares

$$RSS = \sum_{i=1}^n (y_i - \beta_0 - \sum_{j=1}^p \beta_j x_{ij})^2$$

whereas in ridge regression, we estimate $\hat{\beta}^R$ that minimize

$$\sum_{i=1}^n (y_i - \beta - \sum_{j=1}^p \beta_j x_{ij})^2 + \lambda \sum_{j=1}^p \beta_j^2$$

where λ = tuning parameter

$\lambda \sum_{j=1}^p \beta_j^2$ is an l_2 -norm shrinkage penalty that is small if the β are near 0.

Minimizing this objective function subject to this constraint forces $\hat{\beta}^R$ to be closer to 0 than the LS estimates.

- When $\lambda = 0$, we get the LS estimates
- When $\lambda \rightarrow \infty$, the coefficients go to 0.

Because in the l_2 -norm does not contain the intercept (i.e, we don't shrink the intercept), we have to center all the covariates by removing the mean of each column before we do the regression ($\hat{\beta}_0 = \bar{y}$).

$$\hat{\beta}^R = (\mathbf{X}^T \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^T \mathbf{y}$$

To estimate λ , we use cross-validation

- Ridge estimator is not scale-invariant, so coefficients change substantially depending on the scale of the covariates. Hence, before performing ridge regression, we have to scaled and centered covariates (i.e., after removing the mean, divided each column of $\mathbf{X}$ by its standard deviation).
- λ allows for a continuous range of bias-variance trade-offs.
 - As λ increases, the variance decreases but the bias increases. Hence, it works well when LS estimates have high variance.
- Possible even when $p > n$
- Under Bayesian framework, if $y_i \sim N(\beta_0 + x_i^T \beta, \sigma^2)$ is the likelihood and the parameters have the independent prior distributions $\beta_j \sim N(0, \tau^2)$ with known τ^2, σ^2 , then $\lambda = \sigma^2/\tau^2$ (the ratio of regression to prior variances), and the ridge estimate is the posterior mode (and mean, since the posterior is Gaussian)
- The ridge estimates are related to principal component regression. When principal components associated with the predictors, ridge regression shrinks the low variance principal component the most.

2.3.2.2 Lasso regression

- Even though ridge regression shrinks the parameters to 0, unless $\lambda = \infty$, it cant set any parameter exactly equal 0. Hence, Tibshirani (1996) proposed lasso to handle this problem

The lasso coefficients, $\hat{\beta}^L$ are those that minimize the objective function

$$\sum_{i=1}^n (y_i - \beta_0 - \sum_{j=1}^p \beta_j x_{ij})^2 + \lambda \sum_{j=1}^p |\beta_j|$$

The only difference between ridge and lasso is the penalty term:

- Lasso considers l_1 norm penalty (given by $\|\beta\|_1 = \sum_j |\beta_j|$)
- Ridge considers l_2 norm penalty (given by $\|\beta\|_2 = \sqrt{\sum_j \beta_j^2}$) similar to Euclidean distance

When tuning parameter (λ) is large enough, some coefficients are 0 (are dropped from the model)

There is no closed form for the lasso estimates $\hat{\beta}^L$, the choice of λ can be derived from cross-validation

Note:

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- Lasso outperforms ridge when there are redundant predictors, but ridge outperforms lasso when all included predictors are needed
- Lasso shrinks each parameter by approximately the same amount, while ridge shrinks proportionally
- Lasso equivalent Bayesian interpretation is that: if the prior distribution on the parameters β_j are independent double exponential (Laplace) distribution with mean 0 and scale parameter λ , the posterior mode for β is the lasso solution (but not the posterior mean); the Laplace prior puts more “mass” at 0, giving more a priori belief that the coefficients are 0.

2.3.2.3 Comparison

$$\min_{\beta} \sum_{i=1}^n (y_i - \beta_0 - \sum_{j=1}^p \beta_j x_{ij})^2 \text{ subject to } \begin{cases} \sum_{j=1}^p |\beta_j| \leq s \text{ (lasso)} \\ \sum_{j=1}^p \beta_j^2 \leq s \text{ (ridge)} \\ \sum_{j=1}^p I(\beta_j \neq 0) \leq s \text{ (best subset)} \end{cases}$$

All seek to minimize the RSS subject to constraints on the parameters.

In the best subset method, no more than s coefficients can be nonzero, and can't be computed for large p

2.3.2.4 Cross-validation Fitting

Steps:

1. Choose a “grid” of λ values
2. Compute the cross-validation error for each value of λ
3. Select the λ for which the cross-validation error is smallest (one-standard error rule)
4. Refit the model using all available observations for the selected tuning parameter.

2.3.2.5 Generalization

We could generalize the objective function to

$$\sum_{i=1}^n (y_i - \beta_0 - \sum_{j=1}^p \beta_j x_{ij})^2 + \lambda \sum_{j=1}^p |\beta_j|^q$$

for $q \geq 0$, and $q = 0, 1, 2$ correspond to subset selection, lasso, and ridge respectively

We could estimate q from the data, but it isn't worth.

Elastic net penalty (= combination of ridge and lasso) will be an alternative

The penalty component in the objective function is the priori distribution on the β under Bayesian framework (e.g., a normal prior = ridge regression and a Laplace prior = lasso). Hence, we can use **stochastic search variable selection (SSVS)** to consider model selection George and McCulloch (1993)

SSVS is a hierarchical mixture of normal distribution

$$\beta_j | \gamma_j \sim \gamma_j N(0, c_j \tau_j^2) + (1 - \gamma_j) N(0, \tau_j^2)$$

$$\gamma_j \sim^{idd} \text{Bern}(\pi_j)$$

where $\gamma_j = 1$ means the j -th variable is in the model with probability π_j

We would want τ_j to be small so that when $\gamma_j = 0$, it's reasonable to estimate β_j close to 0.

We would want c_j to be large so that we are more likely to get a non-zero β_j when $\gamma_j = 1$ (which is analogous to ridge regression)

An alternative to SSVS produce is **spike and slab** which replaces the second mixture distribution with a delta function at 0 (analogous to lasso)

2.3.3 Dimension Reduction

Let $Z_1, \dots, Z_M$ represent $M < p$ linear combination of the original p predictors:

$$Z_m = \sum_{j=1}^p \phi_{jm} X_j, m = 1, \dots, M$$

for constants $\phi_{1m}, \dots, \phi_{pm}$

Consider the least squared linear regression model

$$y_i = \theta_0 + \sum_{m=1}^M \theta_m z_{im} + \epsilon_i, i = 1, \dots, n$$

Hence, we will try to find an optimal way to choose ϕ_{jm} so that we will have better results than the least squared solution on the original inputs

Special note

$$\begin{aligned}
\sum_{m=1}^M \theta_m z_{im} &= \sum_{m=1}^M \theta_m \sum_{j=1}^p \phi_{jm} x_{ij} \\
&= \sum_{j=1}^p \sum_{m=1}^M \theta_m \phi_{jm} x_{ij} \\
&= \sum_{j=1}^p \beta_j x_{ij}
\end{aligned}$$

Hence,

$$\beta_j = \sum_{m=1}^M \theta_m \phi_{jm}$$

This means the reduced dimension regression is a special case of OLS except that the parameters β_j are constrained.

Hence, when $M = p$ and Z_m are all linearly independent (no constraints), the reduced dimension regression is equivalent to the OLS. The only benefit is that Z_m 's are orthogonal to each other

Similarly to the Shrinkage Methods, the constrained solution is biased, but typically leads to lower variance in the fitted coefficients - especially when $M \ll p$

2.3.3.1 Principal Component Regression

Steps:

1. Find the linear combination of the inputs that accounted for the largest fraction of variance of the inputs
2. Find another linear combination of the inputs that accounted for the next highest amount of variation (subject to being orthogonal to the first)
3. Continue the process

Mathematically, principal components could also be obtained using **singular value decomposition** (SVD) for the centered data matrix

Let $\mathbf{X}$ ($n \times p$) be the centered data matrix, then

$$\mathbf{X} = \mathbf{U}\mathbf{D}\mathbf{V}^T$$

where $\mathbf{U}$ and $\mathbf{V}$ are $n \times p$ and $p \times p$ orthogonal matrices

$\mathbf{D}$ (**singular values**) is $p \times p$ diagonal matrix with diagonal entries $d_1 \geq d_2 \geq \dots \geq d_p \geq 0$

Then,

$$\mathbf{X}^T \mathbf{X} = \mathbf{V} \mathbf{D}^2 \mathbf{V}^T$$

which is the eigen-decomposition of $\mathbf{X}^T \mathbf{X}$ and of the sample covariance matrix $\mathbf{S} = \frac{\mathbf{X}^T \mathbf{X}}{n}$ up to the factor n

And the columns of $\mathbf{V}$ are **eigenvectors** and correspond to the **principal component weights**

Hence, the new variables are

$$\mathbf{z}_1 = \mathbf{X} \mathbf{v}_1, \dots, \mathbf{z}_M = \mathbf{X} \mathbf{v}_M$$

where $\mathbf{v}_m$ is the m -th column of $\mathbf{V}$

These $\mathbf{z}$'s are the principal components.

The elements of $\mathbf{v}_m$ are ϕ 's in the dimension reduction summary

$$var(\mathbf{z}_m) = \frac{d_m^2}{n}$$

Thus, large (small) singular values correspond to directions in the column space of $\mathbf{X}$ that have large (small) variance

Then principle component regression basically construct the first M principal components (i.e., $Z_1, \dots, Z_M$) and uses these components as predictors in the linear regression model fit by least squares

Assumption:

- Directions in which $X_1, \dots, X_p$ have the most variation are the directions that are most associated with Y (not always the case, but typically effective). In other words, these X 's are mostly associated with Y (best predict), but we cannot be certain that X 's variation causes Y 's variation.

Notes:

- While capturing most of the variance from inputs, since $M \ll p$ we don't run into the problem of overfitting
- Another benefit is that the new predictors are orthogonal ($\mathbf{z}_j^T \mathbf{z}_k = 0, j \neq k$), which in the LS estimation we typically run into the problem of collinearity
- Even though PCR uses a low-dimensional set of predictors, no model selection was performed because we only translate all p variables into a linear combination that makes the new PC inputs. Hence, it's closely

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related to Ridge regression than Lasso regression or Best Subset Selection. Seeing Singular Value Decomposition

- $\mathbf{X}\hat{\beta}^R = \sum_{j=1}^p \mathbf{u}_j \frac{d_j^2}{d_j^2 + \lambda} \mathbf{u}_j^T \mathbf{y}$
- where $\mathbf{u}_j$ corresponds to the j -th column of $\mathbf{U}$ from the SVD
- Since LS estimate gives $\mathbf{X}\hat{\beta} = \mathbf{U}\mathbf{U}^T \mathbf{y}$, ridge regression goes to the LS solution when $\lambda = 0$
- The singular vectors that have smaller variation (i.e., smaller d_j^2) are shrunk more than those with larger variation
- Rule of thumb: standardize the predictors before performing the Singular Value Decomposition to generate the principle components.
 - If we don't standardize, the predictors with the largest scale typically account for the most variance, regardless of their importance relative to the output.

2.3.3.2 Partial Least Squares

- Find linear combinations of the data (or directions in the $\mathbf{X}$ matrix) that maximize **variance** and are **orthogonal**
- Find the weight just like [Principle Component Regression], but PCR in a unsupervised way because the choice of these directions is not dependent on Y . Hence, might not provide good predictions of Y
- PLS is a supervised way to do dimension reduction that finds new features $Z_1, \dots, Z_M$ that are linear combinations of the original inputs, but it selects these in a way to account for the covariability with the response Y

Steps to calculate the PLS directions ($Z_m = \sum_{j=1}^p \phi_{jm} X_j$)

1. Standardize the p predictors
2. For Z_1 , set each ϕ_{j1} = the coefficient from the simple linear regression of Y on X_j (this is proportional to the correlation between Y and X_j)
3. Calculate Z_2 , by calculating the residuals of a regression of each variable (separately) on Z_1 (call these e_j), which corresponds to the info remaining that has not been accounted for by Z_1
4. Set ϕ_{j2} = the coefficient from the simple linear regression of Y on each e_j to get Z_2
5. Repeat M to get $Z_1, \dots, Z_M$

Comparison to PCR: Hastie et al. (2001) (chapter 3.5)

The m -th PC direction coefficients $\mathbf{v}_m$ are the α that solve

$$\max_{\alpha} [\text{var}(\mathbf{X})]$$

subject to $\|\alpha\| = 1, \alpha^T \mathbf{S} \mathbf{v}_l = 0, l = 1, \dots, m-1$ (the last condition ensures that each $\mathbf{z}_m = \mathbf{X} \mathbf{v}_m$ is uncorrelated with all of the previous $\mathbf{z}_l$)

where $\mathbf{S}$ is the sample covariance matrix associated with $\mathbf{X}$

The m -th PLS direction $\hat{\phi}_m$ have the α that solves:

$$\max_{\alpha} [\text{corr}^2(\mathbf{y}, \mathbf{X}) \text{var}(\mathbf{X})]$$

subject to $\|\alpha\| = 1, \alpha^T \mathbf{S} \hat{\phi}_l = 0, l = 1, \dots, m-1$

Notes:

- The extra component in PLS helps make the weight more relatable to $\mathbf{Y}$ (but sometimes it's more efficient, but sometimes it isn't)
- Both $\mathbf{X}$ and $\mathbf{Y}$ should be standardized before performing PLS
- The choice of M is made based on Cross-validation
- Although PLS tries to maximize both the correlation to the output and the variance in the inputs, the variance tends to dominate and PLS behaves very similarly to ridge regression and PCR
- The use of the output to supervise the selection of the reduced dimension can reduce bias, but increase variance (bias-variance trade-off), and reduce the added value of the approach relative to PCR
- PLS is closely related to canonical correlation analysis (CCA) in multivariate statistic, which seeks to find a reduced set of linear combinations of inputs and linear combination of outputs that maximize the correlation (e.g., finding the coefficient α, β s.t

$$\max_{\alpha, \beta} [\text{corr}(\mathbf{Y}, \mathbf{X})]$$

where a set of M with orthogonality constraints

- In high-dimensional space ($p \approx n; p > n$), models are easily overfitted (too flexible). Hence, they might appear better than they really are. Hence, to deal with high dimensional data, we use less flexible models (i.e., those that are regularized or constrained), such as Best Subset Selection, Ridge regression, Lasso regression, Principal Component Regression, Partial Least Squares. But we still need to consider their issues
 - Large p always has collinearity; Best Subset Selection or Lasso regression can't guarantee the chosen model is a scientific one

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- p-values are not considered under high-dimensional regressions
- Tuning parameter selection is utmost important for good predictive performance
- The test error tends to increase as the dimensionality of the problem increases (Curse of dimensionality)

Part II

SUPERVISED LEARNING

Part III

1. LOCAL BASED

Chapter 3

Spline Regression

Beyond Linearity (including transformation of variables), we have arbitrary basis expansion (where the basis functions are dependent on X)

3.1 Basis Function Representations

Let the k -th basis function of X be denoted by $b_k(X)$, $k = 1, \dots, K$ which is function that maps from the p -dimensional real numbers to the real line.

Then, the linear functional representation of $f(X)$ given by these basis functions is

$$f(X) = \beta_0 + \sum_{k=1}^K \beta_k b_k(X)$$

Equivalently

$$\mathbf{Y} = \mathbf{B}\beta +$$

where $\mathbf{B}$ is the $n \times (K+1)$ matrix columns given by a vector of 1 for the intercept and $\mathbf{b}_k = (b_k(x_1), \dots, b_k(x_n))'$, $k = 1, \dots, K$

Then, we have

$$\hat{\beta} = (\mathbf{B}^T \mathbf{B})^{-1} \mathbf{B}^T \mathbf{y}$$

if the basis functions are **orthogonal**, then we have a special case $\hat{\beta} = \mathbf{B}^T \mathbf{y}$

Some common basis functions;

- Original linear model: $b_k(X) = X_k, k = 1, \dots, p$
- Polynomials: $b_k(X) = X_j^2; X_j^3; X_j X_m$
- Non-linear transformations: $b_k(X) = \log(X_j); \sqrt{X_j}$
- Step-functions: $b_k(X) = I(c_k \leq X \leq v_k)$, an indicator for a region of X
 1. Breaking the region up into non-overlapping regions and step-function response
 2. Example: for a single X , $b_0(X) = I(X < c_1), \dots, b_{K-1}(X) = I(c_{K-1} \leq X \leq c_K)$ where $c_1, \dots, c_K$ are cut points (i.e., making dummy variables and piece wise constant regression fits)
- Other: sines and cosines, wavelets, empirical basis functions

Another representation

More info can be found on CMU stat

Definition: a k -th order spline is a piecewise polynomial function of degree k , that is continuous and has continuous derivatives of orders $1, \dots, k-1$, at its knot points

Equivalently, a function f is a k -th order spline with knot points at $t_1 < \dots < t_m$ if

- f is a polynomial of degree k on each of the intervals $(-\infty, t_1], [t_1, t_2], \dots, [t_m, \infty)$
- $f^{(j)}$, the j -th derivative of f , is continuous at $t_1, \dots, t_m$ for each $j = 0, 1, \dots, k-1$

A common case is when $k = 3$, called cubic splines. (piecewise cubic functions are continuous, and also continuous at its first and second derivatives)

To parameterize the set of splines, we could use **truncated power basis**, defined as

$$g_1(x) = 1g_2(x) = x \dots g_{k+1}(x) = x^k g_{k+1+j}(x) = (x - t_j)_+^k$$

where $j = 1, \dots, m$ and $x_+ = \max\{x, 0\}$

However, now software typically use B-spline basis.

3.2 Regression Splines

- Spline basis functions are built upon polynomial regression and piecewise constant regression
 - We fit piece wise polynomial regressions, but add some continuity and smoothness constraints.

Example:

$$y_i = \begin{cases} \beta_{01} + \beta_{11} + \beta_{21}x_i^2 + \beta_{31}x_i^3 + \epsilon & \text{if } x_i < c \\ \beta_{02} + \beta_{12} + \beta_{22}x_i^2 + \beta_{32}x_i^3 + \epsilon & \text{if } x_i \geq c \end{cases}$$

where c is a **knot** point

The more flexible we want our model to be, the more knots to be added.

If we have K knots through the range of X , we fit $K + 1$ different (cubic or any or constants) polynomials.

To restrict piece coming together at the knot points, we add the constraint that the fitted curve must be **continuous**.

We constrain the first and second derivatives to be continuous for **smoothness**.

If the polynomials are cubic, then it's cubic spline.

Each constraint that we add reduces the number of parameters to be fit (frees up df), and less complex

A cubic spline with K knots has $K + 4$ df

We can rewrite the cubic spline into the general basis notation

$$b_1(X) = X; b_2(X) = X^2; b_3(X) = X^2; b_4(X) = h(X, \xi_1); b_5(X) = h(X, \xi_2), \dots, b_{K+3}(X) = h(X, \xi_K)$$

where

$$h(x, \xi) = (x - \xi)_+^3 = \begin{cases} (x - \xi)^3 & \text{if } x > \xi \\ 0 & \text{otherwise} \end{cases}$$

and ξ is the knot. Fitting this model with an intercept requires estimating $K + 4$ regression coefficients (i.e., $K + 4$ df)

Notes:

- Splines have high variance at the outer range of the predictors.
 - Natural splines are regression splines with extra constraints on the boundary that mitigates this variance problem by requiring the function to be linear in the regions where X is smaller than the smallest knot, and larger than the largest knot.
 - To decide number and place of knots (K)
 - * To choose number of K , people usually do cross-validation

- * Given K knots, we locate them uniformly in the quantile domain of X
 - Ex: $K = 3$, a knot would be placed at the 25th, 50th, and 75th percentiles of the variable X .
 - Alternatively, we can place knots based on local data abundance.

Another representation

To estimate the regression function $r(X) = E(Y|X = x)$, we can fit a k -th order spline with **knots** at some prespecified locations $t_1, \dots, t_m$

Regression splines are functions of

$$\sum_{j=1}^{m+k+1} \beta_j g_j$$

where

$\beta_1, \dots, \beta_{m+k+1}$ are coefficients $g_1, \dots, g_{m+k+1}$ are the truncated power basis functions for k -th order splines over the knots $t_1, \dots, t_m$

To estimate the coefficients

$$\sum_{i=1}^n (y_i - \sum_{j=1}^m \beta_j g_j(x_i))^2$$

then regression spline is

$$\hat{r}(x) = \sum_{j=1}^{m+k+1} \hat{\beta}_j g_j(x)$$

If we define the basis matrix $G \in R^{n \times (m+k+1)}$ by

$$G_{ij} = g_j(x_i)$$

where $i = 1, \dots, n$, $j = 1, \dots, m + k + 1$

Then,

$$\sum_{i=1}^n (y_i - \sum_{j=1}^m \beta_j g_j(x_i))^2 = \|y - G\beta\|_2^2$$

and the regression spline estimate at x is

$$\hat{r}(x) = g(x)^T \hat{\beta} = g(x)^T (G^T G)^{-1} G^T y$$

3.3 Natural splines

A natural spline of order k , with knots at $t_1 < \dots < t_m$, is a piecewise polynomial function f such that

- f is polynomial of degree k on each of $[t_1, t_2], \dots, [t_{m-1}, t_m]$
- f is a polynomial of degree $(k-1)/2$ on $(-\infty, t_1]$ and $[t_m, \infty)$
- f is continuous and has continuous derivatives of orders $1, \dots, k-1$ at its knots $t_1, \dots, t_m$

Note

natural splines are only defined for odd orders k .

3.4 Smoothing splines

We are interested in a function $g(x)$ and try to minimize $RSS = \sum_{i=1}^n (y_i - g(x_i))^2$

But we wouldn't want to overfit (because $g(x)$ that minimizes this RSS will try to interpolate all of the y_i), then we want a smooth $g(x)$ that minimizes RSS:

$$\sum_{i=1}^n (y_i - g(x_i))^2 + \lambda g''(x)^2 dx$$

where

- λ is a non-negative smoothness
- $g(\cdot)$ is a **smoothing spline**

Notes:

- RSS here is a *loss function*
- second term is a *smoothness penalty term*
 - Second derivative controls the smoothness of the function (smaller in absolute value means smoother)
 - Parameter λ controls the tradeoff between the smoothness and fit
- When $\lambda = 0$, the function is just like linear function (not smooth at all)
- When $\lambda \rightarrow \infty$, then g approaches a straight line that passes through the training data

- The smoothing spline function corresponds to a cubic polynomial basis with knots at the unique values of $x_1, \dots, x_n$ with continuous first and second derivatives at each knot
 1. $g(x)$ that minimizes the smoothing spline constraint is just a natural cubic spline with knots at the unique x observations
 2. λ controls the level of shrinkage for smoothing splines, but there is none in regression spline
 3. At first glance, we might think that smoothing splines might have too many degrees of freedom since we have knots at every data points (i.e., $K = n$). But λ controls roughness and make the **effective degrees of freedom** much smaller (as λ increases from $0 \rightarrow \infty$, df_λ decreases from $n \rightarrow 2$)

Let $\hat{\mathbf{y}}_\lambda$ be the n -vector containing the fitted values for the smoothing spline function for a given λ , then

$$\hat{\mathbf{y}}_\lambda = \mathbf{S}_\lambda \mathbf{y}$$

which means that it is linear in $\mathbf{y}$

According to Hastie et al. (2001) (section 5.4) showed

$$\mathbf{S}_\lambda = \mathbf{N}(\mathbf{N}^T \mathbf{N} + \lambda \mathbf{N})^{-1} \mathbf{N}^T \text{ where } \{N\}_{ij} = b_j(x_i) \text{ and } \{ \}_{jk} = \int b_j''(t) b_k''(t) dt$$

The effective degrees of freedom to be trace of $\mathbf{S}_\lambda$ (**smoother matrix** because it smooths the responses in $\mathbf{y}$) is

$$df_\lambda = \sum_{i=1}^n \{\mathbf{S}_\lambda\}_{ii}$$

$\mathbf{S}_\lambda$ is related to other types of smoothing in Local Regression and Kernel Smoothing

To choose λ , we can use cross-validation (could also leverage the relationship between λ and effective degrees of freedom).

If we can write a **linear predictor**, we can use a simplified LOOCV formula:

$$\begin{aligned} RSS_{CV}(\lambda) &= \sum_{i=1}^n (y_i - \hat{y}_\lambda^{(-i)}(x_i))^2 \\ &= \sum_{i=1}^n \left[\frac{y_i - \hat{y}_\lambda(x_i)}{1 - \{\mathbf{S}_\lambda\}_{ii}} \right]^2 \end{aligned}$$

where we only have to fit the model once for each λ

3.5 Multidimensional Splines

We can extend the above one-dimensional spline models to higher dimensions using **tensor products** to form higher dimensional basis functions

Consider $X \in R^2$ and

- basis functions $b_{1j}(X_1), j = 1, \dots, K_1$ for representing coordinate X_1
- $b_{2k}(X), k = 1, \dots, K_2$ for coordinate X_2

then the *tensor product basis* is:

$$g_{jk}(X) = b_{1j}(X_1)b_{2k}(X_2)$$

where $j = 1, \dots, K_1, k = 1, \dots, K_2$

which can be used as

$$f(X) = \sum_{j=1}^{K_1} \sum_{k=1}^{K_2} \beta_{jk} g_{jk}(X)$$

Then, we seek to minimize

$$\sum_{i=1}^n (y_i - f(x_i))^2 + \lambda J[f]$$

where J is the penalty function for the dimension of the problem

In our example ($X \in R^2$), then

$$J[f] = \int \int [(\frac{\partial^2 f(x)}{\partial x_1^2})^2 + 2(\frac{\partial^2 f(x)}{\partial x_1 \partial x_2})^2 + (\frac{\partial^2 f(x)}{\partial x_2^2})^2] dx_1 dx_2$$

Similar to the one-dimensional case

- $\lambda \rightarrow 0$ gives an interpolating function that goes through all points
- $\lambda \rightarrow \infty$ gives a solution approaching the least squares plane
- We seek a λ that compromises between these extremes

This splines method is known as **thin-plates splines**.

The solution to the thin-plate spline minimization gives

$$f(x) = \beta_0 + \beta^T x + \sum_{j=1}^n \alpha_j b_j(x)$$

where $b_j(x) = \|x - x_j\|^2 \log \|x - x_j\|$ (known as **radial basis functions**)

Additional splines basis functions could be

- B-spline

Local Regression and Kernel Smoothing are related to the spline basis regression formulations.

Another representation

These estimators use a regularized regression over the natural spline basis: placing knots at all points $x_1, \dots, x_n$

For the case of cubic splines, the coefficients are the minimization of

$$\|y - G\beta\|_2^2 + \lambda\beta^T\Omega\beta$$

where $\Omega \in R^{n \times n}$ is the penalty matrix

$$\Omega_{ij} = \int g_i''(t)g_j''(t)dt,$$

and $i, j = 1, \dots, n$

and $\lambda\beta^T\Omega\beta$ is the **regularization term** used to shrink the components of $\hat{\beta}$ towards 0. $\lambda > 0$ is the tuning parameter (or smoothing parameter). Higher value of λ , faster shrinkage (shrinking away basis functions)

Note

smoothing splines have similar fits as kernel regression.

	Smoothing splines	kernel regression
tuning parameter	smoothing parameter λ	bandwidth h

3.6 Application

3.6.1 Example 1

(James et al., 2013, 7.8)

3.6.1.1 Regression Spline Example

```
library(ISLR2)
attach(Wage)
library(splines)
```

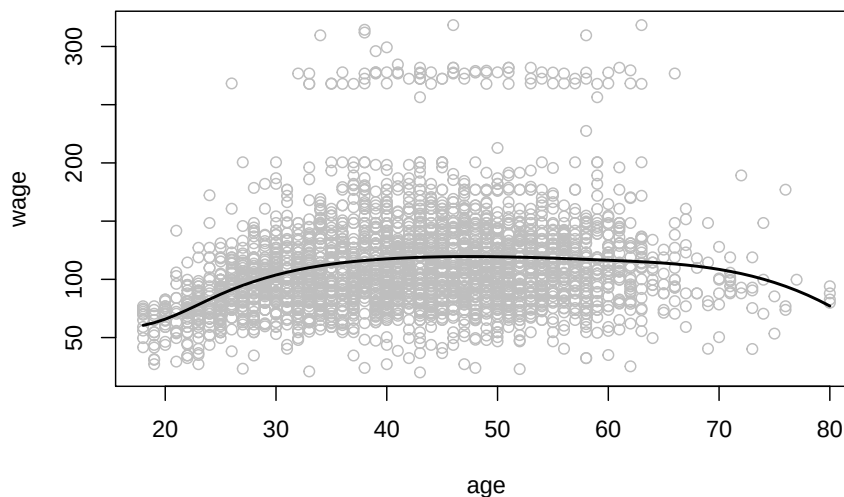
```

# using basis functions for splines with the sepcifeid set of knots
fit <- lm(wage ~ bs(age, knots = c(25, 40, 60)), data = Wage) # 3 knots mean 6 basis function, h

# we want to predict for a grid of values for age
agelims <- range(age)
age.grid <- seq(from = agelims[1], to = agelims[2])

pred <- predict(fit, newdata = list(age = age.grid), se = T)
plot(age, wage, col = "gray")
lines(age.grid, pred$fit, lwd = 2)

```



```

# lines(age.grid, pred$fit + 2 * pred$se, lty = "dashed")
# lines(age.grid, pred$fit - 2 * pred$se, lty = "dashed")

```

Alternatively, we produce knots at uniform quantiles of the data

```
dim(bs(age, knots = c(25, 40, 60)))
```

```
## [1] 3000    6
```

```
dim(bs(age, df = 6))
```

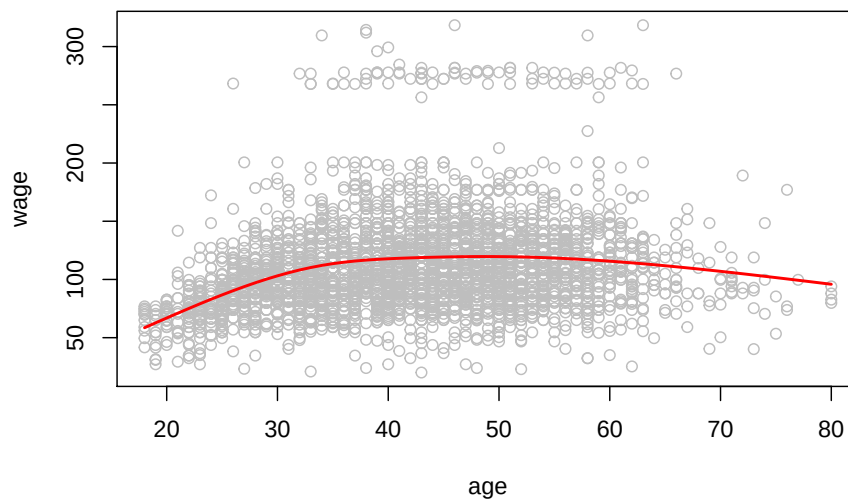
```
## [1] 3000    6
```

```
attr(bs(age, df = 6), "knots")
```

```
##    25%    50%    75%
## 33.75 42.00 51.00
```

3.6.1.2 Natural Spline Example

```
fit2 <- lm(wage ~ ns(age, df = 4), data = Wage) # 4 df
pred2 <- predict(fit2, newdata = list(age = age.grid), se = T)
plot (age , wage , col = "gray")
lines(age.grid, pred2$fit, col = "red", lwd = 2)
```

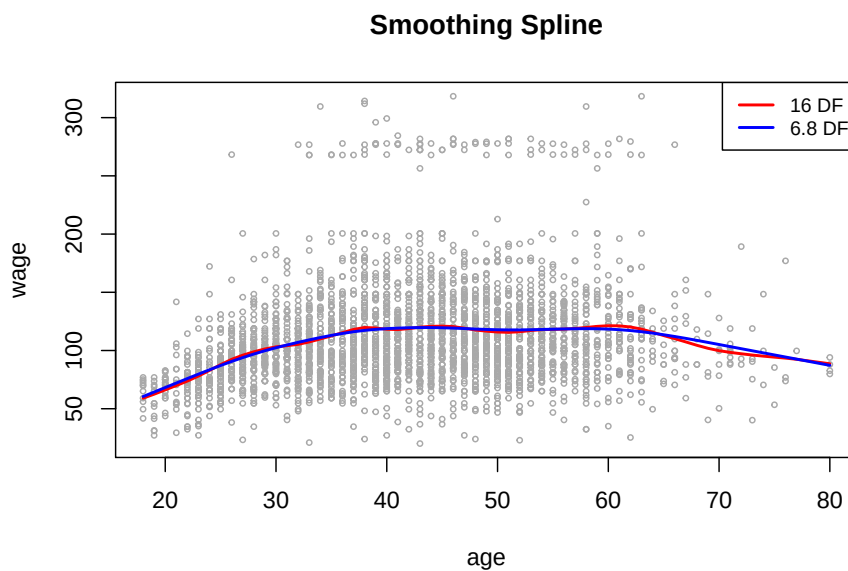


Alternatively, we specify the knots using the `knots` options (similar to Regression Spline Example)

3.6.1.3 Smoothing Spline Example

```
plot(age,
      wage,
      xlim = agelims,
      cex = .5,
      col = "darkgrey")
title("Smoothing Spline")
fit <- smooth.spline(age, wage, df = 16) # we specify 16 df then R will find lambda th
fit2 <- smooth.spline(age, wage , cv = T) # we select lambda (smoothness) based on cro
fit2$df # which result in 6.8 df
```

```
## [1] 6.794596
lines(fit, col = "red", lwd = 2)
lines(fit2, col = "blue", lwd = 2)
legend(
  "topright",
  legend = c("16 DF", "6.8 DF"),
  col = c("red", "blue"),
  lty = 1,
  lwd = 2,
  cex = .8
)
```



3.6.2 plot (Example 2)

```
library(tidyverse)
library(caret)
theme_set(theme_classic())

# Load the data
data("Boston", package = "MASS")
# Split the data into training and test set
set.seed(123)
training.samples <- Boston$medv %>%
  createDataPartition(p = 0.8, list = FALSE)
```

```

train.data <- Boston[training.samples, ]
test.data <- Boston[-training.samples, ]

knots <- quantile(train.data$lstat, p = c(0.25, 0.5, 0.75)) # we use 3 knots at 25,50,

library(splines)
# Build the model
knots <- quantile(train.data$lstat, p = c(0.25, 0.5, 0.75))
model <- lm (medv ~ bs(lstat, knots = knots), data = train.data)
# Make predictions
predictions <- model %>% predict(test.data)
# Model performance
data.frame(
  RMSE = RMSE(predictions, test.data$medv),
  R2 = R2(predictions, test.data$medv)
)

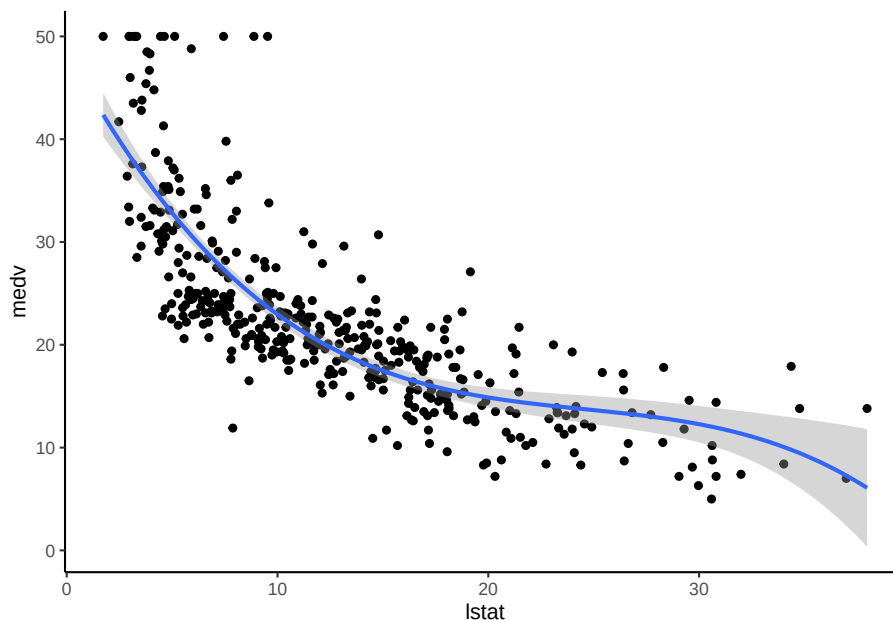
##          RMSE          R2
## 1 5.317372 0.6786367

```

```

ggplot(train.data, aes(lstat, medv) ) +
  geom_point() +
  stat_smooth(method = lm, formula = y ~ splines::bs(x, df = 3))

```



```

attach(train.data)
#fitting smoothing splines using smooth.spline(X,Y,df=...)

```

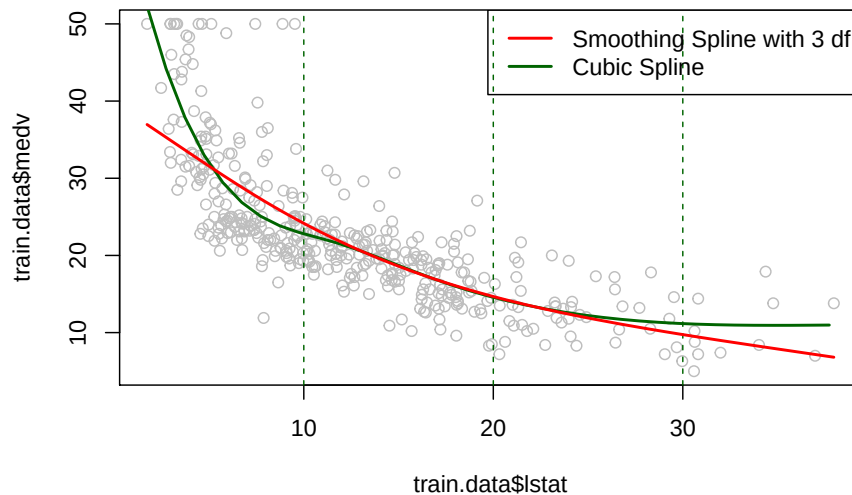


```

fit1<-smooth.spline(train.data$lstat,train.data$medv,df=3 ) # 3 degrees of freedom
#Plotting both cubic and Smoothing Splines
plot(train.data$lstat,train.data$medv,col="grey")
lstat.grid = seq(from = range(lstat)[1], to = range(lstat)[2])
points(lstat.grid,predict(model,newdata = list(lstat=lstat.grid)),col="darkgreen",lwd=2,type="l")
#adding cutpoints
abline(v=c(10,20,30),lty=2,col="darkgreen")
lines(fit1,col="red",lwd=2)

legend("topright",c("Smoothing Spline with 3 df","Cubic Spline"),col=c("red","darkgreen"),lwd=2)

```



Chapter 4

Kernel Smoothing

- also known as kernel regression
- The idea is that we fit a separate, but simple, model at each selected point x_0 by using only those observation that are “close” to x_0 such that $\hat{f}(X)$ is smooth in R^p
- The “localization” is accomplished by a **weighting function** (or **kernel**) $K_\lambda(x_i, x_i)$ that assigns a weight to x_i based on its distance from x_0
 - This kernel depends on a parameter λ that defines the “width” or neighborhood” of the kernel
- This method requires less training than traditional regression models (typically, only λ need be chosen), but are **memory-based methods** in that one needs all the training data any time we want to make a prediction at a location x_0

4.1 One-Dimensional Kernel Smoothers

Under k-nearest neighbor, we try to fit

$$\hat{f}(x) = Ave(y_i | x_i \in N_k(x))$$

where $N_k(x)$ refers to the set of k neighbors nearest the point x in squared distance.

Alternatively, we can consider “**Nadaraya-Watson**” **kernel-weighted average** for our estimate

$$\hat{f}(x_0) = \frac{\sum_{i=1}^n K_\lambda(x_0, x_i) y_i}{\sum_{i=1}^n K_\lambda(x_i, x_i)}$$

where the kernel is given by

$$K_\lambda(x_0, x) = D\left(\frac{|x - x_0|}{\lambda}\right)$$

The difference is that now we have a **continuous** function, as compared to **discrete** under the nearest neighbor fit (which can come in handy in some cases).

Examples of the kernel function

1. Epanechnikov quadratic kernel (has no continuous derivatives at the boundary of its support)

$$D(t) = \begin{cases} \frac{3}{4}(1 - t^2) & \text{if } |t| \leq \lambda \\ 0 & \text{otherwise} \end{cases}$$

2. Tri-Cube kernel (compact, has 2 continuous derivatives at the boundary of its support)

$$D(t) = \begin{cases} (1 - |t|^3)^3 & \text{if } |t| \leq \lambda \\ 0 & \text{otherwise} \end{cases}$$

where $t = |x - x_0|$

3. Gaussian density function (continuously differentiable, but infinite support, could be more taxing on computation):

$$D(t) = \frac{\exp(-\frac{t^2}{2\sigma^2})}{\sqrt{2\pi}\sigma}$$

where σ is the “width”

Alternatively, we can also use **adaptive neighborhoods** where the λ parameter depends on the location x_0 . For example,

$$K_\lambda(x_0, x) = D\left(\frac{|x - x_0|}{h_\lambda(x_0)}\right)$$

Notes:

- Large λ implies lower variance (we average over more observations, larger n), but higher bias (we are approximating $f(x_0)$ with observation further away from x_0)
- An adaptive parameter $h_\lambda(x)$ can keep the bias **constant** but the variance is **inversely proportional** to the local density of observation.
- Biggest problem with kernel smoothers is estimation near the boundaries (primarily due to the asymmetry of the kernels near the boundary); hence, the fits are badly biased near the boundaries, which Local Regression can help.

4.2 Selecting the Kernel Width

λ controls the “width” of kernel K_λ

- Epanechnikov/Tri-Cube: λ is the radius of the support region
- Gaussian: λ is the standard deviation
- Nearest Neighbor: λ is the number of k of nearest neighbors

Notes:

- Smaller width means less averaging, hence higher variance and smaller bias
- Larger width means more averaging, hence less variance and higher bias

To select λ , we typically use Cross-validation

4.3 Kernel Density Estimation

Another use of kernels is to estimate density (which was the original purpose of kernel before kernel regression).

Kernel density estimation is

Assume random sample $x_1, \dots, x_n$ from a probability density $f_X(x)$ and we want to estimate f_X at a point x_0 (assuming one dimension $X \in R$)

The **Parzen estimate** is

$$\hat{f}_X(x_0) = \frac{1}{n\lambda} \sum_{i=1}^n K_\lambda(x_0, x_i)$$

where a choice of kernels is the Gaussian kernel with standard deviation λ , which could be extended to higher dimensional density estimation by using multivariate (Gaussian) kernels.

Non parametric density estimates can be used under the Bayes' theorem

Suppose J classes and fit the nonparametric densities $\hat{f}_j(X), j = 1, \dots, J$ separately for each class

Assume we have prior probability for each class $\hat{\pi}_j$ (which might be from the sample proportions or some previous knowledge), then

$$\hat{P}(G = j|X = x_0) = \frac{\hat{\pi}_j \hat{f}_j(x_0)}{\sum_{k=1}^J \hat{\pi}_k \hat{f}_k(x_0)}$$

For more details, see Hastie et al. (2001) (section 6.6)

Chapter 5

Local Regression

- This method improve bias from Kernel Smoothing at the boundaries by fitting straight lines instead of constants.
- Also known as the **locally weighted linear regression**

It seeks to

$$\min_{\alpha(x_0), \beta(x_0)} \sum_1^n K_\lambda(x_0, x_i) [y_i - \alpha(x_0) - \beta(x_0)x_i]^2$$

and gives the estimate $\hat{f}(x_0) = \hat{\alpha}(x_0) + \hat{\beta}(x_0)x_0$

To fit this, we let $b(x)^T = (1, x)$ then define the $n \times 2$ matrix $\mathbf{B}$ to have i-th row given by $b(x_i)^T$

Then, let $\mathbf{W}(x_0)$ be the $n \times n$ diagonal matrix with the i-th diagonal given by $K_\lambda(x_0, x_i)$. Then

$$\hat{f}(x_0) = b(x_0)^T (\mathbf{B}^T \mathbf{W}(x_0) \mathbf{B})^{-1} \mathbf{B}^T \mathbf{W}(x_0) \mathbf{y} = \sum_{i=1}^n l_i(x_0) y_i$$

Thus, the fitted solution is linear in the y_i with the weights in the linear combination given by $l_i(x_0)$, which are a combination of the kernel weights and least squares equations, which is called **equivalent kernel**

An advantage of this method is that it corrects the boundary bias issues to first order (i.e., the bias is only a function of the second derivatives of $f(x_0)$ and so is small on average).

Thus, there is a small price we pay in that this fit tends to be a little bit biased in regions of curvature in the true function.

We can improve this bias by using **locally polynomial regression** (e.g., **locally quadratic regression**):

$$\min_{\alpha(x_0), \beta_j(x_0), j=1, \dots, d} \sum_{i=1}^n K_\lambda(x_0, x_i) [y_i - \alpha(x_0) - \sum_{j=1}^d \beta_j(x_0) x_i^j]^2$$

which gives the solution $\hat{f}(x_0) = \hat{\alpha}(x_0) + \sum_{j=1}^d \hat{\beta}_j(x_0) x_0^j$

The reduction in bias under locally quadratic fit offset by higher variance

Notes:

- Local linear fits are much better at the boundaries than higher order polynomials in reducing bias, without leading to excessive variance increase.
- Local quadratic fits do not help bias much at the boundaries, but increase the variance a lot
- Local quadratic fits are usually used in reducing bias from curvature in the interior of the domain
- Local polynomials of odd degree dominate those of even degree (based on asymptotic analysis because asymptotically the MSE is dominated by boundary effects).

5.1 Higher Dimensions

To generalize Kernel Smoothing and Local Regression with p -dimensional kernel, we consider local hyperplanes

Example:

- If $d = 1$ (polynomial degree) and $p = 2$, we have $b^T(X) = (1, X_1, X_2)$
- If $d = 2$ and $p = 2$ we have $b^T(X) = (1, X_1, X_2, X_1^2, X_2^2, X_1 X_2)$

then, we solve

$$\min_{\beta(x_0)} \sum_{i=1}^n K_\lambda(x_0, x_i) [y_i - b(x_i)^T \beta(x_0)]^2$$

to fit $\hat{f}(x_0) = b(x_0)^T \hat{\beta}(x_0)$

Then, the kernels of the general form would be

$$K_\lambda(x_0, x) = D\left(\frac{\|x - x_0\|}{\lambda}\right)$$

where $\|\cdot\|$ is the Euclidean norm.

Note: because the Euclidean norm is sensitive to the scale of the variables, we have to standardize the predictors first.

The boundary bias issues are exponentiated under higher dimensions because of the Curse of dimensionality (larger fraction of the data locations are on or near the boundary)

Similar to the one-dimensional case, the local polynomial regression accounts for this boundary bias issue to first order. However, the problem is to balance localness (low bias) and low variance as the dimension increases without the sample size increasing exponentially in p

Hence, we do not see and **should not use smoothing or local polynomial regression in dimensions beyond three.**

Adding structural assumption to the modeling framework can help in cases where the dimension to sample size ratio high.

An example under kernels could be

$$K_{\lambda, \mathbf{A}(x_0, x)} = D\left(\frac{(x - x_0)^T \mathbf{A}(x - x_0)}{\lambda}\right)$$

where $\mathbf{A}$ is a $p \times p$ positive semi-definite matrix, where it can give different weights to the elements of $x - x_0$ and/or recombine the elements of this vector in various ways to shrink some dimensions (“compress”) the points into a smaller region of space

Another way would be to add structure through **varying coefficient models**

Assume p inputs and a regression on the first q variables where the parameters depend on the remaining inputs $Z = (X_{q+1}, X_{q+2}, \dots, X_p)$

$$f(x) = \alpha(Z) + \beta_1(Z)X_1 + \dots + \beta_q(Z)X_q$$

where the coefficients are functions of Z

Under Kernel weighted local regressions, then

$$\min_{\alpha(z_0), \beta(z_0)} \sum_{i=1}^n K_{\lambda}(z_0, z_i) [y_i - \alpha(z_0) - x_{1i}\beta_1(z_0) - \dots - x_{qi}\beta_q(z_0)]^2$$

5.2 Application

Alternative package `locfit`

```
library(ISLR2)
attach(Wage)
library (splines)

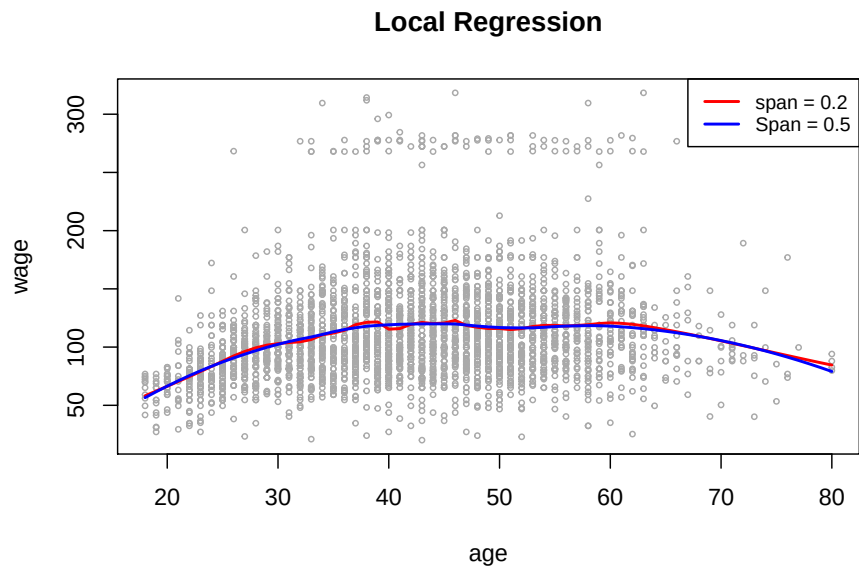
# we want to predict for a grid of values for age
agelims <- range (age)
age.grid <- seq (from = agelims[1], to = agelims [2])

plot(age,
      wage,
      xlim = agelims,
      cex = .5,
      col = "darkgrey")
title("Local Regression")

# each neighborhood consists of (spans) percentage of the observations
fit <- loess(wage ~ age, span = .2, data = Wage) # using spans of 0.2 (i.e., each neigh
fit2 <- loess(wage ~ age, span = .5, data = Wage) # using spans of 0.5

# with larger span, we have smoother fit

lines(age.grid,
      predict(fit, data.frame(age = age.grid)),
      col = "red",
      lwd = 2)
lines(age.grid,
      predict(fit2, data.frame(age = age.grid)),
      col = "blue",
      lwd = 2)
legend(
  "topright",
  legend = c("span = 0.2", "Span = 0.5"),
  col = c("red", "blue"),
  lty = 1,
  lwd = 2 ,
  cex = 0.8
)
```



Part IV

2. TRANSFORMATION BASED

Chapter 6

Generalized Additive Models

- A more flexible method for nonlinear regression models

$$E(Y|X_1, X_2, \dots, X_p) = \alpha + f_1(X_1) + f_2(X_2) + \dots + f_p(X_p)$$

where $f_j(\cdot), j = 1, \dots, p$ are unspecified smooth (nonparametric) functions

Notes:

- We call this model additive because we separate f_j for each X_j
- Complicated functions for each features are allowed, and we can still retain the interpretability

Equivalently, the model can be rewritten as

$$y_i = \alpha + \sum_{j=1}^p f_j(x_{ij}) + \epsilon_i$$

ϵ usually is chosen to follow Gaussian distribution

- If each of the function $f_j(\cdot)$ were written as a basis expansion, then we are back at least squares
- A general approach to fit each function would be a “scatterplot smoother” (e.g., cubic Smoothing splines or Kernel Smoothing) using an algorithm that simultaneously estimates all p-functions
- Transformation of the mean response (e.g., link function) as in Generalized linear models can also be applied.

Example:

Consider two-group classification via logistic regression. We relate the mean of the binary response

$$\mu(X) = P(Y = 1|X)$$

to functions of the predictors via an additive regression model using a logit link function

$$\log\left(\frac{\mu(X)}{1 - \mu(X)}\right) = \alpha + f_1(X_1) + \cdots + f_p(X_p)$$

where each f_j is an unspecified smooth function

In general, we relate the conditional mean response to the additive function of the predictors via a link function, g :

$$g(\mu(X)) = \alpha + f_1(X_1) + \cdots + f_p(X_p)$$

Examples:

- $g(\mu) = \mu$ the identity link
- $g(\mu) = \text{logit}(\mu)$ or $g(\mu) = \text{probit}(\mu)$ as in the glm case for binary and binomial data (probit function is the inverse Gaussian cumulative distribution function: $\text{probit}(\mu) = \Phi^{-1}(\mu)$)
- $g(\mu) = \log(\mu)$: Poisson count data

Function f_j can be nonlinear or linear or any other parametric forms. or a combination of them:

- $g(\mu) = X^T\beta + \alpha_k + f(Z)$: a semi-parametric model, where
 - X is a vector of predictors to be modeled linearly
 - α_k the effect for the k -th level of a qualitative input V
 - the effect of the predictor Z is modeled nonparametrically
- $g(\mu) = f(X) + g_k(Z)$ where
 - k indexes the levels of a qualitative input V and thus creates an interaction term with Z , $g(V, Z) = g_k(Z)$
- $g(\mu) = f(X) + g(Z, W)$: where g is a nonparametric function in two features, Z and W

6.1 Fitting Additive Models

We choose the function f_j to be “smoothers.” Then, we minimize a penalized sum of squares

$$PRSS(\alpha, f_1, \dots, f_p) = \sum_{i=1}^n (y_i - \alpha - \sum_{j=1}^p f_j(x_{ij}))^2 + \sum_{j=1}^p \lambda_j \int f_j''(t_j)^2 dt_j$$

where the $\lambda_j \geq 0$ are tuning parameters

The minimizer is an additive cubic spline model with each of the functions f_j a cubic spline in X_j and with knots at each of the unique values of $x_{ij}, i = 1, \dots, n$

This solution is not unique because α is not identifiable without another constraint. Hence, we add $\sum_{i=1}^n f_j(x_{ij}) = 0 \forall j$ which implies $\hat{\alpha} = \text{ave}(y_i)$

- If $\mathbf{X} = \{x_{ij}\}$ is of full column rank, then the minimizer is unique
- If this isn't the case (i.e., $\mathbf{X}'\mathbf{X}$ is singular) then the **linear portions** of the components f_j **cannot be uniquely determined, but the nonlinear parts can**

To fit the model, we have to use an iterative (backfitting) procedure

Let S_j be the cubic smoothing spline for the j -th function. We apply it to the target residuals $\{y_i - \hat{\alpha} - \sum_{k \neq j} \hat{f}_k(x_{ik})\}_1^n$ as a function of x_{ij} to obtain the new estimate $\hat{f}_j$

The procedure is done for each predictor using the current estimates of the other functions $\hat{f}_k$ when computing the differences.

The process continue until the estimates $\hat{f}_j$ stabilize

6.1.1 The Backfitting Algorithm for Additive Models

Step 1: Initialize

$$\hat{\alpha} = \frac{1}{N} \sum_1^N y_i, \hat{f}_j \equiv 0, \forall i, j$$

Step 2: $j = 1, 2, \dots, p, \dots, 1, 2, \dots, p, \dots$

$$\hat{f}_j \leftarrow S_j[\{y_i - \hat{\alpha} - \sum_{k \neq j} \hat{f}_k(x_{ik})\}_1^N]$$

$$\hat{f}_j \leftarrow \hat{f}_j - \frac{1}{N} \sum_{i=1}^N \hat{f}_j(x_{ij})$$

until the functions $\hat{f}_j$ change less than a prespecified threshold

Notes:

- Any smoother can be used to take the place of S_j such as
 - other univariate regression smoothers (e.g., local polynomial regression and Kernel Smoothing)
 - linear regression operators yielding polynomial fits, piecewise constant fits, parametric spline fits, Fourier series fits
 - smoothers for second or higher-order interactions or periodic smoothers for seasonal effect.
- If we consider the smoother at the training points, it can be represented by the $n \times n$ smoother matrix $\mathbf{S}_j$ which can be used to estimate the df associated with the j -th term from $\text{trace}(\mathbf{S}_j) - 1$

6.2 Fitting Generalized Additive Models

- The optimization criterion here is the penalized log-likelihood
- It maximizes by combining the backfitting procedure with a likelihood maximizer
- The weighted linear regression component is replaced by a weighed backfitting algorithm Hastie et al. (2001) (section 9.2)

6.2.1 Local Scoring Algorithm

Step 1: Compute $\hat{\alpha} = \log\left[\frac{\bar{y}}{1-\bar{y}}\right]$ where $\bar{y} = \text{ave}(y_i)$, the sample proportion of ones, and set $\hat{f}_j \equiv 0 \forall j$

Step 2: Define $\hat{\eta}_i = \hat{\alpha} + \sum_j \hat{f}_j(x_{ij})$ and $\hat{p}_i = \frac{1}{1+\exp(-\hat{\eta}_i)}$

Iterate:

1. Construct the working target variable: $z_i = \hat{\eta}_i + \frac{y_i - \hat{p}_i}{\hat{p}_i(1-\hat{p}_i)}$
2. Construct weights $w_i = \hat{p}_i(1-\hat{p}_i)$
3. Fit an additive model to the targets z_i with weights w_i , using a weighted backfitting algorithm. This gives new estimates $\hat{\alpha}, \hat{f}_j, \forall j$

Step 3: Continue step 2, until the change in the functions falls below a pre-specified threshold.

6.3 Summary

Pros	Cons
we can fit a nonlinear f_j to each X_j and automatically model nonlinear relationships (which we can't do under standard linear regression)	Model is restricted to be additive . Technically, we can include interactions, but we have to make them manually, and we can only do so with low-dimensional interaction terms as additive functions
The nonlinear fits can make more accurate predictions	Difficult to fit in high dimensions (because all variables have to be fit)
Since the model is additive, we can examine the effect of each X_j on Y individually while holding all of the other variables fixed (interpretability)	
The smoothness of the function f_j for the variable X_j can be summarized by df	

An extension of GAM is to include random effects, which is called **Generalized Additive Mixed Models** (GAMMs) (can refer to this book)

6.4 Application

6.4.1 Example 1

(James et al., 2013, 7.8)

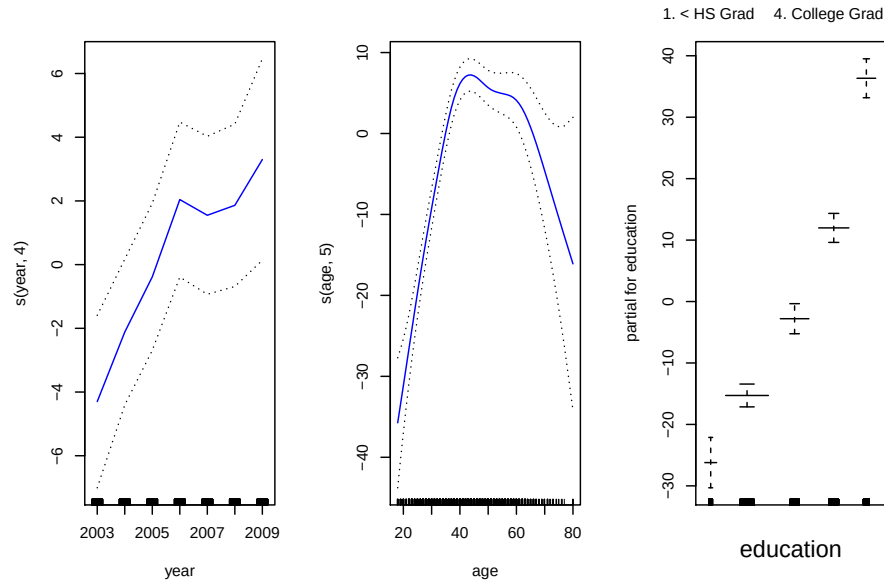
```
library(ISLR2)
attach(Wage)
library(splines)

# we want to predict for a grid of values for age
agelims <- range(age)
age.grid <- seq(from = agelims[1], to = agelims[2])

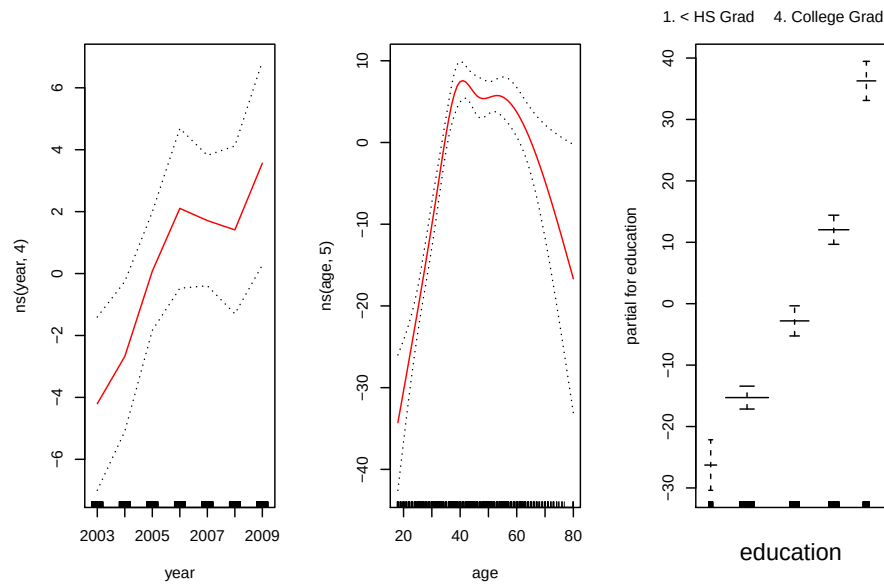
# since gam is just linear regression with transformation of inputs, we can also use lm in R
gam1 <- lm(wage ~ ns(year, 4) + ns(age, 5) + education, data = Wage) # using natural splines

library(gam)
gam.m3 <-
  gam(wage ~ s(year, 4) + s(age, 5) + education, data = Wage) # using smoothing splines

par(mfrow = c(1, 3))
plot(gam.m3, se = T, col = "blue")
```



```
plot.Gam(gam1, se = T, col = "red") # the function of year is rather linear
```



```
gam.m1 <- gam(wage ~ s(age, 5) + education, data = Wage)
gam.m2 <- gam(wage ~ year + s(age, 5) + education, data = Wage)
anova(gam.m1, gam.m2, gam.m3, test = "F") # model 2 is the best one
```

```
## Analysis of Deviance Table
##
## Model 1: wage ~ s(age, 5) + education
## Model 2: wage ~ year + s(age, 5) + education
## Model 3: wage ~ s(year, 4) + s(age, 5) + education
##   Resid. Df Resid. Dev Df Deviance      F    Pr(>F)
## 1      2990      3711731
## 2      2989      3693842  1  17889.2 14.4771 0.0001447 ***
## 3      2986      3689770  3   4071.1  1.0982 0.3485661
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

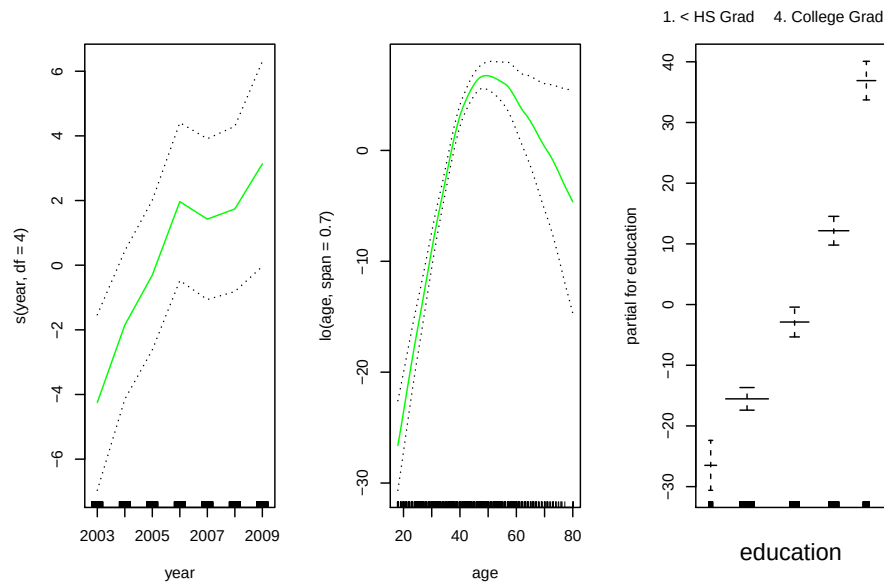
summary(gam.m3)

##
## Call: gam(formula = wage ~ s(year, 4) + s(age, 5) + education, data = Wage)
## Deviance Residuals:
##      Min       1Q   Median       3Q      Max
## -119.43  -19.70   -3.33   14.17  213.48
##
## (Dispersion Parameter for gaussian family taken to be 1235.69)
##
##      Null Deviance: 5222086 on 2999 degrees of freedom
## Residual Deviance: 3689770 on 2986 degrees of freedom
## AIC: 29887.75
##
## Number of Local Scoring Iterations: NA
##
## Anova for Parametric Effects
##           Df Sum Sq Mean Sq F value    Pr(>F)
## s(year, 4)   1   27162    27162  21.981 2.877e-06 ***
## s(age, 5)    1  195338   195338 158.081 < 2.2e-16 ***
## education    4 1069726   267432  216.423 < 2.2e-16 ***
## Residuals  2986 3689770     1236
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Anova for Nonparametric Effects
##           Npar Df Npar F    Pr(F)
## (Intercept)
## s(year, 4)      3  1.086 0.3537
## s(age, 5)       4 32.380 <2e-16 ***
## education
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```

preds <- predict(gam.m2, newdata = Wage)
gam.lo <- gam(wage ~ s(year, df = 4) + lo(age, span = 0.7) + education, data = Wage)
plot.Gam(gam.lo, se = T, col = "green")

```



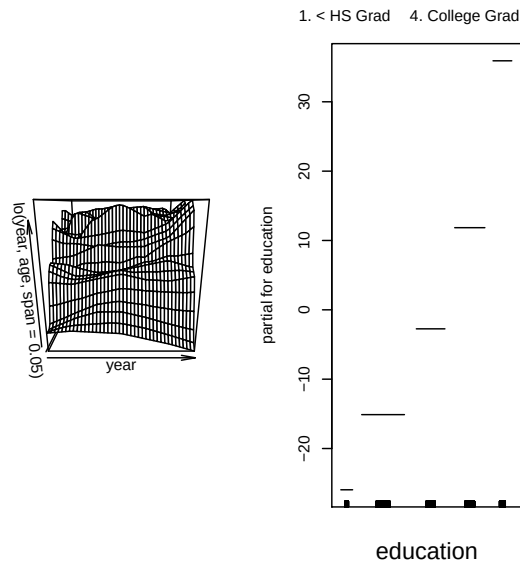
```

# using local regression as building blocks for GAM
gam.lo.i <-
  gam(wage ~ lo(year, age, span = .05) + education, data = Wage)

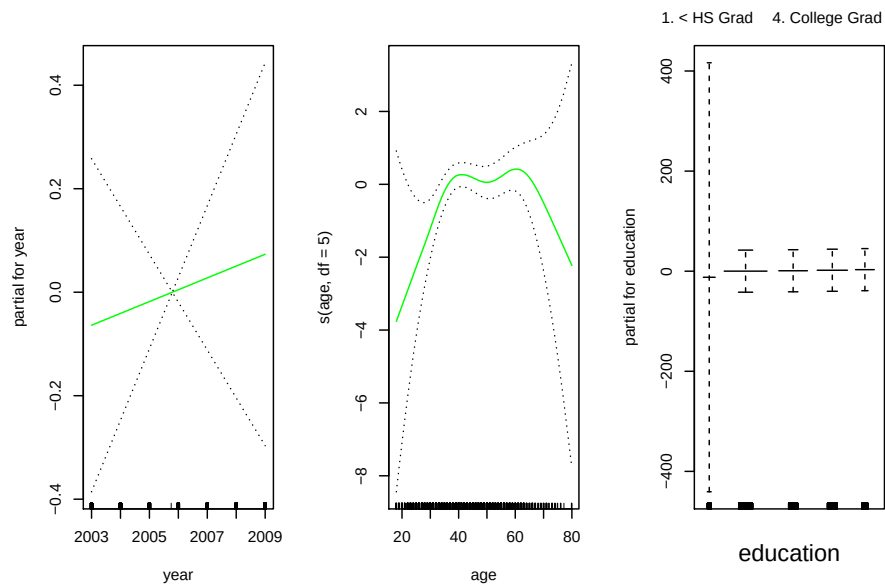
library(akima)
plot(gam.lo.i)

gam.lr <- gam(I(wage > 250) ~ year + s(age, df = 5) + education,
  family = binomial,
  data = Wage)
par(mfrow = c(1, 3))

```



```
plot(gam.lmr, se = T, col = "green")
```

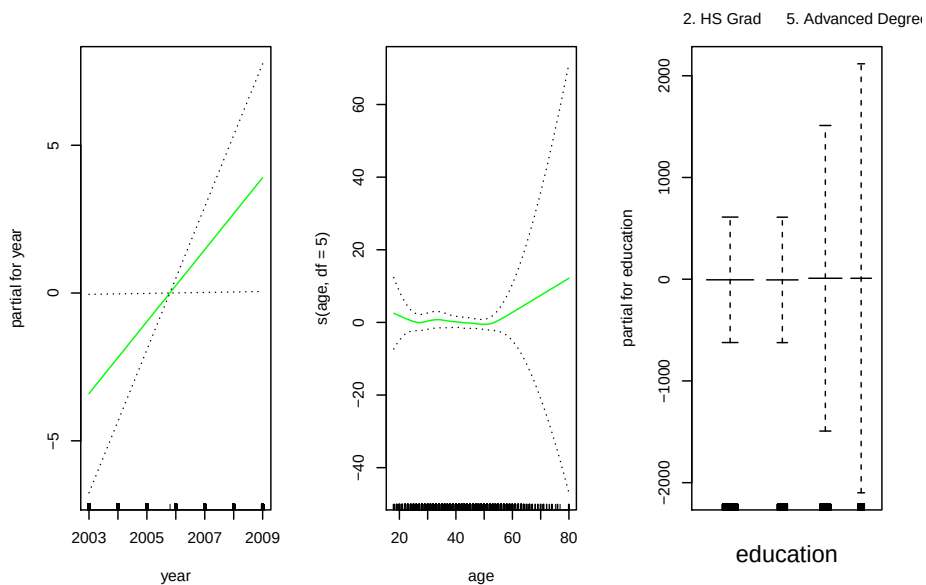


```
table(education, I(wage > 25))
```

```
##
```

```
## education          FALSE TRUE
## 1. < HS Grad       1  267
## 2. HS Grad         2  969
## 3. Some College     3  647
## 4. College Grad     0  685
## 5. Advanced Degree  0  426

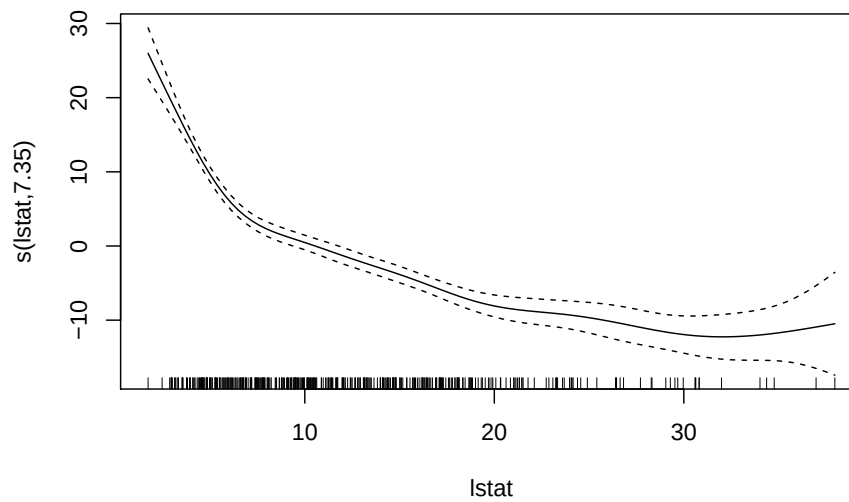
gam.lr.s <- gam(
  I(wage > 25) ~ year + s(age, df = 5) + education,
  family = binomial,
  data = Wage,
  subset = (education != "1. < HS Grad")
)
plot(gam.lr.s, se = T, col = "green")
```



6.4.2 Example 2

To overcome Spline Regression's requirements for specifying the knots, we can use Generalized Additive Models or GAM.

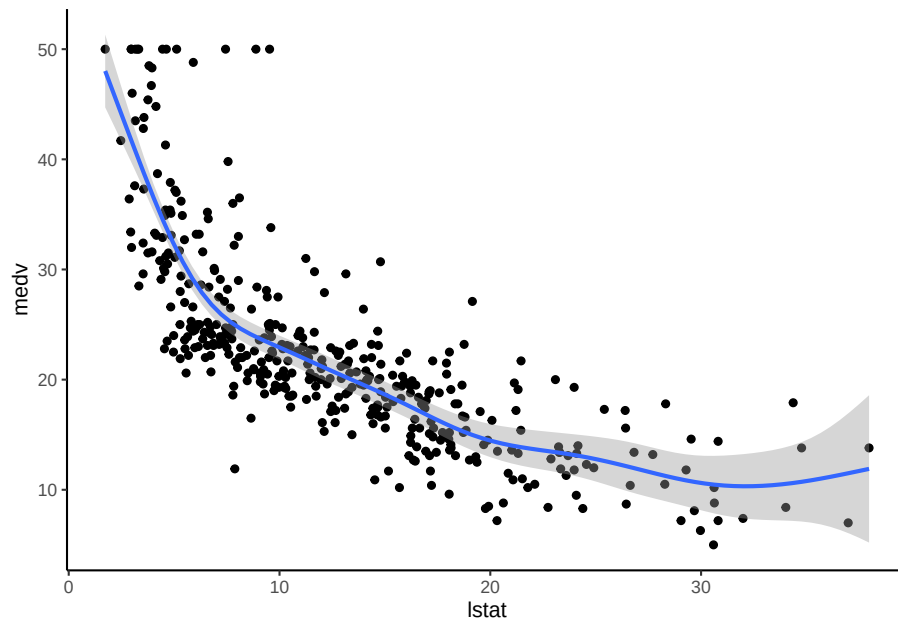
```
library(mgcv)
# Build the model
model <- gam(medv ~ s(lstat), data = train.data)
plot(model)
```

```
# Make predictions
# predictions <- model %>% predict(test.data)
# Model performance
data.frame(
  RMSE = RMSE(predictions, test.data$medv),
  R2 = R2(predictions, test.data$medv)
)
```

```
##          RMSE          R2
## 1 5.317372 0.6786367

ggplot(train.data, aes(lstat, medv) ) +
  geom_point() +
  stat_smooth(method = gam, formula = y ~ s(x))
```



```
detach(train.data)
```

Chapter 7

Gaussian Process Regression

Suggested reading:

- Rasmussen and Williams (2005)
- Wikle et al. (2019) (section 4.2)

Under Gaussian Process (GP), we assume that Y is a random (Gaussian) process that is controlled by a mean function and covariance function.

Response model:

$$z_i = Y_i + \epsilon_i, i = 1, \dots, n$$

Assumptions:

- Y = a latent process specified by a GP
- ϵ_i independent and identically distributed mean-zero observation errors that are independent of Y and have variance σ_ϵ^2
- the errors are assumed to have a normal (Gaussian) distribution $\epsilon_i \sim iidN(0, \sigma_\epsilon^2)$ these errors could be correlated

Consider a **linear predictor** of the form

$$\hat{Y}(x_0) = k_0 + \sum_{i=1}^n w_i z_i$$

where w_i are weights and k_0 is some constant

To pick these weights, we choose “optimal” weight based on mean squared error notion, which requires us to consider the random process that controls Y

7.1 Latent Process Model

Assume that we have the latent process follows the model

$$Y(x) = f(x)$$

where $f(x) \sim GP(m(x), c(x, x'))$ is a random process (specifically a Gaussian Process) and

- $m(x)$ = mean function $E[f(x)]$
- $c(x, x')$ = covariance function $c(x, x') = E[(f(x) - m(x))(f(x') - m(x'))^T]$

The difference between **Gaussian distribution** and a **Gaussian process** is

- Gaussian Distribution: a distribution over vectors
 - Fully specified by a **mean vector** and **covariance matrix**: (e.g., $\mathbf{Y} \sim N(\mu, \Sigma)$ which is a multivariate normal distribution)
- Gaussian Process: a distribution over functions
 - Fully specified by a mean function (m) and covariance function (c) (e.g., $Y \sim GP(m, c)$), where we specify a mean function and covariance function to define a GP)

7.2 Gaussian Processes

- A GP is an infinite-dimensional object
- Allow us to make a prediction anywhere in our x domain assuming that we know the mean and covariance functions (data generating process), we need the covariance between ANY two x locations
- We actually need to consider finite-dimensional distributions because
 - We only have a finite set of data and also we are interested in a finite set of predictions
 - Any finite collection of GP random variables has a joint Gaussian distributions (i.e., a multivariate normal distribution).

7.3 Gaussian Distributions

$$\begin{pmatrix} \mathbf{v} \\ \mathbf{v}_* \end{pmatrix} \sim \text{Gau} \left(\begin{pmatrix} \mu \\ \mu_* \end{pmatrix}, \begin{pmatrix} T & * \\ * & ** \end{pmatrix} \right)$$

where $E(\mathbf{v}) = \mu$, $E(\mathbf{v}_*) = \mu_*$, $\Sigma = \text{cov}(\mathbf{v}, \mathbf{v})$, $\Sigma_{**} = \text{cov}(\mathbf{v}, \mathbf{v}_*)$, $\Sigma_{**} = \text{cov}(\mathbf{v}_*, \mathbf{v}_*)$

Then, partition

$$\mathbf{v}_* | \mathbf{v} \sim \text{Gau}(\mu_* + \Sigma_*^{-1}(\Sigma_{**} - \Sigma_*^{-1} \Sigma_{**}^T) (\mathbf{v} - \mu), \Sigma_* - \Sigma_*^{-1} \Sigma_{**}^T \Sigma_*)$$

7.4 Optimal Linear Prediction

We are interested in predicting $Y(x_0)$ (i.e., at some arbitrary x) given the data $\mathbf{z} \equiv (z_1, \dots, z_n)^T$

Assume that the mean function $\mu(x) \forall x$ (e.g., a constant)

We seek linear predictors that minimize the mean squared prediction error of the form

$$\begin{aligned} \hat{Y}(x_0) &= k(x_0) + \sum_{i=1}^n w_i(x_0) z_i \\ &= k(x_0) + \mathbf{w}_0^T \mathbf{z} \end{aligned}$$

The optimal linear predictor is also the conditional expectation $E(Y(x_0) | \mathbf{z})$.

Hence, we seek the conditional distribution $Y(x_0) | \mathbf{z}$

Rewriting the data model in vector form:

$$\mathbf{z} = \mathbf{Y} + \epsilon, \epsilon \sim \text{Gau}(\mathbf{0}, \mathbf{C}_\epsilon)$$

where $\mathbf{Y}$ and ϵ are ordered in the same way as the data vector $\mathbf{z}$

$$\text{cov}(\mathbf{Y}, \mathbf{Y}) \equiv \mathbf{C}_y$$

$$\text{cov}(\epsilon, \epsilon) \equiv \mathbf{C}_\epsilon$$

$$\text{cov}(\mathbf{z}, \mathbf{z}) = \mathbf{C}_y + \mathbf{C}_\epsilon$$

$$\mathbf{c}_0^T \equiv \text{cov}(Y(x_0), \mathbf{z})$$

$$c_{0,0} \equiv \text{var}(Y(x_0))$$

μ_0 is mean at x_0 and μ is the mean corresponding to the observations.

Then, the joint Gaussian distribution:

$$\begin{pmatrix} Y(x_0) \\ \mathbf{v}_* \end{pmatrix} \sim \text{Gau} \left(\begin{pmatrix} \mu_0 \\ \mathbf{0} \end{pmatrix}, \begin{pmatrix} c_{0,0} & \mathbf{c}_0^T \\ \mathbf{c}_0 & \mathbf{C}_z \end{pmatrix} \right)$$

Using the joint/conditional Gaussian Distributions formula

$$Y(x_0)|\mathbf{z} \sim \text{Gau}(\mu_0 + \mathbf{c}_0^T \mathbf{C}_z^{-1}(\mathbf{z} - \mathbf{z}_0), c_{0,0} - \mathbf{c}_0^T \mathbf{C}_z^{-1} \mathbf{c}_0)$$

The predictor is given by the **mean** of this distribution

$$\hat{Y}(x_0) = \mu_0 + \mathbf{c}_0^T \mathbf{C}_z^{-1}(\mathbf{z} - \mathbf{z}_0)$$

and the prediction **variance** is

$$\sigma_{Y_0}^2 = c_{0,0} - \mathbf{c}_0^T \mathbf{C}_z^{-1} \mathbf{c}_0$$

This predictor is the **optimal predictor** because it minimizes the **mean squared prediction error**:

$$MSPE = E[(Y(x_0) - \hat{Y}(x_0))^2]$$

Hence, the **weights** are given by

$$\mathbf{w}_0 = \mathbf{c}_0' \mathbf{C}_z^{-1}$$

and

$$k(x_0) = \mu_0 - \mathbf{c}_0' \mathbf{C}_z^{-1} \mathbf{c}_0$$

which is an effective kernel (just like Spline Regression and Local Regression)

These formulas can be extended to multiple locations (each location has its own vector of weights). However, if there are a lot of sample locations, there could be computational problem regarding calculating the matrix inverse $\mathbf{C}_z^{-1}$

In cases where you have no measurement errors (i.e., $z_i = y_i$) (equivalently, replacing $\mathbf{C}_z$ with $\mathbf{C}_y$)

7.5 Covariance Functions

To predict at new x locations, we must specify:

1. a mean function (doable)
2. a covariance function between ANY two locations (hard)

In the real world, we hardly ever know the variances and covariances that make up $\mathbf{C}_y, \mathbf{C}_\epsilon, \mathbf{c}_0, c_{0,0}$

A seemingly simple solution is to specify covariance functions in terms of parameters that can give a covariance between any pairs of coordinates and then estimate the parameters, which is the same thing that we did in Linear Mixed Models (e.g., AR models, exponential models)

- **Covariance functions must be positive semi-definite** to ensure that prediction variances are non-negative
- Covariance functions should be realistic (i.e., account for realistic dependencies), which can be difficult sometime. Typically, the assumption that “nearby” observations (similar in the case of Kernel Smoothing) are more dependent is sufficient.

In practice, we usually have to employ a **stationarity assumption**

Example:

If the random process is assumed to be second-order (weakly) stationary then it has constant expectation and a covariance function that can be expressed in terms of the distance between the x locations ($h = x - x'$), we specify

$$c(x, x') = c(h,)$$

In two or more dimensions, if this distance lag does not depend on direction (i.e., $h = ||\mathbf{h}||$), then we call the function **isotropic**

The benefits of stationarity assumption:

- allows for more **parsimonious parameterization** of the covariance function
- provides **pseudo-replication** of dependencies at given lags, which facilitate estimation

To construct covariance functions with this assumption, traditional machine learning would use **separable covariance functions** (e.g., for x_1 and x_2 - 2 dimension prediction)

$$c(h_1; h_2) \equiv c(h_1) \times c_2(h_2)$$

which is valid if both covariance functions are valid

This separability property helps with computation in prediction locations (e.g., matrix inverses we mention above), but it can also be unrealistic in cases like spatio-temporal data

Then to have valid spatial and temporal covariance functions, we can employ the following covariance functions:

- The Matern
- Power exponential: $c(\tau, \theta) = \sigma^2 \exp\{-\frac{|\tau|}{\theta}\}$ where σ^2 is the variance parameter and θ is the dependence parameter
- Gaussian classes

To estimate the parameters in the covariance functions, and in the mean function we can either

1. Under the frequentist perspective, we can use REML, MLE (like Linear Mixed Models) and BLUP for random process
2. Under the Bayesian perspective, we use GP for $f(x)$ as prior distribution, and update this with the observations, $\mathbf{z}$

With a lot of data locations, REML, MLE, and Bayesian can be problematic because of the high-dimensional covariance matrix $\mathbf{C}_z, \mathbf{C}_y$ (which research are still being conducted to improve)

Notes:

- If we model the mean function in terms of covariates ($\mu(x) = \mathbf{w}(x)'\mathbf{m}$ where $\mathbf{w}$ is a vector of covariates. Then, it's Linear Mixed Models with difference that the covariates must be available for ANY x
- In cases where you don't measurement uncertainty in the observation, (ϵ) then the prediction at an observation location will be exactly the same as the observation. And the other way around will filter the uncertainty with the observation
- we can also have non-Gaussian observations conditioned on the GP (like GLMM)

7.6 Application

Other packages: `mlegp`

MacDonald et al. (2015)

Simulation:

$$y(x) = \log(x + .1) + \sin(5\pi x)$$

A Latin Hypercube design is a space-filling design, used in the design of computer experiments.

The Maximin version does space-filling design by maximizing the minimum distance between design points.

```
library(GPfit)
library(lhs)

set.seed(1)
n = 7
x = maximinLHS(n, 1) # draw a Maximin Latin Hypercube Sample
y = matrix(0, n, 1)
for (i in 1:n){
  y[i] = log(x[i] + .1) + sin(5*pi * x[i])
}

GPmodel = GP_fit(x,y)
print(GPmodel)
```

```
##
## Number Of Observations: n = 7
## Input Dimensions: d = 1
##
## Correlation: Exponential (power = 1.95)
## Correlation Parameters:
##      beta_hat
## [1] 1.69817
##
## sigma^2_hat: [1] 0.858342
##
## delta_lb(beta_hat): [1] 0
##
## nugget threshold parameter: 20
```

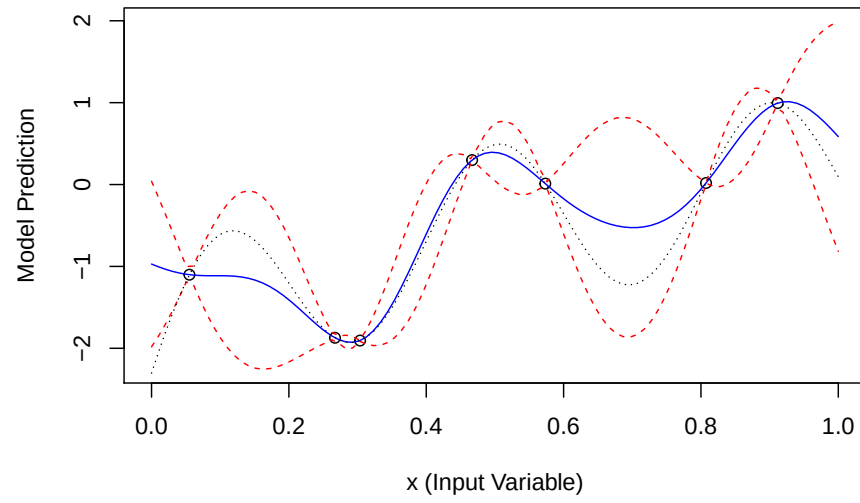
Predict at a grid of new x values

```
xvec <- seq(from = 0, to = 1, length.out = 100)
yvec <- matrix(0, 100, 1)
for (i in 1:100){
  yvec[i] = log(xvec[i] + .1) + sin(5*pi*xvec[i])
}
GPpred <- predict(GPmodel, xvec)
```

```
# data points are circles
# prediction is in the blue
# 95% CIs (yhat + 2*s) are in red
```

```
plot(GPmodel, range= c(0,1), resolution = 100, colors = c("black", "blue", "red"), line_type = c(1,
```

```
# add the true function as a dotted black line  
lines(xvec, yvec, col = "black", lty = 3)
```



Part V

3. TREE BASED

Chapter 8

Classification and Regression Trees

- Segmenting the predictor space into a number of simple regions.
- Develop a collection of splitting rules that make these segments.
- The splitting rules can be thought of as “tree” (therefore, “decision trees”)
- The root of the tree (i.e., the first splitting variable) is typically chosen based on the the smallest sum of squared residuals (SSRs), then the process begins again to choose another branch.
- Decision trees can be used for both classification and regression, ergo the name CART
 - Under regression, the prediction is the average of the partitioned predictor space (i.e., segments)
 - Under classification, we classify based on the most prevalent class in each segment
- Compared to linear models
 - If the underlying relationship is linear, the linear regression/ classification setting works better than CART
 - IF the underlying relationship is nonlinear and complex between the features and the response, the CART model can be better.
- Issues:
 - Categorical predictors can prove to be difficult to implement CART when we have $q > 2$ possible unordered values of the predictors as

there are $2^{q-1} - 1$ possible partitions of the q values into 2 groups. Usually lead to over-fitting.

- Missing predictor values:
 - * For categorical variables, we can make a separate category for “missing”. Or we can construct surrogate variables that mimic the split of the training data using only the non-missing variables
- Non-binary splits: not recommended because it splits the data too quickly and doesn’t leave enough data for the next level of the tree
- Linear Combination Splits: we can make linear combination of the variables to choose the splits (akin to PC-regression), which can help prediction, but reduce interpretation
- Instability of trees: CART has high variance (i.e., a small change in the data can lead to different splits). To handle this problem, we can use “bagging” (i.e., bootstrapping procedure). This can happen even after you prune.
- Lack of Smoothness: CART can lead to non-smooth prediction surface, which can be problematic for regression applications. The MARS procedure can be used instead
- Additive structure: Since CART uses binary structure, it can’t accommodate additive structure, but MARS can
- Trees do not have good predictive accuracy as other regression and classification approaches.

Advantages:

- Easily explainable (to non-statisticians)
- Decision trees may be closer to human decision making than traditional regression and classification
- Trees are easily interpreted (with graphical depiction)
- Trees can handle qualitative predictors without dummy variables

8.1 Regression Trees

Say we have p inputs and a response for each of n observations, $(x_i, y_i), i = 1, \dots, n$ with $x_i = (x_{i1}, \dots, x_{ip})^T$

Suppose that we have a partition of the predictor space into J regions $R_1, \dots, R_J$. Then, our prediction in each region is some constant, c_j . Thus, we could write

the response function as

$$f(x) = \sum_{j=1}^J c_j I(x \in R_j)$$

where $I(x \in R_j)$ is an indicator function (1 when x is in the j -th region, 0 otherwise).

Given the partitions, the estimate $\hat{c}_j$ is the average of y_i in R_j

$$\hat{c}_j = \hat{y}_{R_j} \equiv \text{ave}(y_i | x_i \in R_j)$$

To find the partitions, we seek to divide the predictor space into high-dimensional rectangles (boxes) (i.e., we want to find the boxes $R_1, \dots, R_J$ that minimize the residual sum of squares)

$$\sum_{j=1}^J \sum_{i \in R_j} (y_i - \hat{y}_{R_j})^2$$

where $\hat{y}_{R_j}$ is the mean response for the training observations within the j -th box (e.g., $\hat{c}_j$)

Since we can't consider every possible partition (computationally constraint), we seek binary partitions and specifically, a top down, greedy approach (i.e., at each step the best split is made at that step, once we make a split, we don't go back and change it) known as **recursive binary splitting**.

8.1.1 Recursive Binary Splitting

Step 1: we select the predictors X_j and the cutpoints such that splitting the predictor space into the regions $\{X|X_j < s\}$ and $\{X|X_j \geq s\}$ leads to the maximum reduction in RSS (i.e., for any j and s , we define the regions (half-planes)):

$$R_1(j, s) = \{X|X_j < s\}; R_2(j, s) = \{X|X_j \geq s\}$$

and seek the value of j and s that minimizes

$$\sum_{i: x_i \in R_1(j, s)} (y_i - \hat{y}_{R_1})^2 + \sum_{i: x_i \in R_2(j, s)} (y_i - \hat{y}_{R_2})^2$$

where $\hat{y}_{R_1} = \hat{c}_1$ is the mean response in $R_1(j, s)$

Notes:

- When p is small, it's computationally efficient to find values of j and s .
- After the partition is selected, we repeat the process, looking for the best predictor (j) and best cut point (s) to split one of the two new regions to minimize the RSS, which will give us three regions. We continue to split one of these three regions to minimize the RSS
- The stopping criterion is usually when a region contains fewer than some specified number of observations (because we have to take the averages of these regressions for our mean response)
- The end regressions of the tree ($R_1, \dots, R_J$) are known as **terminal nodes** or **leaves** of the tree
- The points along the tree where the predictor space is split are **internal nodes**
- Regression trees can usually lead to over-fitting on the test set, to reduce this problem, we can prune (i.e., remove leaves) to obtain a **subtree** using cross-validation.

8.1.2 Cost complexity pruning

- also known as **weakest link pruning**

Intuitively,

1. Calculate the total Sum of Squared Residual for each tree
2. Calculate a Tree Score: Tree Score = $SSR + \alpha T$ where
 1. SSR = Sum of Squared Residuals for the tree or sub-tree
 2. T = number of leaves (or Terminal nodes) in the tree or sub-tree
 3. α = tuning parameter (based on cross-validation)
3. Pick tree with lowest tree score

The Tree Complexity Penalty (αT) compensates for the difference in the number of leaves

Mathematically,

We first grow a very large tree T_0 . Instead of considering every possible subtree, we consider a sequence of trees indeed a non-negative tuning parameters, α (i.e., for each value of α there is a subtree T such that $T \in T_0$ that minimizes

$$\sum_{j=1}^{|T|} \sum_{i: x_i \in R_j} (y_i - \hat{y}_{R_j})^2 + \alpha |T|$$

where $|T|$ is the number of terminal nodes

The parameter α controls a tradeoff between the subtree's complexity and how well it fits the training data ($\alpha = 0$ implies $T = T_0$). As α gets larger, the penalty for a larger tree is higher, so it penalizes similarly to the Lasso regression. Moreover, as α increases, the branches get pruned in a nested a predictable fashion. Hence, it's possible to get the whole sequence of subtrees as a function of α (based on cross-validation). After getting α using cross-validation, we can use that value with the full data set to obtain the fitted subtree

8.1.3 Algorithm for regression tree

1. Recursive Binary Splitting for growing a tree with your choice of stopping rule (e.g., observation-based)
2. Use Cost complexity pruning to obtain a sequence of best subtrees as a function of α
3. Use K-Fold Cross-Validation to choose α

```
library(ISLR2)
library(tree)

set.seed(1)
attach(Boston)

train <- sample(1:nrow(Boston), nrow(Boston)/2)
tree.boston <- tree(medv ~ ., Boston, subset = train)

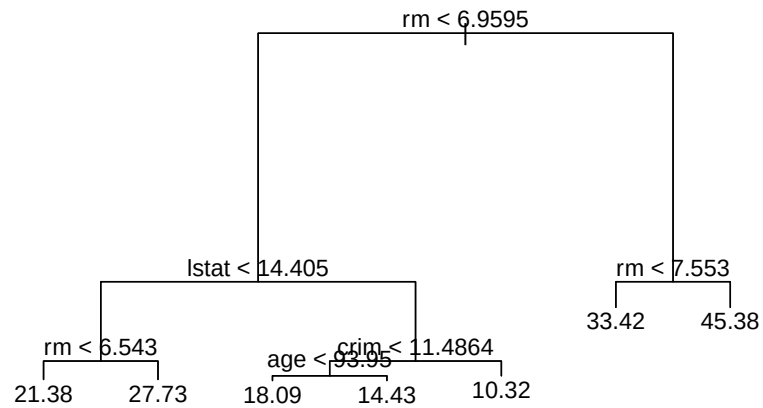
# to fit a bigger tree
# tree.boston <-
#   tree(medv ~ .,
#         Boston,
#         subset = train,
#         control = tree.control(nobs = length(train), mindev = 0))

summary(tree.boston) # only 4 vars have been used (rm, lstat, crim, age)

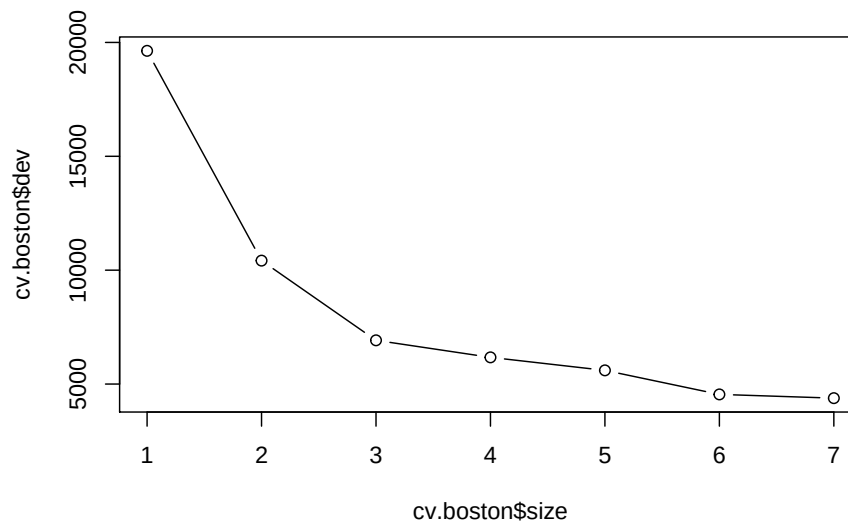
##
## Regression tree:
## tree(formula = medv ~ ., data = Boston, subset = train)
## Variables actually used in tree construction:
## [1] "rm"      "lstat"   "crim"    "age"
## Number of terminal nodes: 7
## Residual mean deviance: 10.38 = 2555 / 246
## Distribution of residuals:
##      Min.   1st Qu.   Median     Mean   3rd Qu.    Max.
## -10.1800  -1.7770   -0.1775    0.0000    1.9230   16.5800

deviance = sum of squared errors for the tree
```

```
plot(tree.boston)
text(tree.boston, pretty = 0)
```

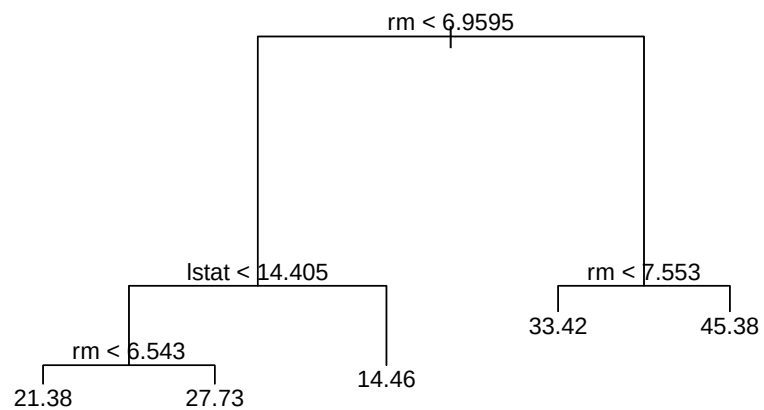


```
cv.boston <- cv.tree(tree.boston) # tree complexity selected by cross-validation
plot(cv.boston$size, cv.boston$dev, type = "b")
```



If we want to prune complex tree

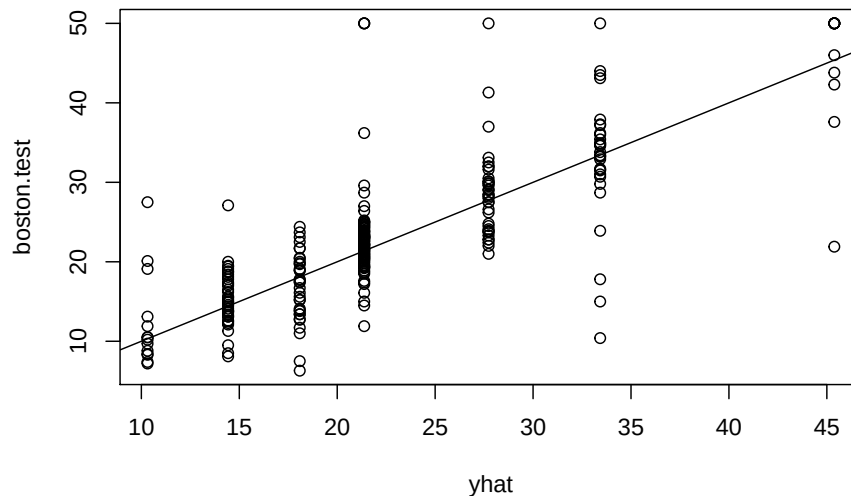
```
prune.boston <- prune.tree(tree.boston, best = 5)
plot(prune.boston)
text(prune.boston, pretty = 0)
```



```

yhat <- predict(tree.boston, newdata = Boston[-train, ])
boston.test <- Boston[-train, "medv"]
plot(yhat, boston.test)
abline(0, 1)

```



```

mean((yhat - boston.test) ^ 2) # Mean squared error

```

```
## [1] 35.28688
```

The square root of the MSE is 5.9, meaning on average we predict within \$5.9k of the true median home value.

8.2 Classification Trees

When we have a qualitative response we can use classification tree.

The main difference from Regression Trees is that we predict the response based on the most commonly occurring class in the region associated with that terminal node.

We may also be interested in the class proportion for each node when interpreting a classification tree

We grow a classification tree in the same manner as the Regression Trees, with Recursive Binary Splitting. However, instead of using RSS we use other measure related to the classification summary.

First we define the proportion of class k observation in the j -th node as

$$\hat{p}_{jk} = \frac{1}{n_j} \sum_{x_i \in R_j} I(y_i = k)$$

we classify an observation in node j to class k with the maximum value of $\hat{p}_{jk}$

The three measures for growing tree:

1. Classification error rate: $1 - \max_k(\hat{p}_{jk})$
 - measures the fraction of the training observations in that region that do not belong to the most common class
2. Gini index: $\sum_{k=1}^K \hat{p}_{jk}(1 - \hat{p}_{jk})$
 - measures the total variance across the K classes
3. Cross-Entropy or Deviance: $-\sum_{k=1}^K \hat{p}_{jk} \log(\hat{p}_{jk})$
 - related to an information measures and the deviance

The Gini index and Cross-Entropy both take small values when all $\hat{p}_{jk}$ are close to 1 or 0. Hence, we say they measure node purity, which are more sensitive to node probabilities than the classification error rate. Hence, we typically use the last two to grow the tree.

Rule of thumb: use Gini and Cross-Entropy to build and prune a tree and use classification error rate for pruning if prediction accuracy of the pruned tree is the goal

```
library(tree)
library(ISLR2)
attach(Carseats)
High <- factor(ifelse(Sales <= 8, "No", "Yes")) # transform Sales (continuous) to binary variable

Carseats <- data.frame(Carseats, High)

tree.carseats <- tree(High ~ . - Sales, Carseats) # using all variables but Sales

summary(tree.carseats) # training error rate of 9%
```

```
##
## Classification tree:
## tree(formula = High ~ . - Sales, data = Carseats)
## Variables actually used in tree construction:
## [1] "ShelveLoc" "Price" "Income" "CompPrice" "Population"
## [6] "Advertising" "Age" "US"
## Number of terminal nodes: 27
## Residual mean deviance: 0.4575 = 170.7 / 373
## Misclassification error rate: 0.09 = 36 / 400
```

```
# tree.carseats
```

Deviance formula:

$$-2 \sum_m \sum_k n_{mk} \log \hat{p}_{mk}$$

where

- n_{mk} = number of observations in the m-th terminal node in the k-th class

Smaller deviance is preferable (good fit to the training data).

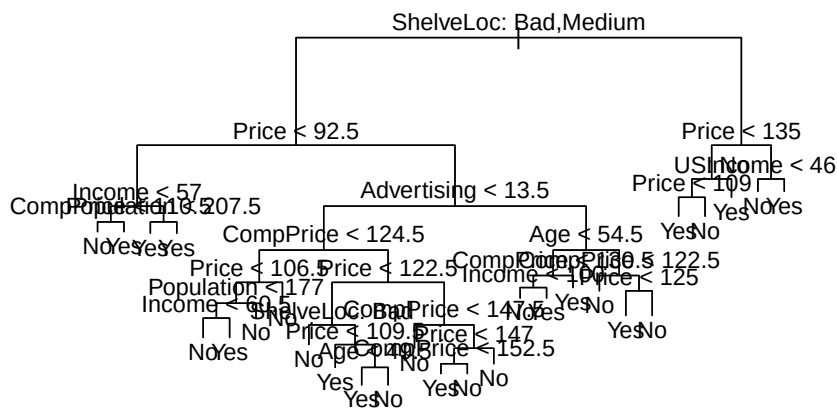
Residual mean deviance formula

$$\frac{-2 \sum_m \sum_k n_{mk} \log \hat{p}_{mk}}{n - |T_0|}$$

where $|T_0|$ = number of terminal nodes

```
plot(tree.carseats)
```

```
text(tree.carseats, pretty = 0) # shelving location is the most important indicator
```



Evaluation on both training and test data

```
set.seed(1)
```

```
train <- sample(1:nrow(Carseats), 200) # index 200 training datapoints
```

```

# get the test data
Carseats.test <- Carseats[-train,]

# get actual prediction
High.test <- High[-train]
tree.carseats <- tree(High ~ . - Sales, Carseats, subset = train)

tree.pred <- predict(tree.carseats, Carseats.test, type = "class") # return actual class prediction

table(tree.pred, High.test)

##           High.test
## tree.pred No Yes
##           No  84  37
##           Yes 35  44

# correct prediction rate
(table(tree.pred, High.test)[1, 1] + table(tree.pred, High.test)[2, 2])/nrow(Carseats.test) # 64%

## [1] 0.64

Tree with pruning

set.seed(7)
cv.carseats <- cv.tree(tree.carseats, FUN = prune.misclass) # cross-validation to get the optimal
# prune.misclass = classification error rate guides the cross-validation and pruning process
# default = deviance guides the cross-validation and pruning process

cv.carseats

## $size
## [1] 20 18 10  8  6  4  2  1
##
## $dev
## [1] 61 62 60 54 52 51 83 83
##
## $k
## [1] -Inf  0.0  0.5  1.5  2.0  4.0 12.0 19.0
##
## $method
## [1] "misclass"
##
## attr(,"class")
## [1] "prune"          "tree.sequence"

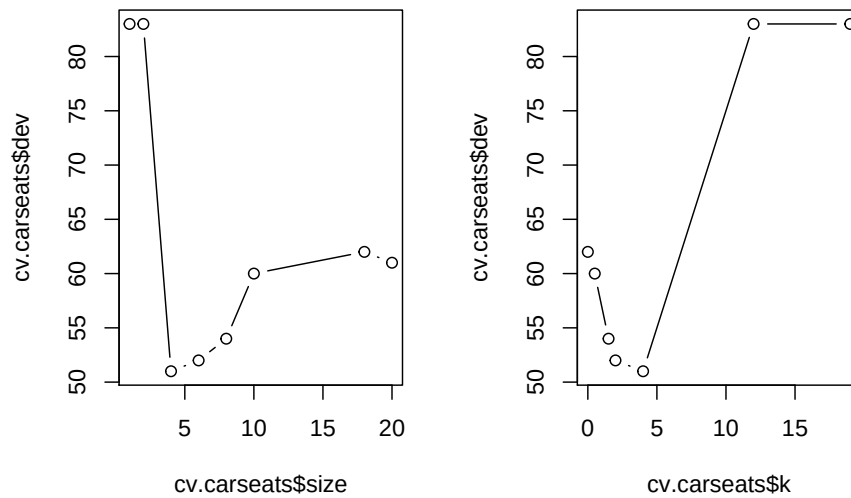
size = number of terminal nodes of each tree

k =  $\alpha$  = cost-complexity parameter.

```

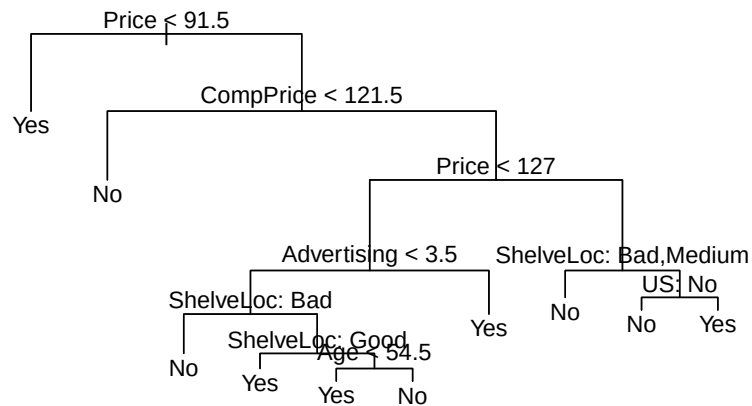
dev = number of cross-validation errors

```
par(mfrow = c(1, 2))
plot(cv.carseats$size, cv.carseats$dev, type = "b")
plot(cv.carseats$k, cv.carseats$dev, type = "b")
```



It seems like 9 tree nodes have the lowest number of cv errors (dev in this case). Hence, we will pick this tree

```
set.seed(7)
prune.carseats <- prune.misclass(tree.carseats, best = 9)
plot(prune.carseats)
text(prune.carseats, pretty = 0)
```

```
tree.pred <- predict(prune.carseats, Carseats.test, type = "class")
table(tree.pred, High.test)
```

```
##           High.test
## tree.pred No  Yes
##           No  89  25
##           Yes 30  56
```

```
(table(tree.pred, High.test)[1, 1] + table(tree.pred, High.test)[2, 2]) /
  nrow(Carseats.test)
```

```
## [1] 0.725
```

We can see slight improvement of prediction rate from 64 to over 70 percent

We can still increase the classification accuracy but with the price of tree complexity

Chapter 9

Multivariate Adaptive Regression Splines

- Also known as MARS
- an adaptive procedure for regression when there are a large number of input variables.
- Can also be viewed as a generalization of stepwise linear regression or a modification of the CART procedure that improves its performance in the regression setting
- uses expansions in the piecewise linear basis function
- To mimic CART, the MARS procedure makes the following changes:
 - Replace the piece wise linear basis function by step functions (e.g, $I(x - t > 0); I(x - t) \leq 0$)
 - When a model term is in a multiplication by a candidate term, it gets replaced by the interaction, and hence it not available for further interaction
 - With these changes, the MARS forward procedure is similar to the CART tree-growing algorithm (i.e., multiplying a step function by a pair of reflected step functions is equivalent to splitting a node at the step. The second restriction is a node may not be split more than once (leaning to the binary tree structure)). Hence, MARS is more powerful because it allows for additive structures because it's not limited to the binary tree structure
- MARS is not used that much today anymore.

Consider basis function of the form (“+” means “positive part” with a not at t)

$$(x - t)_+ \begin{cases} x - t & \text{if } x > t \\ 0 & \text{otherwise} \end{cases}$$

and

$$(t - x)_+ \begin{cases} t - x & \text{if } x < t \\ 0 & \text{otherwise} \end{cases}$$

These are linear splines and the pair is **reflected pair**.

We form such reflected pairs for each input X_j with knots at each observed value x_{ij} of that input variable.

The collection of basis function is

$$C = \{(X_j - t)_+, (t - X_j)_+\}_{t \in \{x_{1j}, \dots, x_{nj}\}, j = 1, \dots, p}$$

there can be as many as $2np$ basis functions (if all of the input values are distinct), which can be large.

Consider a model fo the form

$$f(X) = \beta_0 + \sum_{m=1}^M \beta_m h_m(X)$$

where each $h_m(X)$ is a function in C , or a product of two or more such functions.

Similar to forward stepwise linear regression, we try to build a final model using these functions of the inputs (instead of the original inputs)

For a given choice of h_m , we estimate the β_m using linear squares (i.e., minimizing the residual sum of squares). Hence, the choice of $h_m(X)$ is very important

First, we choose the constant function $h_0(X) = 1$ and all of the functions in C are candidates.

At each stage, we consider a new basis function pair all products of a function h_m in the existing model set M with one of the reflected pairs in C , i.e., the

$$\hat{\beta}_{M+1} h_l(X) \times (X_j - t)_+ + \hat{\beta}_{M+2} h_l(X) \times (t - X_j)_+, h_l \in M$$

that produces the largest decrease in training error

h_l here is some basis function that is already in the model

Then, these are added to the model set M

Example:

First, consider adding to the model a function of the form $\beta_1(X_j - t)_+ + \beta_2(t - X_j)_+ : t \in \{x_{ij}\}$ (multiplication by the constant function produces the function itself)

Assume that for a particular data set the best choice is $\hat{\beta}_1(X_2 - x_{72})_+ + \hat{\beta}_2(x_{72} - X_2)_+$ we then add this to the set M

Then, we consider adding a pair of products of the form

$$h_m(X) \times (X_j - t)_+$$

$$h_m(X) \times (t - X_j)_+, t \in \{x_{ij}\}$$

where for h_m we have the choices

$$h_0(X) = 1, h_1(X) = (X_2 - x_{72})_+$$

or

$$h_2(X) = (x_{72} - X_2)_+$$

For example, the third choice produces functions such as $(X_1 - x_{51})_+ \times (x_{72} - X_2)_+$

The procedure continues until a pre-specified number of terms is in the model, which will be a large model that overfits the data.

Hence, we use backward deletion (i.e., the term that leads to the smallest increase in residual squared error is deleted at each stage). Thus, we have an estimated “best” model $\hat{f}_\lambda$ of each size (number of terms) λ

We could use cross-validation to choose the best model, but it’s computationally intensive. Instead, we use generalized cross-validation, which is defined as

$$GCV(\lambda) = \frac{\sum_{i=1}^n (y_i - \hat{f}_\lambda(x_i))^2}{(1 - M(\lambda)/n)^2}$$

where $M(\lambda)$ is the effective number of parameters in the model. (i.e., accounting for λ and the number of parameters used in selecting the optimal position for the knots).

Typically, we use $M(\lambda) = r + cK$ where

- r = number of linearly independent basis functions

- K = number of knots selected in the forward process
- $c = 3$ (typically) or 2 if the model is restricted to be additive

Advantages

- A major advantage of these piecewise linear basis functions is that they are “local” (i.e., non-zero in only a small portion of the features space, even when multiplied together), which facilitates model fitting in high dimensions.
- A computational advantage because it takes on the order of n operations to try very knot in each step of the process (for each of the $j = 1, \dots, p$ variables)
- If higher order products are in the model, lower order components must also be included in the model. (in reality, we don’t have to, but it’s a good modeling practice)
- MARS can be extended to classification settings (e.g, setting the response variable to 0 and 1) but it’s artificial. There are specific MARS-like algorithms designed for classification

Disadvantages

- Each input can occur at most one time in a product. Thus, **higher order powers of an input are not allowed**. If you include it, then it can produce unstable fits, particularly at the boundaries
 - One can also restrict products to be of a certain order (e.g, only second-degree or third-degree)

Chapter 10

Bagging

- Bootstrapping can help tree methods (e.g., CART)
- Bagging is a special of random forest with $m = p$ where m is the number of input variables (used in a tree), and p is the number of available variables.
- Since CART and MARS suffer from high variance (because if we split the training data into 2 samples at random and fit both halves, the results can be quite different), we use **bootstrap aggregation** (i.e., **bagging**), which is a general-proposed procedure to reduce variance
- If the procedure has low variance, it will yield similar results if applied repeatedly to distinct data sets (e.g., as in linear regression when n is larger relative to p)
- The idea of bagging is that **averaging reduces variance** (somewhat like measurement error). To reduce variation, we take many training sets, and build a separate prediction model, and average them.

To execute bagging, we calculate $\hat{f}^1(X), \hat{f}^2(X), \dots, \hat{f}^B(X)$ using B separate training sets, and average them

$$\hat{f}_{bag}(X) = \frac{1}{B} \sum_{b=1}^B \hat{f}^b(x)$$

However, in reality, we don't have access to multiple training sets, so we use bootstrap samples from our (single) training set (i.e., we generate B bootstrap sample - **sampling randomly with replacement**). Hence, $\hat{f}_{bag}(x)$ is the result of averaging the fits from these B bootstrap samples.

In the case of regression trees with bagging, we construct B regression trees and average the resulting predictions, but the trees are grown large with no

pruning. Hence, each tree has high variance but low bias and averaging then reduces the variance. In practice, **100** samples is sufficient to reduce the variance significantly.

Notes:

- Bagging can be used for classification, where we record the class predicted by each of the B trees for a given test observation and then choose the category that occurs the most often.
- We can also estimate the error associated with a bagged model **without having to perform cross-validation**.
 - The observations not used in the bootstrap samples (for each sample) are the out-of-bag (OOB) observations.
 - We predict the response for the i -th observation using each of the trees in which the observation was OOB. Then, we average to get a single prediction, and get OOB MSE (or classification error)
 - OOB error is a valid estimate of the test error for the bagged model because the response for each observation is predicted using only the trees that were fit without using that observation.

Disadvantages

- Although bagging can improve predictive power (relative to a single tree), it reduced interpretation (i.e., when we average over many different trees, the variable splits (or basis function included) can change dramatically, preventing interpretation). Thus, we can't know which variable is important
 - To deal with this problem, we can keep track of (to get measure of variable importance in the tree)
 - * the total amount that the residual sum of squares is decreased due to splits over a given predictor averaged over all B trees (in the case of regression), or
 - * add up the total amount that the Gini index is decreased by splits over a given predictor, averaged over all B trees (in case of classification).

```
library(randomForest)
library(ISLR2)
library(tree)

attach(Boston)

set.seed(1)
```



```

train <- sample(1:nrow(Boston), nrow(Boston)/2)
boston.test <- Boston[-train, "medv"]

bag.boston <-
  randomForest(
    medv ~ .,
    data = Boston,
    subset = train,
    mtry = 12, # all 12 predictors should be used for each split of the tree
    importance = T
  ) # we could use random forest because bagging is a special case of random forest with m = p
bag.boston

```

```

##
## Call:
## randomForest(formula = medv ~ ., data = Boston, mtry = 12, importance = T, subset = train)
##           Type of random forest: regression
##           Number of trees: 500
## No. of variables tried at each split: 12
##
##           Mean of squared residuals: 10.99681
##           % Var explained: 85.69

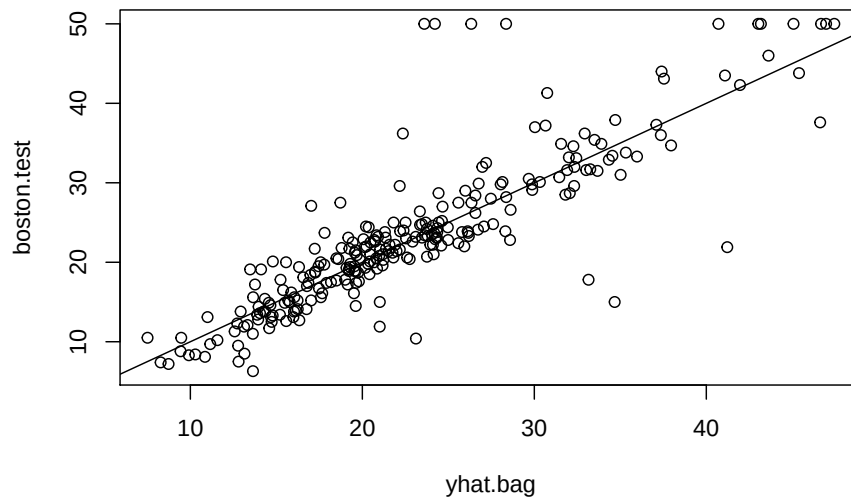
```

Bagging performance

```

yhat.bag <- predict(bag.boston, newdata = Boston[-train, ])
plot(yhat.bag, boston.test)
abline(0, 1)

```



```
mean((yhat.bag - boston.test) ^ 2) # test set MSE
```

```
## [1] 23.04225
```

To change the number of grown trees

```
bag.boston <-
  randomForest(
    medv ~ . ,
    data = Boston,
    subset = train,
    mtry = 12,
    ntree = 25 # specify number of grown tree
  )
yhat.bag <- predict(bag.boston, newdata = Boston[-train,])
mean((yhat.bag - boston.test) ^ 2)
```

```
## [1] 23.63749
```

randomForest() uses

- $\frac{p}{3}$ variables to build a random forest of Regression Trees
- $\sqrt{p}$ variables to build a random forest of Classification Trees

```
set.seed(1)
rf.boston <-
  randomForest(
```

```

    medv ~ . ,
    data = Boston,
    subset = train,
    mtry = 6,
    # use 6 variables
    importance = T
  )
yhat.rf <- predict(rf.boston, newdata = Boston[-train, ])
mean((yhat.rf - boston.test) ^ 2)

```

```
## [1] 19.62021
```

Random forest outperforms bagging (20 < 25)

To o see variable importance

```
importance(rf.boston)
```

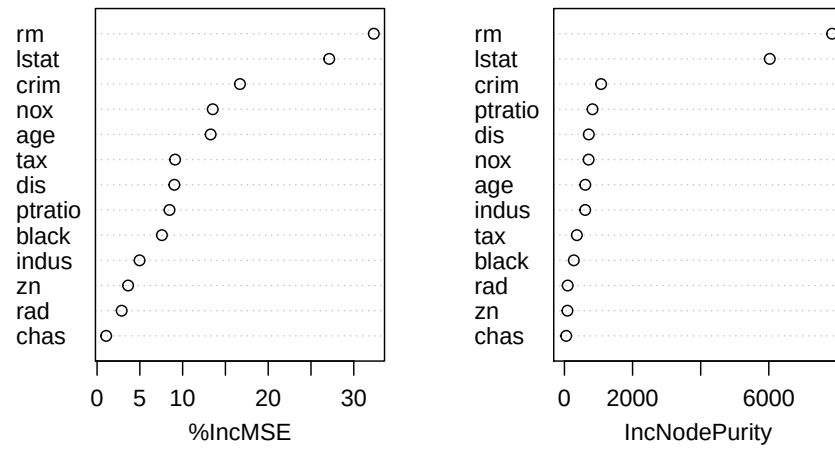
```
##           %IncMSE IncNodePurity
## crim      16.697017      1076.08786
## zn         3.625784         88.35342
## indus      4.968621        609.53356
## chas       1.061432         52.21793
## nox       13.518179        709.87339
## rm        32.343305       7857.65451
## age       13.272498        612.21424
## dis        9.032477        714.94674
## rad        2.878434         95.80598
## tax        9.118801        364.92479
## ptratio    8.467062        823.93341
## black      7.579482        275.62272
## lstat     27.129817       6027.63740
```

2 columns:

1. mean decrease of accuracy in predictions on OOB samples
2. the total decrease in node impurity after splitting and averaged over all trees
 - Regression Trees based on the training RSS
 - Classification Trees based on the deviance

```
varImpPlot(rf.boston) #rm and lstate are most important
```

rf.boston



Chapter 11

Boosting

- Similar to Bagging but more powerful, where we can improve the predictions from decision trees
- It was designed for classification problems (but now we can also apply to regression settings).
- In the machine learning world, this method combines the output of many “weak” learners to produce a power “committee”
- A classic boosting algorithm for classification is “AdaBoost.M1” (Freund and Schapire, 1997)
- It can be considered as one of the best off-the-shelf classifiers
- Under boosting, trees are *not grown independently*, but rather **sequentially**
- There are no bootstrap samples with boosting because we build trees on the residuals in a step-by-step manner
- In the literature, “fitting the data hard” means we fit a single large decision tree to the data, but we will **learn slowly** by fitting a decision tree to the residuals (not Y) from the model, which helps to improve the part of the tree that did not perform well in the existing tree structure.
- We add this new decision tree into the fitted function with a shrinkage parameter (λ) to slow the process down to add more trees to areas that don’t fit well (also helps prevent overfitting). Hence, each tree after will depend strongly on the trees that have already been grown.

11.1 Regression Boosting

Algorithm for boosting regression trees

1. Set $\hat{f}(x) = 0$ and $r_i = y_i \forall i$ (in the training set)
2. For $b = 1, \dots, B$ repeat
 1. Fit a tree $\hat{f}^b$ with d splits ($d+1$ terminal nodes) to the training data (X, r)
 2. Update $\hat{f}$ by adding in a shrunk version of the new tree $\hat{f}(x) \leftarrow \hat{f}(x) + \lambda \hat{f}^b(x)$
 3. Update the residuals $r_i \leftarrow r_i - \lambda \hat{f}^b(x_i)$
3. Final boosted model $\hat{f}(x) = \sum_{b=1}^B \lambda \hat{f}^b(x)$

Boosting has 3 tuning parameters:

1. B : the number of trees. boosting can overfit if B is too large, we can use Cross-validation Fitting to select B
2. λ : the shrinkage parameter, which controls the rate at which boosting learns. Typical values are 0.01 or 0.001 (best choice depends on the data at hand). The smaller the λ , the larger B must be to achieve good performance
3. d : the number of splits in each tree. $d = 1$ works well (surprisingly), which we call the tree a “stump.” With $d = 1$, the boosted ensemble is fitting an additive model since each term only involves one variable. More generally, d is the interaction depth and controls the interaction order of the boosted model (since d splits can involve at most d variables).

11.2 Classification Boosting

For classification problem, consider a two-class classification code such that $Y \in \{-1, 1\}$ and a vector of predictor variables, X .

We have a classifier $G(X)$ that produces prediction that selects either -1 or 1. the error rate on the training sample is

$$\bar{err} = \frac{1}{n} \sum_{i=1}^n I(y_i \neq G(x_i))$$

A weak classifier is one whose error rate is only slightly better than random guessing. Now, boosting sequentially applies a weak classification algorithm to produce a sequence of weak classifiers, $G_m(x), m = 1, \dots, M$ (M is above B)

The predictions are then combined through a **weighted majority vote** to produce the final prediction

$$G(x) = \text{sign}\left(\sum_{m=1}^M \alpha_m G_m(x)\right)$$

where the weights $\alpha_m, m = 1, \dots, M$ are computed by the boosting algorithm and give highest influence to the best classifiers

At each boosting step, weights $w_1, \dots, w_n$ are applied to the training observations $(x_i, y_i), i = 1, \dots, n$.

To start, weights are set equal, $w_i = \frac{1}{n}$ so there is no preference for the first classification.

For each successive iteration, the observation weights at iteration m are modified such that those observations that were misclassified by the classifier $G_{m-1}(x)$ have their weights increased, those that were correctly classified have their weights decreased

As iterations continues, observations that are difficult to classify correctly receive ever-increasing influence. Each successive classifier is forced to concentrate on those training observations that are missed by the previous classifiers in the sequence

Why boosting works so well?

From the perspective of a basis function expansion (3.1:

$$f(x) = \sum_{m=1}^M \beta_m b(x; \gamma_m)$$

where

- β_m are the expansion coefficients
- $b(x, \gamma)$ are simple functions of the multivariate inputs x and parameters γ

then, boosting is a way of fitting such an additive expansion where the basis functions are the individual classifiers $G_m(x) \in \{-1, 1\}$

To fit this model, we minimize some loss function averaged over the training data (e.g, squared error loss).

$$\min_{\{\beta_m, \gamma_m\}_1^M} \sum_{i=1}^n L(y_i, \sum_{m=1}^M \beta_m b(x_i; \gamma_m))$$

For most loss functions $L(y, f(x))$ and basis functions $b(x; \gamma)$, the numerical optimization algorithms are usually computationally intensive.

In some cases, a solution is to solve the subproblem of fitting just a single basis function

$$\min_{\{\beta, \gamma\}} \sum_{i=1}^n L(y_i, \beta b(x_i; \gamma))$$

Then, we approximate the more general minimization problem by sequentially adding new basis functions to the expansion without adjusting the parameters and coefficients that have already been added. This process is called **forward stagewise additive modeling** (i.e., at each iteration m , we solve for the optimal basis function $b(x, \gamma_m)$ and coefficient β_m to add to the current expansion $f_{m-1}(x)$, which gives $f_m(x)$ and we repeat the process while previously added terms remain constant)

Example: squared error loss

$$L(y, f(x)) = (y - f(x))^2$$

then

$$\begin{aligned} L(y_i, f_{m-1}(x_i) + \beta_m b(x_i; \gamma_m)) &= (y_i - f_{m-1}(x_i) - \beta_m b(x_i; \gamma_m))^2 \\ &= (r_{im} - \beta_m b(x_i; \gamma_m))^2 \end{aligned}$$

where $r_{im} = y_i - f_{m-1}(x)$ is the residual of the current model on the i -th observation.

The term $\beta_m b(x; \gamma_m)$ that best fits the current residuals is added to the expansion at each step

11.3 AdaBoost

Ideas of AdaBoost

1. It combines “weak learners” (e.g., stumps) to make classifications
2. Weights of stumps are different
3. Each stump is made sequentially based on previous stump’s mistakes.

AdaBoost.M1

1. Initialize the observation weights $w_i = \frac{1}{N}, i = 1, \dots, N$
2. For $m = 1$ to M
 1. Fit a classifier $G_m(x)$ to the training data with weights w_i
 2. Calculate $err_m = \frac{\sum_{i=1}^N w_i I(y_i \neq G_m(x_i))}{\sum_{i=1}^N w_i}$
 3. Calculate $\alpha_m = \log(\frac{1-err_m}{err_m})$
 4. Set $w_i \leftarrow w_i \times \exp(\alpha_m \times I(y_i \neq G_m(x_i))), i = 1, \dots, N$
3. $G(x) = \text{sign}(\sum_{m=1}^M \alpha_m G_m(x))$

AdaBoost.M1 is equivalent to forward stagewise modeling using the loss function (Hastie et al., 2001, 10.4)

$$L(y, f(x)) = \exp(-yf(x))$$

which is the “exponential loss” (closely related to the binomial negative log-likelihood (deviance), although they have different properties in finite data (real-world) settings)

This loss function is attractive because it facilitates computation

There are more loss functions. Typically we might choose a loss function because

- it’s robust to certain properties (e.g., outliers)
- it facilitates computation

When a loss function is differentiable, we can make use of the gradient of the loss function to improve the optimization, which is the basis of gradient boosting (“gradient boosting models (GBM)”); formerly, Multiple Additive Regression Trees (MART)).

11.4 Gradient Boosting

- The algorithm XGBoost has become one of the standard implementations of gradient boosting, which is present in both R and Python.
 - It uses Newton-Raphson for its optimization (i.e., a second order Taylor’s approximation for the loss function to make the connection) instead of gradient descent
 - It improves efficiency for more complicated settings than the original gradient boosting algorithms
- Additional regularization and built-in cross-validation and handling of missing data are standard
- Compare to AdaBoost: Gradient Boost makes a single leaf, representing an initial guess for the Weights of all of the samples. Similar to AdaBoost, a new tree is made based on previous tree’s errors and gradient boost scales the trees (but by the same amount). Unlike AdaBoost, each tree can be larger than a stump.
- Scaling the tree by the Learning Rate makes a small step in the right direction

11.4.1 Gradient Boost Regression

Input: Data $\{(x_i, y_i)\}_{i=1}^n$ and a differentiable Loss Function $L(y_i, F(x))$ (e.g., Regression: squared residual: $0.5(\text{Observed} - \text{Predicted})^2$)

Step 1: Initialize model with a constant value: $F_0(x) = \underset{\gamma}{\operatorname{argmin}} \sum_{i=1}^n L(y_i, \gamma)$

Step 2: for $m = 1 \rightarrow M$ (number of trees):

1. Compute the (pseudo) residual: $r_{im} = -[\frac{\partial L(y_i, F(x_i))}{\partial F(x_i)}]_{F(x)=F_{m-1}(x)}$ for $i = 1, \dots, n$ (where $\frac{\partial L(y_i, F(x_i))}{\partial F(x_i)}$ is the gradient, ergo the name gradient boost)
2. Fit a regression tree to the r_{im} values and create terminal regions R_{jm} for $j = 1, \dots, J_m$
3. For $j = 1, \dots, J_m$ compute $\gamma_{jm} = \underset{\gamma}{\operatorname{argmin}} \sum_{x_i \in R_{ij}} L(y_i, F_{m-1}(x_i) + \gamma)$
4. Update $F_m(x) = F_{m-1}(x) + \nu \sum_{j=1}^{J_m} \gamma_{jm} I(x \in R_{jm})$ where ν is the learning rate (0 to 1) (smaller value means slower learning rate)

Output: $F_M(x)$

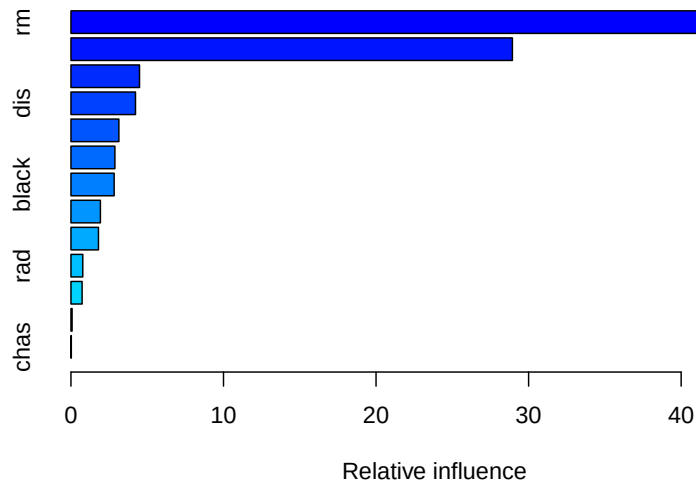
```
library(gbm)
library(ISLR2)
library(tree)

attach(Boston)

set.seed(1)
train <- sample(1:nrow(Boston), nrow(Boston)/2)
boston.test <- Boston[-train, "medv"]

boost.boston <-
  gbm(
    medv ~ .,
    data = Boston[train, ],
    distribution = "gaussian", # because this is regression setting
    # distribution = "bernoulli", # use this for classification
    n.trees = 5000, # number of trees
    interaction.depth = 4 # limits the depth of each tree
  )

summary(boost.boston) #rm and lstate are most important
```



```
##          var      rel.inf
## rm          rm 48.13967682
## lstat      lstat 28.93851185
## crim       crim  4.49413146
## dis        dis  4.23182696
## age        age  3.14221169
## nox        nox  2.88094283
## black      black 2.83238772
## ptratio    ptratio 1.93050932
## tax        tax  1.80427054
## rad        rad  0.77569461
## indus      indus 0.73110525
## zn         zn   0.07442923
## chas       chas  0.02430170
```

11.4.2 Gradient Boost Classification

Input: Data $\{(x_i, y_i)\}_{i=1}^n$ and a differentiable Loss Function $L(y_i, F(x))$ (e.g., Classification: negative log likelihood: $-\sum_{i=1}^N \mathbf{y}_i \times \log(\mathbf{p}) + (1 - \mathbf{y}_i) \times \log(1 - \mathbf{p})$ where y_i is the observed value, equivalently, $-\text{Observed} \times \log(\text{odds}) + \log(1 + \exp(\log(\text{odds})))$ which is differentiable)

Step 1: Initialize model with a constant value: $F_0(x) = \underset{\gamma}{\operatorname{argmin}} \sum_{i=1}^n L(y_i, \gamma)$

where γ is a $\log(\text{odds})$ value

Step 2: for $m = 1 \rightarrow M$ (number of trees):

1. Compute the (pseudo) residual: $r_{im} = -[\frac{\partial L(y_i, F(x_i))}{\partial F(x_i)}]_{F(x)=F_{m-1}(x)}$ for $i = 1, \dots, n$ (where $\frac{\partial L(y_i, F(x_i))}{\partial F(x_i)}$ is the gradient, ergo the name gradient boost)
2. Fit a regression tree to the r_{im} values and create terminal regions R_{jm} for $j = 1, \dots, J_m$
3. For $j = 1, \dots, J_m$ compute $\gamma_{jm} = \underset{\gamma}{\operatorname{argmin}} \sum_{x_i \in R_{ij}} L(y_i, F_{m-1}(x_i) + \gamma)$
4. Update $F_m(x) = F_{m-1}(x) + \nu \sum_{j=1}^{J_m} \gamma_{jm} I(x \in R_{jm})$ where ν is the learning rate (0 to 1) (smaller value means slower learning rate)

Output: $F_M(x)$

$$\frac{\sum \text{Residual}_i}{\sum [\text{Previous Prob}_i \times (1 - \text{Previous Prob}_i)]}$$

11.5 Light Gradient Boosting

Reference: Machado et al. (2019) and here

Based on new techniques:

1. Gradient-based One Side Sampling (GOSS)
2. Exclusive Feature Bundling (EFB)

Advantages:

- Faster and higher efficiency
- Lower Memory
- Better accuracy
- Can be used with parallel, distributed computing, GPU learning
- Can handle big data.
- An improvement on XGBoost

Function Reference

Example by lightgbm

```
library(lightgbm)
data(agaricus.train, package='lightgbm')
train <- agaricus.train
dtrain <- lgb.Dataset(train$data, label = train$label)
model <- lgb.cv(
  params = list(
    objective = "regression"
    , metric = "l2"
```

```

    )
    , data = dtrain
)

```

```

## [LightGBM] [Warning] Auto-choosing row-wise multi-threading, the overhead of testing was 0.002
## You can set `force_row_wise=true` to remove the overhead.
## And if memory is not enough, you can set `force_col_wise=true`.
## [LightGBM] [Info] Total Bins 214
## [LightGBM] [Info] Number of data points in the train set: 4342, number of used features: 107
## [LightGBM] [Warning] Auto-choosing row-wise multi-threading, the overhead of testing was 0.001
## You can set `force_row_wise=true` to remove the overhead.
## And if memory is not enough, you can set `force_col_wise=true`.
## [LightGBM] [Info] Total Bins 214
## [LightGBM] [Info] Number of data points in the train set: 4342, number of used features: 107
## [LightGBM] [Warning] Auto-choosing row-wise multi-threading, the overhead of testing was 0.000
## You can set `force_row_wise=true` to remove the overhead.
## And if memory is not enough, you can set `force_col_wise=true`.
## [LightGBM] [Info] Total Bins 214
## [LightGBM] [Info] Number of data points in the train set: 4342, number of used features: 107
## [LightGBM] [Info] Start training from score 0.481345
## [LightGBM] [Warning] No further splits with positive gain, best gain: -inf
## [LightGBM] [Info] Start training from score 0.491248
## [LightGBM] [Warning] No further splits with positive gain, best gain: -inf
## [LightGBM] [Info] Start training from score 0.473745
## [LightGBM] [Warning] No further splits with positive gain, best gain: -inf
## [1] "[1]: valid's 12:0.20343+0.000764127"
## [LightGBM] [Warning] No further splits with positive gain, best gain: -inf
## [LightGBM] [Warning] No further splits with positive gain, best gain: -inf
## [LightGBM] [Warning] No further splits with positive gain, best gain: -inf
## [1] "[2]: valid's 12:0.165717+0.00102951"
## [LightGBM] [Warning] No further splits with positive gain, best gain: -inf
## [LightGBM] [Warning] No further splits with positive gain, best gain: -inf
## [LightGBM] [Warning] No further splits with positive gain, best gain: -inf
## [1] "[3]: valid's 12:0.134931+0.00122249"
## [LightGBM] [Warning] No further splits with positive gain, best gain: -inf
## [LightGBM] [Warning] No further splits with positive gain, best gain: -inf
## [LightGBM] [Warning] No further splits with positive gain, best gain: -inf
## [1] "[4]: valid's 12:0.110007+0.00139105"
## [LightGBM] [Warning] No further splits with positive gain, best gain: -inf
## [LightGBM] [Warning] No further splits with positive gain, best gain: -inf
## [LightGBM] [Warning] No further splits with positive gain, best gain: -inf
## [1] "[5]: valid's 12:0.0897115+0.00148545"
## [LightGBM] [Warning] No further splits with positive gain, best gain: -inf
## [LightGBM] [Warning] No further splits with positive gain, best gain: -inf
## [LightGBM] [Warning] No further splits with positive gain, best gain: -inf

```

```

## [1] "[6]: valid's 12:0.0733307+0.00158646"
## [LightGBM] [Warning] No further splits with positive gain, best gain: -inf
## [LightGBM] [Warning] No further splits with positive gain, best gain: -inf
## [LightGBM] [Warning] No further splits with positive gain, best gain: -inf
## [1] "[7]: valid's 12:0.0600474+0.00163927"
## [LightGBM] [Warning] No further splits with positive gain, best gain: -inf
## [LightGBM] [Warning] No further splits with positive gain, best gain: -inf
## [LightGBM] [Warning] No further splits with positive gain, best gain: -inf
## [1] "[8]: valid's 12:0.0492893+0.00168788"
## [LightGBM] [Warning] No further splits with positive gain, best gain: -inf
## [LightGBM] [Warning] No further splits with positive gain, best gain: -inf
## [LightGBM] [Warning] No further splits with positive gain, best gain: -inf
## [1] "[9]: valid's 12:0.0405069+0.00168261"
## [LightGBM] [Warning] No further splits with positive gain, best gain: -inf
## [LightGBM] [Warning] No further splits with positive gain, best gain: -inf
## [1] "[10]: valid's 12:0.0333282+0.0017056"
## [LightGBM] [Warning] No further splits with positive gain, best gain: -inf
## [LightGBM] [Warning] No further splits with positive gain, best gain: -inf
## [1] "[11]: valid's 12:0.0275167+0.0017165"
## [LightGBM] [Warning] No further splits with positive gain, best gain: -inf
## [LightGBM] [Warning] No further splits with positive gain, best gain: -inf
## [1] "[12]: valid's 12:0.0228061+0.00173197"
## [LightGBM] [Warning] No further splits with positive gain, best gain: -inf
## [LightGBM] [Warning] No further splits with positive gain, best gain: -inf
## [1] "[13]: valid's 12:0.0189156+0.00175123"
## [LightGBM] [Warning] No further splits with positive gain, best gain: -inf
## [LightGBM] [Warning] No further splits with positive gain, best gain: -inf
## [1] "[14]: valid's 12:0.0157203+0.00167966"
## [LightGBM] [Warning] No further splits with positive gain, best gain: -inf
## [LightGBM] [Warning] No further splits with positive gain, best gain: -inf
## [1] "[15]: valid's 12:0.0131337+0.00163128"
## [LightGBM] [Warning] No further splits with positive gain, best gain: -inf
## [1] "[16]: valid's 12:0.0110208+0.00159361"
## [LightGBM] [Warning] No further splits with positive gain, best gain: -inf
## [1] "[17]: valid's 12:0.00928227+0.00149819"
## [LightGBM] [Warning] No further splits with positive gain, best gain: -inf
## [1] "[18]: valid's 12:0.00786465+0.00142062"
## [1] "[19]: valid's 12:0.00671564+0.00134583"
## [1] "[20]: valid's 12:0.00578033+0.0012865"
## [1] "[21]: valid's 12:0.00499955+0.00121312"
## [LightGBM] [Warning] No further splits with positive gain, best gain: -inf
## [1] "[22]: valid's 12:0.00435034+0.00116644"
## [1] "[23]: valid's 12:0.00382413+0.00113199"
## [LightGBM] [Warning] No further splits with positive gain, best gain: -inf
## [1] "[24]: valid's 12:0.00340385+0.001089"
## [LightGBM] [Warning] No further splits with positive gain, best gain: -inf

```

```
## [1] "[25]: valid's 12:0.00303964+0.00105824"
## [1] "[26]: valid's 12:0.00274592+0.00103256"
## [LightGBM] [Warning] No further splits with positive gain, best gain: -inf
## [1] "[27]: valid's 12:0.0025083+0.0010174"
## [1] "[28]: valid's 12:0.00229756+0.000995728"
## [1] "[29]: valid's 12:0.00213457+0.000974289"
## [1] "[30]: valid's 12:0.00199983+0.000959303"
## [1] "[31]: valid's 12:0.00187945+0.000936821"
## [1] "[32]: valid's 12:0.00178762+0.000926426"
## [1] "[33]: valid's 12:0.00169532+0.000905434"
## [1] "[34]: valid's 12:0.00162788+0.000891321"
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## [1] "[36]: valid's 12:0.00148941+0.000848408"
## [1] "[37]: valid's 12:0.00144201+0.000833432"
## [1] "[38]: valid's 12:0.00138472+0.000815984"
## [1] "[39]: valid's 12:0.00133625+0.000799245"
## [1] "[40]: valid's 12:0.00129534+0.000783908"
## [1] "[41]: valid's 12:0.00123877+0.000756225"
## [1] "[42]: valid's 12:0.00120462+0.000739414"
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## [1] "[45]: valid's 12:0.00108563+0.000673361"
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## [1] "[68]: valid's 12:0.000644114+0.000430402"
## [1] "[69]: valid's 12:0.000633505+0.000422865"
```

```
## [1] "[70]: valid's 12:0.000623016+0.000416683"
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## [1] "[72]: valid's 12:0.000598924+0.000403649"
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## [1] "[74]: valid's 12:0.000579913+0.000391408"
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## [1] "[78]: valid's 12:0.000537647+0.000367313"
## [1] "[79]: valid's 12:0.000528014+0.000359727"
## [1] "[80]: valid's 12:0.000518134+0.000354966"
## [1] "[81]: valid's 12:0.000507917+0.000348403"
## [1] "[82]: valid's 12:0.000500364+0.000343793"
## [1] "[83]: valid's 12:0.000491649+0.000338621"
## [1] "[84]: valid's 12:0.000484727+0.000334855"
## [1] "[85]: valid's 12:0.000474592+0.000327903"
## [1] "[86]: valid's 12:0.000467556+0.000322853"
## [1] "[87]: valid's 12:0.000458133+0.000317051"
## [1] "[88]: valid's 12:0.000450222+0.000312505"
## [1] "[89]: valid's 12:0.000443537+0.000308884"
## [1] "[90]: valid's 12:0.00043687+0.000304464"
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## [1] "[92]: valid's 12:0.000422767+0.00029546"
## [1] "[93]: valid's 12:0.000415459+0.000291728"
## [1] "[94]: valid's 12:0.0004087+0.00028792"
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## [1] "[97]: valid's 12:0.000391296+0.000278707"
## [1] "[98]: valid's 12:0.000386102+0.000275288"
## [1] "[99]: valid's 12:0.000379684+0.000271936"
## [1] "[100]: valid's 12:0.000373031+0.000267905"
```

Example by Geeksforgeek (python)

Data link

```
# library(reticulate)
# py_install("lightgbm")
# py_install("pandas")
# py_install("xgboost")

#importing standard libraries
import numpy as np
import pandas as pd
from pandas import Series, DataFrame

#import lightgbm and xgboost
```



```

import lightgbm as lgb
import xgboost as xgb

#loading our training dataset 'adult.csv' with name 'data' using pandas
data=pd.read_csv('images/adult.data',header=None)

#Assigning names to the columns
data.columns=['age','workclass','fnlwgt','education','education-num','marital_Status','occupation']

#glimpse of the dataset
data.head()

# Label Encoding our target variable

##      age      workclass  fnlwgt  ...  hours_per_week  native_country  Income
## 0    39      State-gov   77516  ...             40    United-States  <=50K
## 1    50  Self-emp-not-inc   83311  ...             13    United-States  <=50K
## 2    38      Private  215646  ...             40    United-States  <=50K
## 3    53      Private  234721  ...             40    United-States  <=50K
## 4    28      Private  338409  ...             40           Cuba  <=50K
##
## [5 rows x 15 columns]
from sklearn.preprocessing import LabelEncoder,OneHotEncoder
l=LabelEncoder()
l.fit(data.Income)

## LabelEncoder()
l.classes_

## array([' <=50K', ' >50K'], dtype=object)
data.Income=Series(l.transform(data.Income)) #label encoding our target variable
data.Income.value_counts()

#One Hot Encoding of the Categorical features

## 0    24720
## 1     7841
## Name: Income, dtype: int64
one_hot_workclass=pd.get_dummies(data.workclass)
one_hot_education=pd.get_dummies(data.education)
one_hot_marital_Status=pd.get_dummies(data.marital_Status)
one_hot_occupation=pd.get_dummies(data.occupation)

```

```

one_hot_relationship=pd.get_dummies(data.relationship)
one_hot_race=pd.get_dummies(data.race)
one_hot_sex=pd.get_dummies(data.sex)
one_hot_native_country=pd.get_dummies(data.native_country)

#removing categorical features
data.drop(['workclass','education','marital_Status','occupation','relationship','race']

#Merging one hot encoded features with our dataset 'data'
data=pd.concat([data,one_hot_workclass,one_hot_education,one_hot_marital_Status,one_hot

#removing duplicate columns
_, i = np.unique(data.columns, return_index=True)
data=data.iloc[:, i]

#Here our target variable is 'Income' with values as 1 or 0.
#Separating our data into features dataset x and our target dataset y
x=data.drop('Income',axis=1)
y=data.Income

#Imputing missing values in our target variable
y.fillna(y.mode()[0],inplace=True)

#Now splitting our dataset into test and train
from sklearn.model_selection import train_test_split
x_train,x_test,y_train,y_test=train_test_split(x,y,test_size=.3)

#The data is stored in a DMatrix object
#label is used to define our outcome variable
dtrain=xgb.DMatrix(x_train,label=y_train)
dtest=xgb.DMatrix(x_test)

#setting parameters for xgboost
parameters={'max_depth':7, 'eta':1, 'silent':1,'objective':'binary:logistic','eval_metr

#training our model
num_round=50
from datetime import datetime
start = datetime.now()
xg=xgb.train(parameters,dtrain,num_round)

```

```

## [13:34:03] WARNING: D:\bld\xgboost-split_1634712635879\work\src\learner.cc:576:
## Parameters: { "silent" } might not be used.
##
## This could be a false alarm, with some parameters getting used by language bindings but
## then being mistakenly passed down to XGBoost core, or some parameter actually being used
## but getting flagged wrongly here. Please open an issue if you find any such cases.

stop = datetime.now()

#Execution time of the model
execution_time_xgb = stop-start
execution_time_xgb

#datetime.timedelta( , , ) representation => (days , seconds , microseconds)
#now predicting our model on test set

## datetime.timedelta(microseconds=876667)
ypred=xg.predict(dtest)
ypred

#Converting probabilities into 1 or 0

## array([0.05985198, 0.32874027, 0.30227152, ..., 0.5249348 , 0.5778189 ,
##        0.04279852], dtype=float32)
for i in range(0,9769):
    if ypred[i]>=.5:      # setting threshold to .5
        ypred[i]=1
    else:
        ypred[i]=0

#calculating accuracy of our model
from sklearn.metrics import accuracy_score
accuracy_xgb = accuracy_score(y_test,ypred)
accuracy_xgb

# Light GBM

## 0.8585320913092436

train_data=lgb.Dataset(x_train,label=y_train)
#setting parameters for lightgbm

```

[illegible]


```

        ypred2[i]=0

#calculating accuracy
accuracy_lgbm = accuracy_score(ypred2,y_test)
accuracy_lgbm

## 0.8581226328180981

y_test.value_counts()

## 0      7395
## 1      2374
## Name: Income, dtype: int64
from sklearn.metrics import roc_auc_score

#calculating roc_auc_score for xgboost
auc_xgb = roc_auc_score(y_test,ypred)
auc_xgb

#calculating roc_auc_score for light gbm.

## 0.7625557296677495

auc_lgbm = roc_auc_score(y_test,ypred2)
auc_lgbm

## 0.7577092208640711

comparison_dict = {
    "accuracy score":(accuracy_lgbm,accuracy_xgb),
    "auc score":(auc_lgbm,auc_xgb),
    "execution time":(execution_time_lgbm,execution_time_xgb)
}

#Creating a dataframe 'comparison_df' for comparing the performance of Lightgbm and xgboost
comparison_df = DataFrame(comparison_dict)
comparison_df.index= ['LightGBM','xgboost']
comparison_df

##           accuracy score  auc score  execution time
## LightGBM           0.858123    0.757709 0 days 00:00:00.160570
## xgboost             0.858532    0.762556 0 days 00:00:00.876667

```

11.6 XGBoost

- Extreme Gradient Boosting
-

11.7 Partial Dependence Plots

- It can help interpret parameter effects from tree-based models (e.g, Bagging and Boosting)

Consider a subvector X_S of $l < p$ of the input predictor space X , indexed by $S \subset \{1, \dots, p\}$

Let C be the complement such that $S \cup C = \{1, \dots, p\}$

We expect that $f(X) = f(X_S, X_C)$ and define the partial dependence of $f(X)$ on X_S to be

$$f_S(X_S) = E_{X_C} f(X_S, X_C)$$

which is the marginal average of f

It can serve as a useful description of the effect of the chosen subset on $f(X)$ when elements of X_S do not have strong interactions with elements of X_C

We can estimate this expectation by

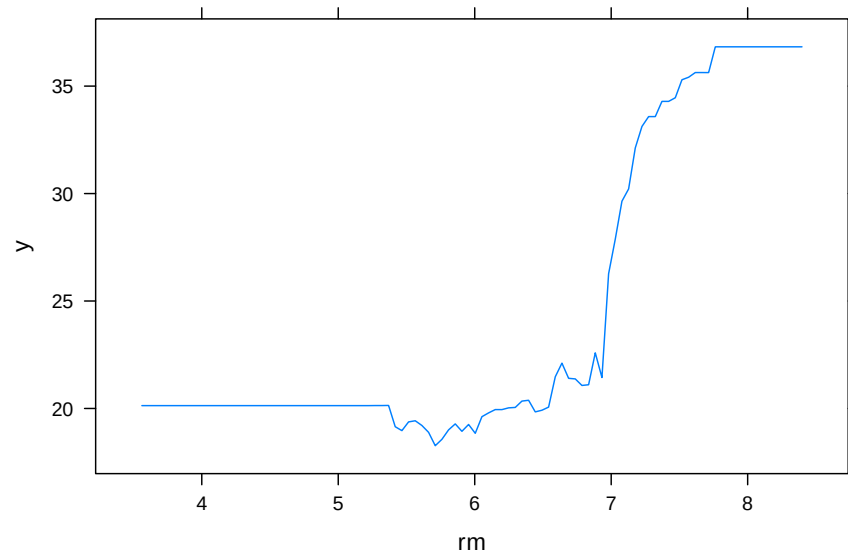
$$\bar{f}_S(X_S) = \frac{1}{n} \sum_{i=1}^n f(X_S, x_{iC})$$

where $\{x_{1C}, \dots, x_{nC}\}$ are the values of X_C occurring in the training data

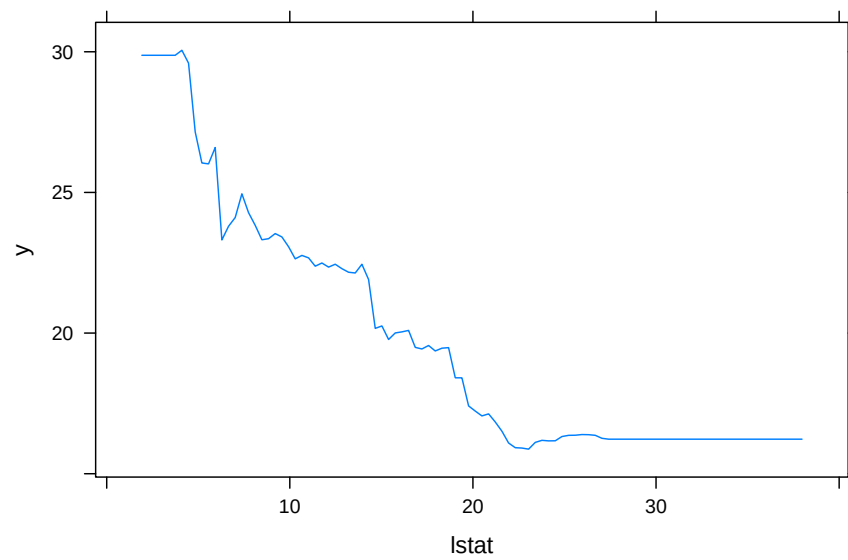
- This can be computationally intensive to evaluate (but, for decision trees $\bar{f}_S(X_S)$ can be computed efficiently from the tree itself).
- The interpretation of the partial dependence plots is that they represent the effect of X_S on $f(X)$ after accounting for the average effects of the other variables X_C on $f(X)$
- You can't assume that this present the effect of X_S *ignoring* the effects of X_C (unless $X_S \perp X_C$)

Partial dependence plots for the 2 most important variables, showing marginal effect of the variables on the response after integrating out the other variables (i.e., conditioned on other variables)

```
plot(boost.boston, i = "rm") # median house prices increase with rm
```



```
plot(boost.boston, i = "lstat") # median house prices decrease with lstate
```



Examine test MSE


```
attach(Boston)
set.seed(1)
train <- sample(1:nrow(Boston), nrow(Boston)/2)

yhat.boost <-
  predict(boost.boston, newdata = Boston[-train, ], n.trees = 5000)
mean((yhat.boost - boston.test) ^ 2)
```

```
## [1] 19.37033
```

The above analysis used $\lambda = 0.001$ (shrinkage parameter)

But we can try other values

```
boost.boston <-
  gbm(
    medv ~ .,
    data = Boston[train, ],
    distribution = "gaussian",
    n.trees = 5000,
    interaction.depth = 4,
    shrinkage = 0.2,
    verbose = F
  )
yhat.boost <-
  predict(boost.boston, newdata = Boston[-train, ], n.trees = 5000)
mean((yhat.boost - boston.test) ^ 2)
```

```
## [1] 18.94809
```

Lower shrinkage parameter (faster learning) might lead to lower test MSE

Chapter 12

Random Forests

- Similar to Bagging since both are bootstrap-based procedure (based on committee of learning trees)
- Different from Bagging because the trees are effectively decorrelated
 - At each split in the tree-building process (for each bootstrap sample), we select the splitting rule from a randomly selected relatively small subset of the original input variables. Hence, different variables are used in the tree building process across the bootstrap sample. Thus, the trees are less dependent on each other (and dominant covariates), reducing the variance of the prediction
- This variance reduction is beneficial because trees are inherently noisy, but if grown fairly deep, have relatively low bias. Hence, averaging many low-biased independently distributed trees together gives relatively unbiased and low variance predictions/classification.

The variance of an average of correlated variables is larger than the average of the same number of independent variables (i.e., the effective degrees of freedom are less for correlated data)

Hence, an average of B identically distributed random variables, each with variance σ^2 that have pairwise correlation ρ is

$$\rho\sigma^2 + \sigma^2 \frac{1-\rho}{B}$$

As B increases, there is still limiting variance $\rho\sigma^2$

If we reduce ρ , then we could get a smaller variance for the average. Hence, random forests reduces the inherent dependence in Bagging

Intuitive steps:

1. Create bootstrapped datasets (with replacement)
2. Create a decision tree (1) using the bootstrapped dataset (2) using a random subset of variables at each step.
3. Repeat step 1 and 2
4. To make prediction, we run the data through all the created decision trees, and see the average, or majority vote (just like Bagging- using aggregated bootstrapped data)

Procedure

We randomly select a smaller number of input variables, m , where $m \leq p$

Typically, we use (rules of thumb, no scientific backing - simulation study):

- $m = \sqrt{p}$ for classification
- $m = \frac{p}{3}$ for regression

After B trees are grown, $\{T(x, \Theta_b)\}_1^B$, the random forest regression predictor is

$$\hat{f}_{rf}^B(x) = \frac{1}{B} \sum_{b=1}^B T(x; \Theta_b)$$

where Θ_b is the cut points associated with the b-th tree

Algorithm for Random Forest (both regression and classification) (Hastie et al., 2001, Alg 15.1)

1. For $b = 1$ to B
 1. Draw a bootstrap sample $\mathbf{Z}^*$ of size N from the training data
 2. Grow a random-forest tree T_b to the bootstrapped data, by recursively repeating the following steps for each terminal node of the tree, until the minimum node size n_{min} is reached:
 1. Select m variables at random from the p variables
 2. Pick the best variable/split-point among the m
 3. Split the node into two daughter nodes
2. Output the ensemble of trees $\{T_b\}_1^B$

To predict a new point x

- Regression: $\hat{f}_{rf}^B(x) = \frac{1}{B} \sum_{b=1}^B T_b(x)$
- Classification: Let $\hat{C}_b(x)$ be the class prediction of the b-th random-forest tree. Then $\hat{C}_{rf}^B(x) = \text{majority vote } \{\hat{C}_b(x)\}_1^B$

- In general, Gradient Boosting is comparable Random Forests and Random Forests is usually better than Bagging. But in practice, this isn't always the case.
- Best method is based on the type of structure in the process (e.g, degree of linearity/non-linearity, dependence in the covariates, relative importance of specific covariates, etc.)
- To get out-of-bag (OOB) samples, for each observation pair (x_i, y_i) , its random forest predictor is the average of the trees corresponding to bootstrap samples in which this observation did **not** appear
- OOB error (i.e., comparison between OOB prediction to the actual response) is effectively a cross-validation error estimate. Once this error stabilizes, then the model is trained.

Variable Importance

- Since random forests typically include more variables (due to the random selection), it's more likely to recognize additional "important" variables
- Variable importance is determined by a measure of the important in the split-criterion for each variable as aggregated across each tree.
- Variable importance can also be determined by the variable's prediction strength (based on OOB prediction). When the b -th tree is grown, the OOB samples are passed down the tree and the prediction accuracy is taken. Then, the value for the j -th variable are randomly permuted in the OOB samples, and the accuracy is computed. The decrease in accuracy is averaged over all trees and gives a measure of importance for variable j .
 - This does not measure the effect on prediction if this variable was not available because the refitted model would likely contain surrogate variables to take its place

Notes:

- Proximity Plots: indicate which observations are effectively close together in terms of the random forest classifier (typically only provide info on where the points lie relative to the decision boundary)
- Overfitting: Random forests can be robust to overfitting if the number of relevant variables in X is large relative to p
- Bias: The bias in random forests is the same as the bias of the individual tree, with only improvement in MSE due to variance reduction
- Similar to Nearest Neighbor Classifiers (e.g., k -nearest neighbor)
- Packages: `randomForest`, `cforest`, `ParallelForest`, `ranger`, `Rborist`
- Random Forests are good with general nonlinear nonparametric prediction

12.1 Application

Example by Josh Starmer

```
load package
library(ggplot2)
library(cowplot)
library(randomForest)
```

NOTE: The data used in this demo comes from the UCI machine learning repository. <http://archive.ics.uci.edu/ml/index.php> Specifically, this is the heart disease data set. <http://archive.ics.uci.edu/ml/datasets/Heart+Disease>

```
url <- "http://archive.ics.uci.edu/ml/machine-learning-databases/heart-disease/processed/pt1.dat"
data <- read.csv(url, header=FALSE)
```

Reformat the data so that it is 1) Easy to use (add nice column names) 2) Interpreted correctly by randomForest.

```
head(data) # you see data, but no column names
```

```
##   V1 V2 V3  V4  V5 V6 V7  V8 V9 V10 V11 V12 V13 V14
## 1 63  1  1 145 233  1  2 150  0 2.3   3 0.0 6.0   0
## 2 67  1  4 160 286  0  2 108  1 1.5   2 3.0 3.0   2
## 3 67  1  4 120 229  0  2 129  1 2.6   2 2.0 7.0   1
## 4 37  1  3 130 250  0  0 187  0 3.5   3 0.0 3.0   0
## 5 41  0  2 130 204  0  2 172  0 1.4   1 0.0 3.0   0
## 6 56  1  2 120 236  0  0 178  0 0.8   1 0.0 3.0   0
```

```
colnames(data) <- c(
  "age",
  "sex", # 0 = female, 1 = male
  "cp", # chest pain
        # 1 = typical angina,
        # 2 = atypical angina,
        # 3 = non-anginal pain,
        # 4 = asymptomatic
  "trestbps", # resting blood pressure (in mm Hg)
  "chol", # serum cholestoral in mg/dl
  "fbs", # fasting blood sugar if less than 120 mg/dl, 1 = TRUE, 0 = FALSE
  "restecg", # resting electrocardiographic results
        # 1 = normal
        # 2 = having ST-T wave abnormality
        # 3 = showing probable or definite left ventricular hypertrophy
  "thalach", # maximum heart rate achieved
  "exang", # exercise induced angina, 1 = yes, 0 = no
  "oldpeak", # ST depression induced by exercise relative to rest
```

```

"slope", # the slope of the peak exercise ST segment
# 1 = upsloping
# 2 = flat
# 3 = downsloping
"ca", # number of major vessels (0-3) colored by fluoroscopy
"thal", # this is short of thalium heart scan
# 3 = normal (no cold spots)
# 6 = fixed defect (cold spots during rest and exercise)
# 7 = reversible defect (when cold spots only appear during exercise)
"hd" # (the predicted attribute) - diagnosis of heart disease
# 0 if less than or equal to 50% diameter narrowing
# 1 if greater than 50% diameter narrowing
)

# head(data) # now we have data and column names

# str(data) # this shows that we need to tell R which columns contain factors
# it also shows us that there are some missing values. There are "?"s
# in the dataset.

## First, replace "?"s with NAs.
data[data == "?"] <- NA

## Now add factors for variables that are factors and clean up the factors
## that had missing data...
data[data$sex == 0,]$sex <- "F"
data[data$sex == 1,]$sex <- "M"
data$sex <- as.factor(data$sex)

data$cp <- as.factor(data$cp)
data$fbs <- as.factor(data$fbs)
data$restecg <- as.factor(data$restecg)
data$exang <- as.factor(data$exang)
data$slope <- as.factor(data$slope)

data$ca <- as.integer(data$ca) # since this column had "?"s in it (which
# we have since converted to NAs) R thinks that
# the levels for the factor are strings, but
# we know they are integers, so we'll first
# convert the strings to integers...
data$ca <- as.factor(data$ca) # ...then convert the integers to factor levels

data$thal <- as.integer(data$thal) # "thal" also had "?"s in it.
data$thal <- as.factor(data$thal)

```

```
## This next line replaces 0 and 1 with "Healthy" and "Unhealthy"
data$hd <- ifelse(test=data$hd == 0, yes="Healthy", no="Unhealthy")
data$hd <- as.factor(data$hd) # Now convert to a factor

str(data) ## this shows that the correct columns are factors and we've replaced

## 'data.frame':    303 obs. of  14 variables:
## $ age      : num  63 67 67 37 41 56 62 57 63 53 ...
## $ sex      : Factor w/ 2 levels "F","M": 2 2 2 2 1 2 1 1 2 2 ...
## $ cp      : Factor w/ 4 levels "1","2","3","4": 1 4 4 3 2 2 4 4 4 4 ...
## $ trestbps: num  145 160 120 130 130 120 140 120 130 140 ...
## $ chol     : num  233 286 229 250 204 236 268 354 254 203 ...
## $ fbs     : Factor w/ 2 levels "0","1": 2 1 1 1 1 1 1 1 1 2 ...
## $ restecg : Factor w/ 3 levels "0","1","2": 3 3 3 1 3 1 3 1 3 3 ...
## $ thalach  : num  150 108 129 187 172 178 160 163 147 155 ...
## $ exang    : Factor w/ 2 levels "0","1": 1 2 2 1 1 1 1 2 1 2 ...
## $ oldpeak  : num  2.3 1.5 2.6 3.5 1.4 0.8 3.6 0.6 1.4 3.1 ...
## $ slope    : Factor w/ 3 levels "1","2","3": 3 2 2 3 1 1 3 1 2 3 ...
## $ ca      : Factor w/ 4 levels "0","1","2","3": 1 4 3 1 1 1 3 1 2 1 ...
## $ thal    : Factor w/ 3 levels "3","6","7": 2 1 3 1 1 1 1 1 3 3 ...
## $ hd      : Factor w/ 2 levels "Healthy","Unhealthy": 1 2 2 1 1 1 2 1 2 2 ...

## "?"s with NAs because "?" no longer appears in the list of factors
## for "ca" and "thal"
```

Now we are ready to build a random forest.

NOTE: For most machine learning methods, you need to divide the data manually into a “training” set and a “test” set. This allows you to train the method using the training data, and then test it on data it was not originally trained on.

In contrast, Random Forests split the data into “training” and “test” sets for you. This is because Random Forests use bootstrapped data, and thus, not every sample is used to build every tree. The “training” dataset is the bootstrapped data and the “test” dataset is the remaining samples. The remaining samples are called the “Out-Of-Bag” (OOB) data.

```
set.seed(42)

## impute any missing values in the training set using proximities
data.imputed <- rfImpute(hd ~ ., data = data, iter=6)

## ntree      OOB      1      2
## 300: 17.49% 12.80% 23.02%
## ntree      OOB      1      2
## 300: 16.83% 14.02% 20.14%
## ntree      OOB      1      2
```



```
## 300: 17.82% 13.41% 23.02%
## ntree 00B 1 2
## 300: 17.49% 14.02% 21.58%
## ntree 00B 1 2
## 300: 17.16% 12.80% 22.30%
## ntree 00B 1 2
## 300: 18.15% 14.63% 22.30%

## NOTE: iter = the number of iterations to run. Breiman says 4 to 6 iterations
## is usually good enough. With this dataset, when we set iter=6, OOB-error
## bounces around between 17% and 18%. When we set iter=20,
# set.seed(42)
# data.imputed <- rfImpute(hd ~ ., data = data, iter=20)
## we get values a little better and a little worse, so doing more
## iterations doesn't improve the situation.
##
## NOTE: If you really want to micromanage how rfImpute(),
## you can change the number of trees it makes (the default is 300) and the
## number of variables that it will consider at each step.

## Now we are ready to build a random forest.

## NOTE: If the thing we're trying to predict (in this case it is
## whether or not someone has heart disease) is a continuous number
## (i.e. "weight" or "height"), then by default, randomForest() will set
## "mtry", the number of variables to consider at each step,
## to the total number of variables divided by 3 (rounded down), or to 1
## (if the division results in a value less than 1).
## If the thing we're trying to predict is a "factor" (i.e. either "yes/no"
## or "ranked"), then randomForest() will set mtry to
## the square root of the number of variables (rounded down to the next
## integer value).

## In this example, "hd", the thing we are trying to predict, is a factor and
## there are 13 variables. So by default, randomForest() will set
## mtry = sqrt(13) = 3.6 rounded down = 3
## Also, by default random forest generates 500 trees (NOTE: rfImpute() only
## generates 300 trees by default)
model <- randomForest(hd ~ ., data=data.imputed, proximity=TRUE)

## RandomForest returns all kinds of things
model # gives us an overview of the call, along with...

##
## Call:
## randomForest(formula = hd ~ ., data = data.imputed, proximity = TRUE)
```

```
##                               Type of random forest: classification
##                               Number of trees: 500
## No. of variables tried at each split: 3
##
## OOB estimate of error rate: 16.83%
## Confusion matrix:
##      Healthy Unhealthy class.error
## Healthy    142      22  0.1341463
## Unhealthy   29     110  0.2086331
```

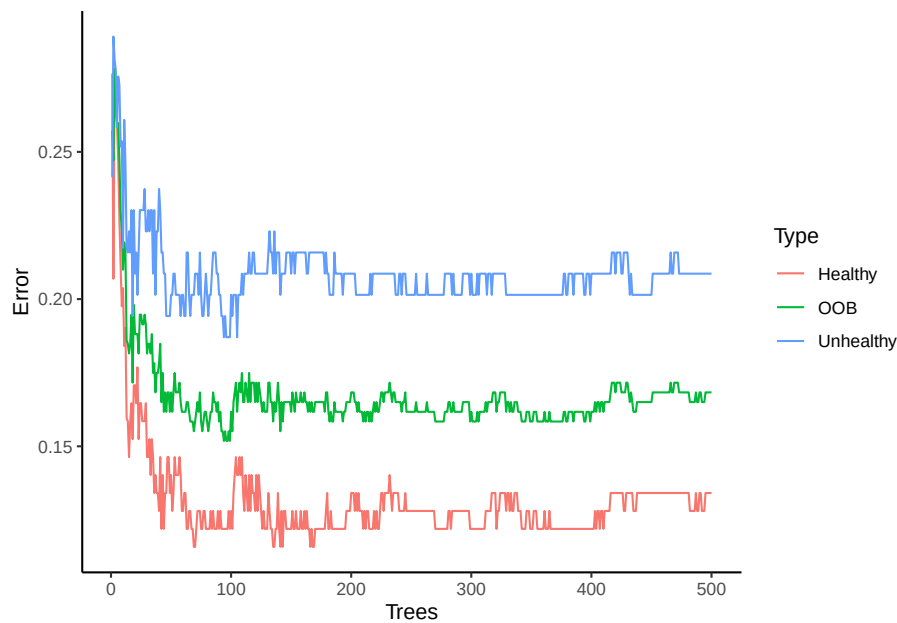
```
# 1) The OOB error rate for the forest with ntree trees.
# In this case ntree=500 by default
# 2) The confusion matrix for the forest with ntree trees.
# The confusion matrix is laid out like this:
```

	Healthy	Unhealthy
Healthy	142	22
Unhealthy	29	110

Healthy | Number of healthy people | Number of healthy people | | correctly called “healthy” | incorrectly called “unhealthy” | | by the forest. | by the forest | ————— Unhealthy | Number of unhealthy people | Number of unhealthy people | | incorrectly called | correctly called “unhealthy” | | “healthy” by the forest | by the forest | —————

```
## Now check to see if the random forest is actually big enough...
## Up to a point, the more trees in the forest, the better. You can tell when
## you've made enough when the OOB no longer improves.
oob.error.data <- data.frame(
  Trees=rep(1:nrow(model$err.rate), times=3),
  Type=rep(c("OOB", "Healthy", "Unhealthy"), each=nrow(model$err.rate)),
  Error=c(model$err.rate[, "OOB"],
    model$err.rate[, "Healthy"],
    model$err.rate[, "Unhealthy"]))

ggplot(data=oob.error.data, aes(x=Trees, y=Error)) +
  geom_line(aes(color=Type))
```



```
# ggsave("oob_error_rate_500_trees.pdf")

## Blue line = The error rate specifically for calling "Unheathly" patients that
## are OOB.
##
## Green line = The overall OOB error rate.
##
## Red line = The error rate specifically for calling "Healthy" patients
## that are OOB.

## NOTE: After building a random forest with 500 tress, the graph does not make
## it clear that the OOB-error has settled on a value or, if we added more
## trees, it would continue to decrease.
## So we do the whole thing again, but this time add more trees.

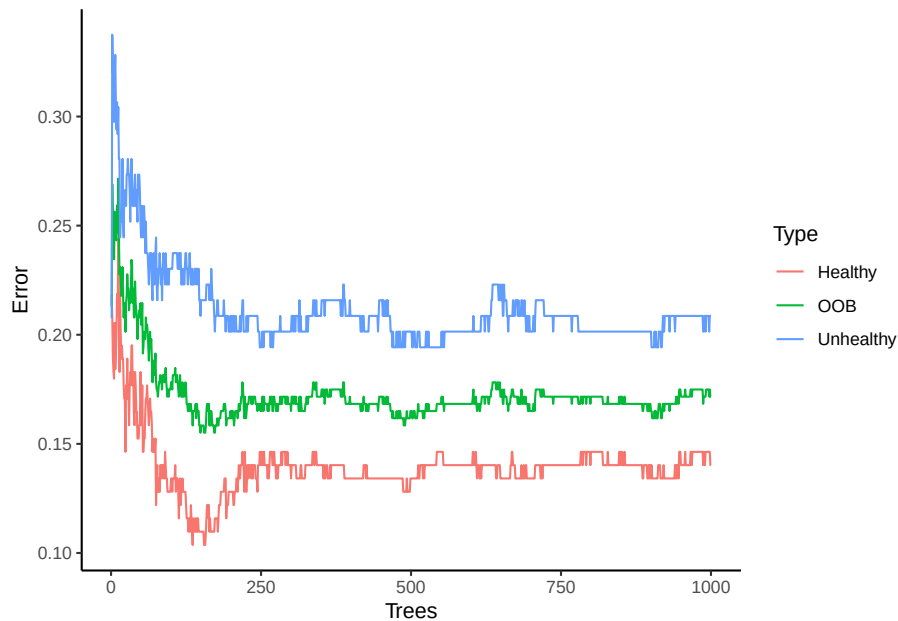
model <- randomForest(hd ~ ., data=data.imputed, ntree=1000, proximity=TRUE)
model

##
## Call:
## randomForest(formula = hd ~ ., data = data.imputed, ntree = 1000,      proximity = TRUE)
##              Type of random forest: classification
##              Number of trees: 1000
## No. of variables tried at each split: 3
##
```

```
##          OOB estimate of error rate: 17.16%
## Confusion matrix:
##          Healthy Unhealthy class.error
## Healthy      141        23  0.1402439
## Unhealthy     29        110  0.2086331

oob.error.data <- data.frame(
  Trees=rep(1:nrow(model$err.rate), times=3),
  Type=rep(c("OOB", "Healthy", "Unhealthy"), each=nrow(model$err.rate)),
  Error=c(model$err.rate[, "OOB"],
    model$err.rate[, "Healthy"],
    model$err.rate[, "Unhealthy"]))

ggplot(data=oob.error.data, aes(x=Trees, y=Error)) +
  geom_line(aes(color=Type))
```



```
# ggsave("oob_error_rate_1000_trees.pdf")

## After building a random forest with 1,000 trees, we get the same OOB-error
## 16.5% and we can see convergence in the graph. So we could have gotten
## away with only 500 trees, but we wouldn't have been sure that number
## was enough.

## If we want to compare this random forest to others with different values for
## mtry (to control how many variables are considered at each step)...
```

```

oob.values <- vector(length=10)
for(i in 1:10) {
  temp.model <- randomForest(hd ~ ., data=data.imputed, mtry=i, ntree=1000)
  oob.values[i] <- temp.model$err.rate[nrow(temp.model$err.rate),1]
}
oob.values

## [1] 0.1749175 0.1782178 0.1716172 0.1782178 0.1683168 0.1881188 0.1815182
## [8] 0.1980198 0.1782178 0.1848185

## find the minimum error
min(oob.values)

## [1] 0.1683168

## find the optimal value for mtry...
which(oob.values == min(oob.values))

## [1] 5

## create a model for proximities using the best value for mtry
model <- randomForest(hd ~ .,
                      data=data.imputed,
                      ntree=1000,
                      proximity=TRUE,
                      mtry=which(oob.values == min(oob.values)))

## Now let's create an MDS-plot to show how the samples are related to each
## other.
##
## Start by converting the proximity matrix into a distance matrix.
distance.matrix <- as.dist(1-model$proximity)

mds.stuff <- cmdscale(distance.matrix, eig=TRUE, x.ret=TRUE)

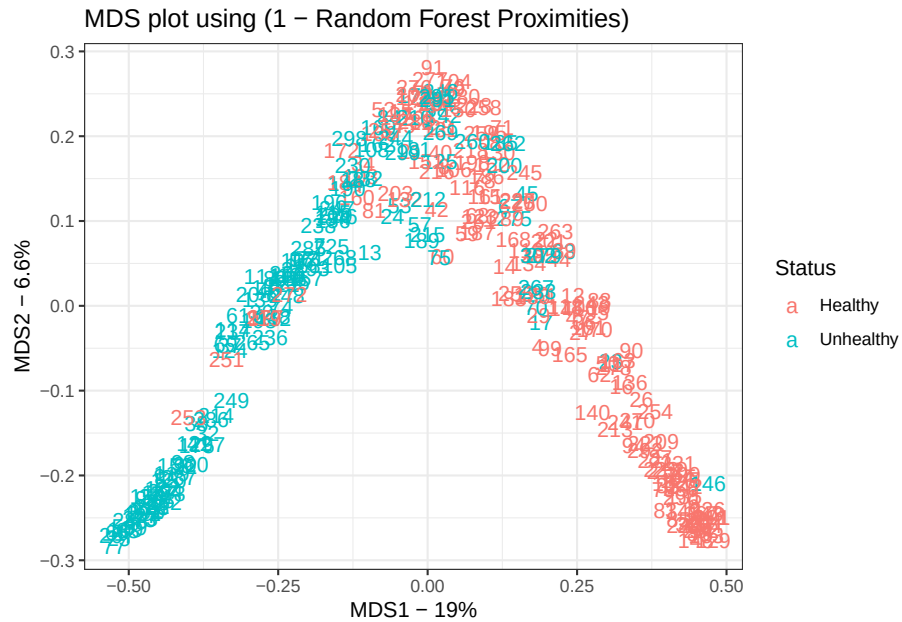
## calculate the percentage of variation that each MDS axis accounts for...
mds.var.per <- round(mds.stuff$eig/sum(mds.stuff$eig)*100, 1)

## now make a fancy looking plot that shows the MDS axes and the variation:
mds.values <- mds.stuff$points
mds.data <- data.frame(Sample=rownames(mds.values),
                      X=mds.values[,1],
                      Y=mds.values[,2],
                      Status=data.imputed$hd)

ggplot(data=mds.data, aes(x=X, y=Y, label=Sample)) +
  geom_text(aes(color=Status)) +
  theme_bw() +

```

```
xlab(paste("MDS1 - ", mds.var.per[1], "%", sep="")) +
ylab(paste("MDS2 - ", mds.var.per[2], "%", sep="")) +
ggtitle("MDS plot using (1 - Random Forest Proximities)")
```



```
# ggsave(file="random_forest_mds_plot.pdf")
```

Chapter 13

Bayesian Additive Regression Trees

- also known as BART (Sparapani et al., 2021)
- is related to Bagging and Random Forests, and Boosting because
 - Each tree is constructed in a random manner (similar to Bagging and Random Forests)
 - Each tree tries to capture signal not yet accounted for by the current model (similar to Boosting)

Let K be the number of regression trees, B the number of (MCMC) iterations for which the BART algorithm will be run

Let $\hat{f}_k^b(x)$ represent the prediction at x for the k -th regression tree used in the b -th iteration.

Then, the K trees from that iteration will be summed as

$$\hat{f}^b(x) = \sum_{k=1}^K \hat{f}_k^b(x), b = 1, \dots, B$$

In the first iteration, all trees are initialized to have a single root node with $\hat{f}_k^1(x) = (1/nK) \sum_{i=1}^n y_i$, which is the mean of the response values divided by the total number of trees.

Thus,

$$\hat{f}^1(x) = \sum_{k=1}^K \hat{f}_k^1(x) = \frac{1}{n} \sum_{i=1}^n y_i \text{ (mean response)}$$

In each later iteration, BART updates each of the K trees, one at a time

In the b -th iteration, to update the k -th tree, we subtract from each response value the predictions from all but the k -th tree to obtain **a partial residual**:

$$r_i = y_i - \sum_{k' < k} \hat{f}_{k'}^b(x_i) - \sum_{k' > k} f_{k'}^{b-1}(x_i), i = 1, \dots, n$$

Instead of fitting a new tree to this partial residual, BART randomly chooses a perturbation to the tree in the previous iteration ($\hat{f}_k^{b-1}$) from a set of possible perturbations (this is where MCMC comes in), favoring ones that improve the fit to the partial residual. There are 2 components to this perturbation:

- change the structure of the tree by adding or pruning branches
- change the prediction in each terminal node of the tree

The output of BART is a collection of prediction models:

$$\hat{f}^b(x) = \sum_{k=1}^K \hat{f}_k^b(x), b = 1, \dots, B$$

where we don't include the results from the iterations in the MCMC burn-in period (say L burn-in samples):

$$\hat{f}^b(x) = \frac{1}{B-L} \sum_{b=L+1}^B \hat{f}_k^b(x)$$

Since this method is under the Bayesian paradigm, we can use the posterior samples to get distributional estimates of the predictions

BART algorithm (James et al., 2013) Alg 8.3

1. Let $\hat{f}_1^1(x) = f_2^1(x) = \dots = \hat{f}_K^1(x) = \frac{1}{nK} \sum_{i=1}^n y_i$
2. Compute $\hat{f}^1(x) = \sum_{k=1}^K \hat{f}_k^1(x) = \frac{1}{n} \sum_{i=1}^n y_i$
3. For $b = 2, \dots, B$
 1. For $k = 1, \dots, K$
 1. For $i = 1, \dots, n$, compute the current partial residual (r_i
 2. Fit a new tree $\hat{f}_k^b(x)$ to r_i by randomly perturbing the k -th tree from the previous iteration, $\hat{f}_k^{b-1}(x)$ (favor perturbations that improve the fit)
2. Compute $\hat{f}^b(x) = \sum_{k=1}^K \hat{f}_k^b(x)$

4. Compute the mean after L burn-in samples $\hat{f}(x) = \frac{1}{B-L} \sum_{b=L+1}^B \hat{f}^b(x)$

Notes

- Step 3.1.2 prevents overfitting since it limits how "hard we fit the data in each iteration
- The individual trees are quite small. we limit the tree size to avoid overfitting the data (especially when we grew very large trees)
- Each time we randomly perturb a tree to fit the residuals, we are drawing a new tree from a posterior distribution
- When we use BART, we have to specify
 - the number trees K
 - the number of interactions B
 - the number of burn-in interactions L
- We typically choose large values for $B = 1000$ and $K = 200$, and moderate value for $L = 100$
- BART is a good out-of-box tool (i.e., minimal tuning)

Example by (James et al., 2013, Ch 8.3)

```
library(BART)
library(ISLR2)
library(tree)

set.seed(1)
attach(Boston)

train <- sample(1:nrow(Boston), nrow(Boston)/2)
# tree.boston <- tree(medu ~ ., Boston, subset = train)

x <- Boston[,1:12]
y <- Boston[, "medv"]
xtrain <- x[train,]
ytrain <- y[train]
xtest <- x[-train,]
ytest <- y[-train]

bartfit <- gbart(xtrain, ytrain, x.test = xtest)

## *****Calling gbart: type=1
## *****Data:
## data:n,p,np: 253, 12, 253
## y1,yn: 0.213439, -5.486561
## x1,x[n*p]: 0.109590, 210.970000
```

```

## xp1,xp[np*p]: 0.027310, 396.900000
## *****Number of Trees: 200
## *****Number of Cut Points: 100 ... 100
## *****burn,nd,thin: 100,1000,1
## *****Prior:beta,alpha,tau,nu,lambda,offset: 2,0.95,0.795495,3,4.42346,21.7866
## *****sigma: 4.765364
## *****w (weights): 1.000000 ... 1.000000
## *****Dirichlet:sparse,theta,omega,a,b,rho,augment: 0,0,1,0.5,1,12,0
## *****printevery: 100
##
## MCMC
## done 0 (out of 1100)
## done 100 (out of 1100)
## done 200 (out of 1100)
## done 300 (out of 1100)
## done 400 (out of 1100)
## done 500 (out of 1100)
## done 600 (out of 1100)
## done 700 (out of 1100)
## done 800 (out of 1100)
## done 900 (out of 1100)
## done 1000 (out of 1100)
## time: 3s
## trcnt,tecnt: 1000,1000

# test error
yhat.bart <- bartfit$yhat.test.mean
mean((ytest - yhat.bart)^2)

## [1] 21.95357

# how many times each variable appeared in the collection of trees
ord <- order(bartfit$varcount.mean, decreasing = T)
bartfit$varcount.mean[ord]

##      rm      nox      tax      rad      chas      age      black      indus      dis p r t r a t i o
## 25.037 23.264 21.029 20.280 19.647 19.031 18.760 18.024 17.756 17.609
##      zn      crim
## 14.677 11.516

```

Example by the package's authors Sparapani et al. (2021)

y = mdev = the median value of owner-occupied homes

x1 = rm = the average number of rooms

x2 = lstat = the percent of the population that is lower status

```

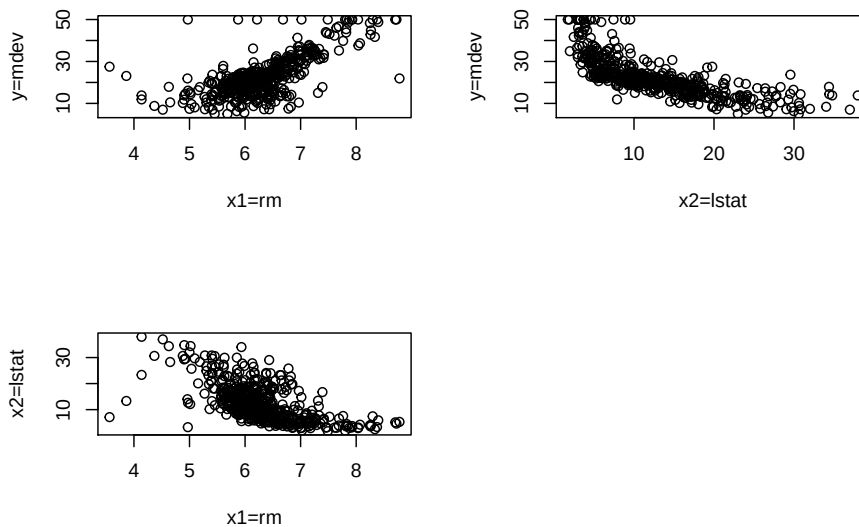
library(MASS)
x <- Boston[,c(6,13)]

```

```

y <- Boston$medv
# head(cbind(x,y))
par(mfrow = c(2,2))
plot(x[, 1], y, xlab = "x1=rm", ylab = "y=medv")
plot(x[, 2], y, xlab = "x2=lstat", ylab = "y=medv")
plot(x[, 1], x[, 2], xlab = "x1=rm", ylab = "x2=lstat")
par(mfrow = c(1, 1))

```



```

library("BART")
set.seed(99)
nd <- 200 # number of posterior draws
burn <- 50
# use wbart for continuous variables
post <- wbart(x, y, nskip = burn, ndpost = nd)

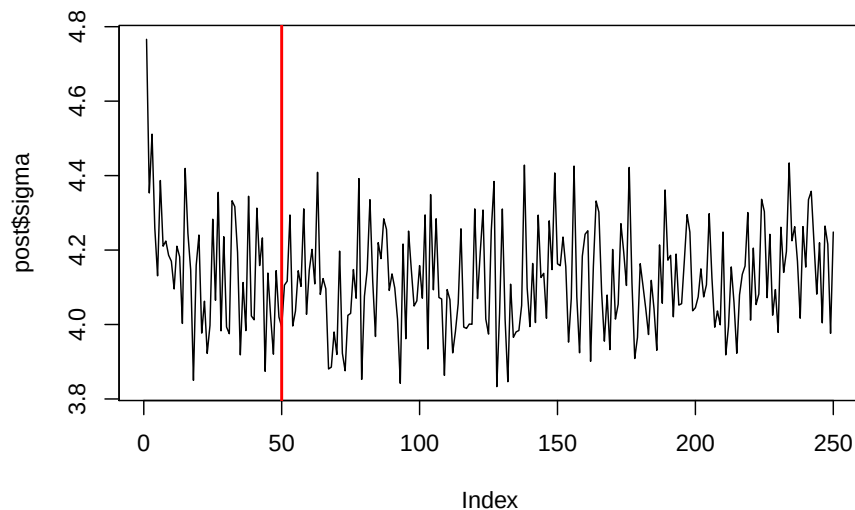
## *****Into main of wbart
## *****Data:
## data:n,p,np: 506, 2, 0
## y1,yn: 1.467194, -10.632806
## x1,x[n*p]: 6.575000, 7.880000
## *****Number of Trees: 200
## *****Number of Cut Points: 100 ... 100
## *****burn and ndpost: 50, 200
## *****Prior:beta,alpha,tau,nu,lambda: 2.000000,0.950000,0.795495,3.000000,5.979017
## *****sigma: 5.540257

```

```
## *****w (weights): 1.000000 ... 1.000000
## *****Dirichlet:sparse,theta,omega,a,b,rho,augment: 0,0,1,0.5,1,2,0
## *****nkeeptrain,nkeeptest,nkeepstme,nkeepreedraws: 200,200,200,200
## *****printevery: 100
## *****skiptr,skipte,skipteme,skiptreedraws: 1,1,1,1
##
## MCMC
## done 0 (out of 250)
## done 100 (out of 250)
## done 200 (out of 250)
## time: 1s
## check counts
## trcnt,tecnt,temecnt,treedrawscnt: 200,0,0,200
```

Convergence assessment

```
plot(post$sigma, type = "l")
abline(v = burn, lwd = 2, col = "red")
```

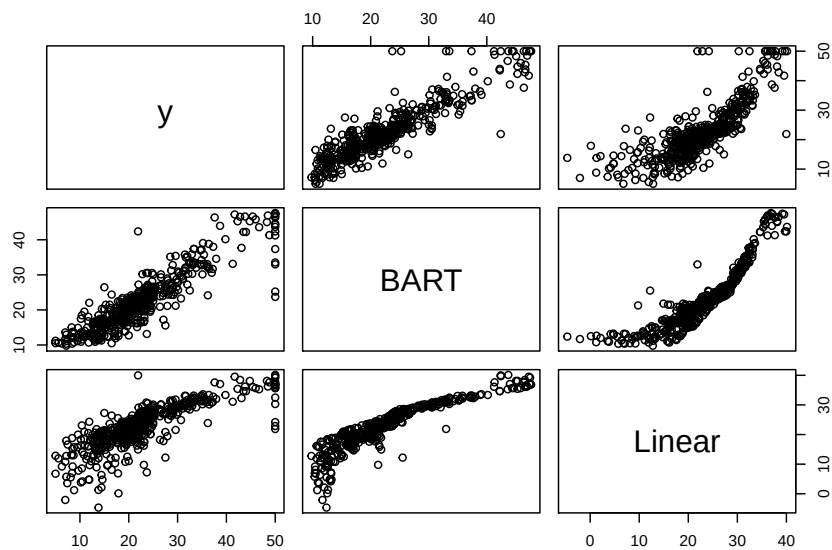


good convergence

Prediction comparison with lm

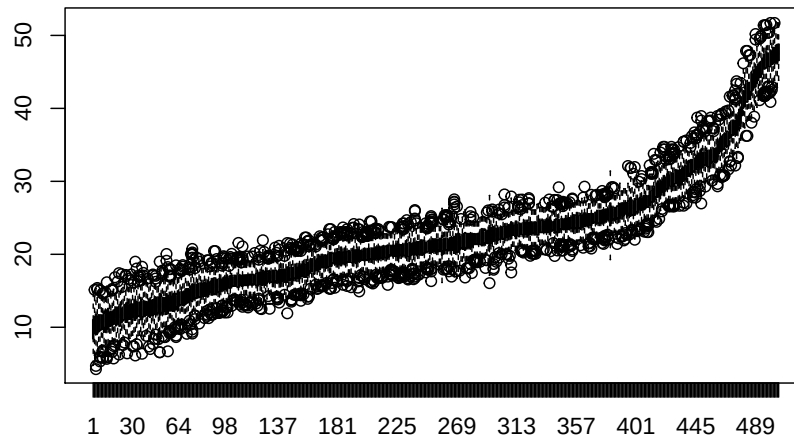
```
lmf <- lm(y~., data.frame(x, y))
fitmat <- cbind(y, post$yhat.train.mean, lmf$fitted.values)
colnames(fitmat) <- c("y", "BART", "Linear")
cor(fitmat)
```

```
##           y      BART   Linear
## y      1.0000000 0.9051200 0.7991005
## BART   0.9051200 1.0000000 0.8978003
## Linear 0.7991005 0.8978003 1.0000000
pairs(fitmat)
```



Prediction

```
i <- order(post$yhat.train.mean)
boxplot(post$yhat.train[, i])
```



```

n <- length(y)
set.seed(14)
i <- sample(1:n, floor(0.75 * n))
x.train <- x[i, ]; y.train = y[i]
x.test <- x[-i, ]; y.test = y[-i]
cat("training sample size = ", length(y.train), "\n")

## training sample size = 379
cat("testing sample size = ", length(y.test), "\n")

## testing sample size = 127

set.seed(99)
post1 <- wbart(x.train, y.train, x.test)

## *****Into main of wbart
## *****Data:
## data:n,p,np: 379, 2, 127
## y1,yn: 14.101319, -7.098681
## x1,x[n*p]: 7.206000, 12.120000
## xp1,xp[np*p]: 6.421000, 23.970000
## *****Number of Trees: 200
## *****Number of Cut Points: 100 ... 100
## *****burn and ndpost: 100, 1000
## *****Prior:beta,alpha,tau,nu,lambda: 2.000000,0.950000,0.795495,3.000000,6.258099
## *****sigma: 5.668084

```

```

## *****w (weights): 1.000000 ... 1.000000
## *****Dirichlet:sparse,theta,omega,a,b,rho,augment: 0,0,1,0.5,1,2,0
## *****nkeeptrain,nkeeptest,nkeeptestme,nkeepreedraws: 1000,1000,1000,1000
## *****printevery: 100
## *****skiptr,skipte,skipteme,skiptreedraws: 1,1,1,1
##
## MCMC
## done 0 (out of 1100)
## done 100 (out of 1100)
## done 200 (out of 1100)
## done 300 (out of 1100)
## done 400 (out of 1100)
## done 500 (out of 1100)
## done 600 (out of 1100)
## done 700 (out of 1100)
## done 800 (out of 1100)
## done 900 (out of 1100)
## done 1000 (out of 1100)
## time: 3s
## check counts
## trcnt,tecnt,temecnt,treedrawscnt: 1000,1000,1000,1000
length(post1$yhat.test.mean)

## [1] 127
Thinning (to speed things up)
set.seed(4)
post3 <-
  wbart(
    x.train,
    y.train,
    nskip = 1000,
    ndpost = 10000,
    nkeeptrain = 0,
    nkeeptest = 0,
    nkeeptestmean = 0,
    nkeepreedraws = 200
  )

## *****Into main of wbart
## *****Data:
## data:n,p,np: 379, 2, 0
## y1,yn: 14.101319, -7.098681
## x1,x[n*p]: 7.206000, 12.120000
## *****Number of Trees: 200
## *****Number of Cut Points: 100 ... 100

```

```
## *****burn and ndpost: 1000, 10000
## *****Prior:beta,alpha,tau,nu,lambda: 2.000000,0.950000,0.795495,3.000000,6.258099
## *****sigma: 5.668084
## *****w (weights): 1.000000 ... 1.000000
## *****Dirichlet:sparse,theta,omega,a,b,rho,augment: 0,0,1,0.5,1,2,0
## *****nkeeptrain,nkeeptest,nkeepstme,nkeepreedraws: 0,0,0,200
## *****printevery: 100
## *****skiptr,skipte,skipteme,skiptreedraws: 10001,10001,10001,50
##
## MCMC
## done 0 (out of 11000)
## done 100 (out of 11000)
## done 200 (out of 11000)
## done 300 (out of 11000)
## done 400 (out of 11000)
## done 500 (out of 11000)
## done 600 (out of 11000)
## done 700 (out of 11000)
## done 800 (out of 11000)
## done 900 (out of 11000)
## done 1000 (out of 11000)
## done 1100 (out of 11000)
## done 1200 (out of 11000)
## done 1300 (out of 11000)
## done 1400 (out of 11000)
## done 1500 (out of 11000)
## done 1600 (out of 11000)
## done 1700 (out of 11000)
## done 1800 (out of 11000)
## done 1900 (out of 11000)
## done 2000 (out of 11000)
## done 2100 (out of 11000)
## done 2200 (out of 11000)
## done 2300 (out of 11000)
## done 2400 (out of 11000)
## done 2500 (out of 11000)
## done 2600 (out of 11000)
## done 2700 (out of 11000)
## done 2800 (out of 11000)
## done 2900 (out of 11000)
## done 3000 (out of 11000)
## done 3100 (out of 11000)
## done 3200 (out of 11000)
## done 3300 (out of 11000)
## done 3400 (out of 11000)
## done 3500 (out of 11000)
```



```
## done 3600 (out of 11000)
## done 3700 (out of 11000)
## done 3800 (out of 11000)
## done 3900 (out of 11000)
## done 4000 (out of 11000)
## done 4100 (out of 11000)
## done 4200 (out of 11000)
## done 4300 (out of 11000)
## done 4400 (out of 11000)
## done 4500 (out of 11000)
## done 4600 (out of 11000)
## done 4700 (out of 11000)
## done 4800 (out of 11000)
## done 4900 (out of 11000)
## done 5000 (out of 11000)
## done 5100 (out of 11000)
## done 5200 (out of 11000)
## done 5300 (out of 11000)
## done 5400 (out of 11000)
## done 5500 (out of 11000)
## done 5600 (out of 11000)
## done 5700 (out of 11000)
## done 5800 (out of 11000)
## done 5900 (out of 11000)
## done 6000 (out of 11000)
## done 6100 (out of 11000)
## done 6200 (out of 11000)
## done 6300 (out of 11000)
## done 6400 (out of 11000)
## done 6500 (out of 11000)
## done 6600 (out of 11000)
## done 6700 (out of 11000)
## done 6800 (out of 11000)
## done 6900 (out of 11000)
## done 7000 (out of 11000)
## done 7100 (out of 11000)
## done 7200 (out of 11000)
## done 7300 (out of 11000)
## done 7400 (out of 11000)
## done 7500 (out of 11000)
## done 7600 (out of 11000)
## done 7700 (out of 11000)
## done 7800 (out of 11000)
## done 7900 (out of 11000)
## done 8000 (out of 11000)
## done 8100 (out of 11000)
```

```

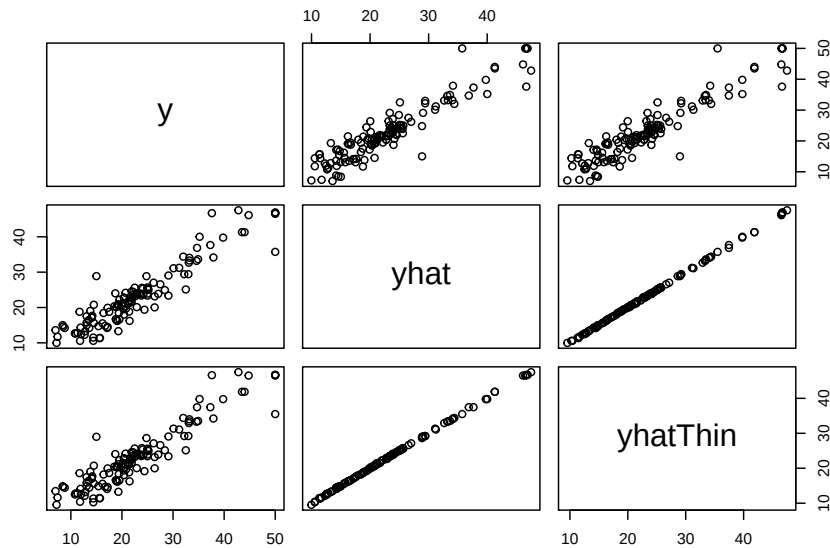
## done 8200 (out of 11000)
## done 8300 (out of 11000)
## done 8400 (out of 11000)
## done 8500 (out of 11000)
## done 8600 (out of 11000)
## done 8700 (out of 11000)
## done 8800 (out of 11000)
## done 8900 (out of 11000)
## done 9000 (out of 11000)
## done 9100 (out of 11000)
## done 9200 (out of 11000)
## done 9300 (out of 11000)
## done 9400 (out of 11000)
## done 9500 (out of 11000)
## done 9600 (out of 11000)
## done 9700 (out of 11000)
## done 9800 (out of 11000)
## done 9900 (out of 11000)
## done 10000 (out of 11000)
## done 10100 (out of 11000)
## done 10200 (out of 11000)
## done 10300 (out of 11000)
## done 10400 (out of 11000)
## done 10500 (out of 11000)
## done 10600 (out of 11000)
## done 10700 (out of 11000)
## done 10800 (out of 11000)
## done 10900 (out of 11000)
## time: 23s
## check counts
## trcnt,tecnt,temecnt,treedrawscnt: 0,0,0,200

yhatthin <- predict(post3, x.test)

## *****In main of C++ for bart prediction
## tc (threadcount): 1
## number of bart draws: 200
## number of trees in bart sum: 200
## number of x columns: 2
## from x,np,p: 2, 127
## ***using serial code

fmat <- cbind(y.test, post1$yhat.test.mean, apply(yhatthin, 2, mean))
colnames(fmat) <- c("y", "yhat", "yhatThin")
pairs(fmat)

```



Friedman's partial dependence function to see variable importance

```
x.train <- as.matrix(Boston[i,-14])
set.seed(12)
post4 <- wbart(x.train, y.train)

## *****Into main of wbart
## *****Data:
## data:n,p,np: 379, 13, 0
## y1,yn: 14.101319, -7.098681
## x1,x[n*p]: 0.550070, 12.120000
## *****Number of Trees: 200
## *****Number of Cut Points: 100 ... 100
## *****burn and ndpost: 100, 1000
## *****Prior:beta,alpha,tau,nu,lambda: 2.000000,0.950000,0.795495,3.000000,4.636409
## *****sigma: 4.878720
## *****w (weights): 1.000000 ... 1.000000
## *****Dirichlet:sparse,theta,omega,a,b,rho,augment: 0,0,1,0.5,1,13,0
## *****nkeeptrain,nkeepertest,nkeepertestme,nkeepreedraws: 1000,1000,1000,1000
## *****printevery: 100
## *****skiptr,skipte,skipteme,skiptreedraws: 1,1,1,1
##
## MCMC
## done 0 (out of 1100)
## done 100 (out of 1100)
## done 200 (out of 1100)
```

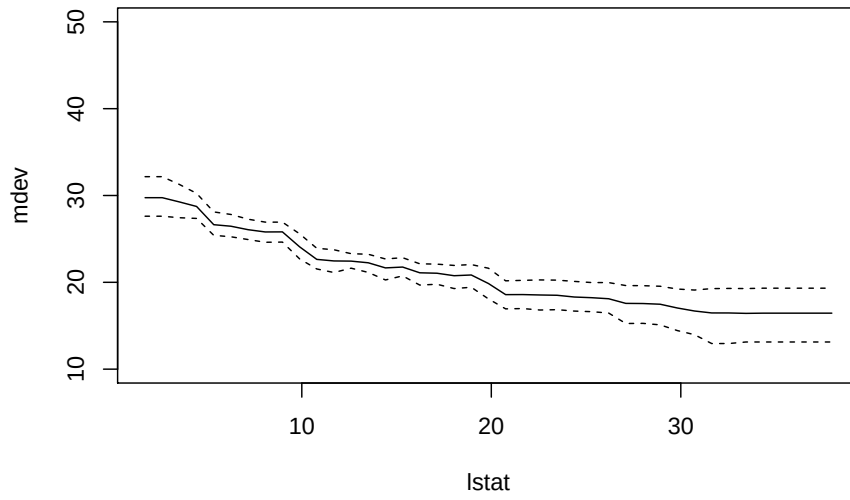
```
## done 300 (out of 1100)
## done 400 (out of 1100)
## done 500 (out of 1100)
## done 600 (out of 1100)
## done 700 (out of 1100)
## done 800 (out of 1100)
## done 900 (out of 1100)
## done 1000 (out of 1100)
## time: 3s
## check counts
## trcnt,tecmt,temecnt,treedrawscnt: 1000,0,0,1000

H <- length(y.train)
L <- 41
x <- seq(min(x.train[, 13]), max(x.train[, 13]), length.out = L)
x.test <- cbind(x.train[, -13], x[1])
for (j in 2:L)
  x.test <- rbind(x.test, cbind(x.train[, -13], x[j]))
pred <- predict(post4, x.test)
```

```
## *****In main of C++ for bart prediction
## tc (threadcount): 1
## number of bart draws: 1000
## number of trees in bart sum: 200
## number of x columns: 13
## from x,np,p: 13, 15539
## ***using serial code

partial <- matrix(nrow = 1000, ncol = L)
for (j in 1:L) {
  h <- (j - 1) * H + 1:H
  partial[, j] <- apply(pred[, h], 1, mean)
}
plot(
  x,
  apply(partial, 2, mean),
  type = "l",
  ylim = c(10, 50),
  xlab = "lstat",
  ylab = "mdev"
)

lines(x, apply(partial, 2, quantile, probs = 0.025), lty = 2)
lines(x, apply(partial, 2, quantile, probs = 0.975), lty = 2)
```



For examples on categorical variables, see (Sparapani et al., 2021)

Part VI

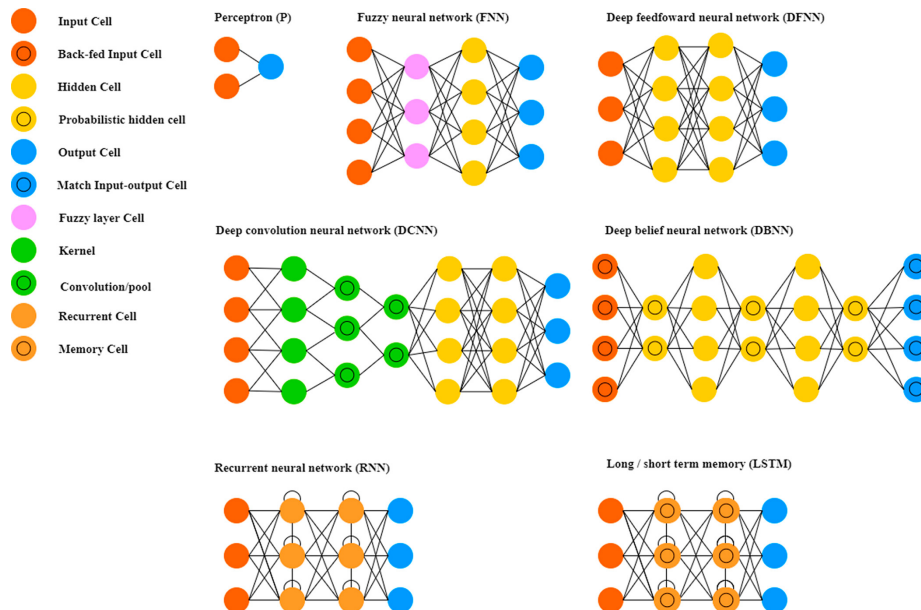
4. NETWORK BASED - DEEP LEARNING

Chapter 14

Neural Networks

Readings:

- (James et al., 2013, Ch 10) with (statlearning.com) and online course
- (Hastie et al., 2001, Ch 11)
- Deep Learning (Goodfellow et al., 2016)
- Deep Learning with R (Chollet, 2018)
- Abdar et al. (2021)
- Samek et al. (2021)



(Jiao et al., 2020, p. 321)

Notes:

- The basic idea behind neural networks (NN) is to find nonlinear functions of linear combinations of inputs to model a response.
- NN include a large class of models. Even the simple version of these models is a form of “universal approximator” (i.e., can represent any function)
- NN is a multi-stage regression or classification model that can be represented by a network diagram
 - In the artificial intelligence community, they want to come up with models that mimic the way neurons and synapses create intelligence in the human brain

Refresher on Non-linear regression

A model with the form of

$$Y_i = f(\mathbf{x}_i; \boldsymbol{\beta}) + \eta_i, i = 1, \dots, n$$

where

- $f(\mathbf{x}_i; \boldsymbol{\beta})$ is a nonlinear function (known up to parameters, $\boldsymbol{\beta}$) relating to $E(Y_i)$ to the independent variables $\mathbf{x}_i$
- $\mathbf{x}_i$ is a $p \times 1$ vector of independent variables (fixed)
- $\boldsymbol{\beta}$ is a $q \times 1$ vector of parameters
- η_i 's are iid variables mean 0 and variance σ^2

The objective function that we seek to minimize is a loss function

Ex: for continuous data, we can use squared-error loss (e.g., residual sums of squares)

$$Q(f(\mathbf{x}); y; \boldsymbol{\beta}) = \sum_{i=1}^n \{y_i - f(\mathbf{x}_i; \boldsymbol{\beta})\}^2$$

and

$$\hat{\boldsymbol{\beta}} = \operatorname{argmin}(Q(f(\mathbf{x}); y; \boldsymbol{\beta}))$$

Estimation

To minimize numerically based on a Taylor expansion and iterative algorithm of the form

$$\gamma^{(j+1)} = \gamma^{(j)} - \epsilon_j \mathbf{A}_j \frac{\partial Q(\hat{\theta}^{(j)})}{\partial}$$

where

- $\mathbf{A}_j$ = some positive definite matrix
- $\frac{\partial Q(\hat{\theta}^{(j)})}{\partial}$ (i.e., $\nabla_{\theta} Q(\hat{\theta}^{(j)})$) is the gradient of the objective function $Q(\cdot)$ at $\hat{\theta}$
- ϵ_j = learning rate

Estimation via Optimization (this matrix controls the gradient - control all parameters at the same time and stop the model from over shooting, but it is very expensive to calculate; hence, in modern machine learning we usually just control for the learning rate)

- Gradient Descent: $\mathbf{A}_j \equiv \mathbf{I}_p$
- Gauss-Newton: $\mathbf{A}_j \equiv [\mathbf{F}(\hat{\theta}^{(j)})' \mathbf{F}(\hat{\theta}^{(j)})]^{-1}$ where $\mathbf{F}(\theta) \equiv \frac{\partial f(\mathbf{x}_i; \cdot)}{\partial \theta_j}$ ($n \times p$ Jacobian matrix)
- Levenberg-Marquardt: $\mathbf{A}_j \equiv [\mathbf{F}(\hat{\theta}^{(j)})' \mathbf{F}(\hat{\theta}^{(j)}) + \tau \mathbf{I}]^{-1}$ (Ridge penalty: L_2 regularization)
- Newton-Raphson: $\mathbf{A}_j \equiv [\frac{\partial^2 Q(\hat{\theta}^{(j)})}{\partial \theta \partial \theta'}]^{-1}$ ($p \times p$ Hessian matrix)

In general, optimization algorithms use an objective function (e.g., loss function) that consists of a sum over the training data

In “big data” problems (large n) and **computation of gradient** can be taxing (i.e., a large number of floating point operations that must be used for gradient calculation). Plus, we can also have problem with **redundant data** in the training sample.

14.1 Stochastic Gradient Descent

- It’s an optimization technique denoted as SGD
- SGD can fix the “big data” problem by considering minimizing the expected loss, which requires estimates of the expected loss.
 - This can be done with averages of small samples (min-batches) of the training data (assuming the input $\mathbf{x}_i$ come from the same underlying distribution)
- Considering the expectation of the gradient by randomly sampling a small number of samples from the training set (without replacement) and taking the average turns out to be a very effective solution to this problem. Thus, the method, stochastic gradient descent is “stochastic” because it is based

on samples and expectation, but it can also be used in an *online scenario* where data are received **sequentially**.

- Smaller batch sizes offer more regularization and more flexibility (generalization) outside the training sample

Basic Stochastic Gradient Descent (SGD) Algorithm (Goodfellow et al. 2016, Ch 18)

- **Require:** Learning rate ϵ
- **Require:** Initial parameters $^{(j)}$
 - **while** Stopping criterion not met **do**
 - * Sample a mini-batch of m inputs $\{\mathbf{x}_1, \dots, \mathbf{x}_m\}$ and targets $\{\mathbf{y}_1, \dots, \mathbf{y}_m\}$
 - * Set $\mathbf{g} = 0$
 - * **for** $i = 1$ to m **do**
 - Compute gradient estimate: $\mathbf{g} \leftarrow \mathbf{g} + \frac{1}{m} \nabla_{\theta} Q(f(\mathbf{x}_i; \theta), \mathbf{y}_i)$
 - * **end for**
 - * Apply update: $^{(j+1)} \leftarrow ^{(j)} - \epsilon \mathbf{g}$
 - **end while**

Note: one cycle of several min-batches that correspond to all training observations is called an epoch. Hence, the number of epochs is the number of complete passes through the training data.

Using previous update (direction and magnitude) to make the new update a convex combination of the previous update and the current gradient, it gives a more stable update, which is known as momentum

Basic Stochastic Gradient Descent (SGD) with Momentum Algorithm (Goodfellow et al. 2016, Ch 8)

- **Require:** Learning rate ϵ , momentum parameter ν
- **Require:** Initial parameters $^{(j)}$ and “velocity” $\mathbf{v}$
 - **while** Stopping criterion not met **do**
 - * Sample a mini-batch of m inputs $\{\mathbf{x}_1, \dots, \mathbf{x}_m\}$ and targets $\{\mathbf{y}_1, \dots, \mathbf{y}_m\}$
 - * Set $\mathbf{g} = 0$
 - * **for** $i = 1$ to m **do**
 - Compute gradient estimate: $\mathbf{g} \leftarrow \mathbf{g} + \frac{1}{m} \nabla_{\theta} Q(f(\mathbf{x}_i; \theta), \mathbf{y}_i)$
 - * **end for**

- * Compute velocity update: $\mathbf{v} \leftarrow \nu \mathbf{v} - \epsilon \mathbf{g}$
- * Apply update: $\mathbf{w}^{(j+1)} \leftarrow \mathbf{w}^{(j)} - \mathbf{v}$
- end while

The learning rate parameter is the most important tuning parameter in SGD.

Ways to adapt the learning rate (e.g., Goodfellow et al. 2016, Chp. 8):

- AdaGrad: adapts the learning rates for each parameter by scaling them inversely proportional to the sum of all historical squared
- RMSProp: modifies AdaGrad by changing the squared gradient accumulation into an exponentially weighted moving average of squared gradients (2nd moments)
- Adam: combination of RMSProp and momentum that updates both first and second-order moments with exponential weighting and performs a bias correction.

Challenges and modifications (Gradient Descent Algorithms):

- Convergence (failure to converge: convergence to local minima)
- Starting values
- Regularization if there are many parameters (shrink $\hat{\theta}$ towards 0)
 - L_2 penalty (Ridge regression): add $\lambda \sum_{j=1}^q \theta_j^2$ to the objective function, $Q(\theta)$
 - L_1 penalty (Lasso regression): add $\lambda \sum_{j=1}^q |\theta_j|$ to objective function, $Q(\theta)$
 - Others
- Recall a neural network is a multi-stage nonlinear regression or classification model that can be represented by a network diagram
- Essentially, we get parameter estimates like we would with nonlinear regression (there are just a LOT more to estimate and we call them “weights”). But in this case, we can make it more efficient

14.2 Simple Single Layer Neural Network

- A simple type of neural network is a “single hidden layer feed forward network (FNN)” or “single layer perceptron.”
 - Input $\mathbf{x}$ (p-dimensional)
 - Output $\mathbf{y}$ (k-dimensional) (typically, $k = 1$ in most nonlinear regression and binary classification)

14.2.1 Hidden Layer

$$z_j = g(\alpha_{j0} + \sum_{i=1}^p \alpha_{ji}x_i), j = 1, \dots, J$$

where

- z_j is the hidden variable
- α_{j0} is the “bias” (offset) parameter
- $\{\alpha_{ji}\}$ are the ”weight parameters
- $g(\cdot)$ is the activation function

A classic activation function is the **sigmoid** (logistic) function:

$$g(a) = \frac{1}{1 + \exp(-a)}$$

Other options include:

- Hyperbolic tangent
- Radial basis function
- Rectified linear unit (universal use nowadays)
- Softmax
- Identity

Note:

- The hidden layer is just transformations of the input variables - similar to Basis Function Representations
- These transformations are crucial since the format is very flexible and we “learn” the most useful ones

It simplifies notation to assume $x_0 = 1$ then

$$z_j = g(\sum_{i=1}^p \alpha_{ji}x_i), j = 1, \dots, J$$

Typical activation functions include

- Sigmoid (logistic)

$$\sigma(z) = \frac{1}{1 + \exp(-z)}$$

- Hyperbolic Tangent

$$\tanh(z) = \frac{\exp(z) - \exp(-z)}{\exp(z) + \exp(-z)}$$

- Rectified Linear Unit (universal use nowadays because it enforces sparse activations and yields a simple gradient (e.g., it can be computed and stored more efficiently), the offset parameter in z can shift in the inflection point away from zero)

$$ReLU(z) = \max(0, z)$$

14.2.2 Output Layer

$$y_k = h(\beta_{k0} + \sum_{j=1}^J \beta_{kj} z_j), k = 1, \dots, K$$

where

- y_k is the output variable
- $h(\cdot)$ is another activation function (typically, the identity function for nonlinear regression or a sigmoid or softmax function for classification problems)

As with the Hidden Layer, we can write

$$y_k = h(\sum_{j=0}^J \beta_{kj} z_j), k = 1, \dots, K$$

In summary, the single layer feed-forward NN can be written as (combining Hidden Layer and Output Layer):

$$y_k(\mathbf{x}; \alpha, \beta) = h(\sum_{j=0}^J \beta_{kj} g(\sum_{i=0}^p \alpha_{ji} x_i))$$

Similar to traditional nonlinear regression, to estimate the parameters (“train the network”) we select an objective function $Q(\alpha, \beta)$ (e.g., squared error, cross-entropy) and then seek a gradient-based approach to obtain parameter estimates of the α, β

Hence, NN is just a complicated nonlinear transformation of the inputs

14.2.3 Discrimination Problem

14.2.3.1 Two Class Discrimination

When the output has $(0, 1)$, the standard neural net implementation includes one hidden layer and one output unit with a sigmoidal transformation, $h(\cdot)$:

$$y_l = h(\beta_0 + \sum_{j=1}^J \beta_j z_{jl}) = h(\sum_{j=0}^J \beta_j z_{jl})$$

The hidden layer is the same, but the objective function can be changed to **cross-entropy loss** (the negative binomial log-likelihood), where $\tilde{y}$ is the observed response:

$$Q(\cdot, \cdot, \{\mathbf{x}_l, \tilde{y}_l\}) = - \sum_l \tilde{y}_l + (1 - \tilde{y}_l) \log(1 - y_l)$$

14.2.3.2 Multiclass Discrimination

When $K > 2$ category classification, one has K output functions:

$$o_{kl} = \sum_{j=0}^J \beta_{kj} z_{jl}, k = 1, \dots, K$$

softmax function is used to transform these to the classification output (the same transformation in multinomial logistic regression);

$$y_{kl} = h(o_{kl}) = \frac{\exp(o_{kl})}{\sum_{k'} \exp(o_{k'l})}$$

The objective function is given by cross-entropy (negative multinomial log-likelihood):

$$Q(\cdot) = - \sum_l \sum_k \tilde{y}_{kl} \log(y_{kl})$$

Example by (James et al., 2013, Ch. 10)

```
library(ISLR2)
Gitters <- na.omit(Hitters)
n <- nrow(Gitters)
set.seed(13)
ntest <- trunc(n/3)
testid <- sample(1:n, ntest)
```


Get prediction error of the linear model

```
lfit <- lm(Salary ~ ., data = Gitters[-testid,])
lpred <- predict(lfit, Gitters[testid,])
with(Gitters[testid,], mean(abs(lpred - Salary)))

## [1] 254.6687

x <- scale(model.matrix(Salary ~ . -1, data = Gitters))
y <- Gitters$Salary

library(glmnet)
cvfit <- cv.glmnet(x[-testid,], y[-testid], type.measure = "mae")
cpred <- predict(cvfit, x[testid,], s = "lambda.min")
mean(abs(y[testid] - cpred))

## [1] 252.2994

# library(keras)
# modnn <- keras_model_sequential() %>%
#   layer_dense(units = 50, activation = "relu", input_shape = ncol(x)) %>%
#   layer_dropout(rate = 0.4) %>%
#   layer_dense(units = 1)
```

14.3 Deep Neural Networks

14.3.1 Backpropagation

- used in the Stochastic Gradient Descent to train the parameters
- It applies the chain rule to calculate the gradient (thank to the hierarchical/compositional nature of the model)

$$\frac{\partial Q}{\partial \alpha_{ji}} = \frac{\partial Q}{\partial y_k} \frac{\partial y_k}{\partial z_j} \frac{\partial z_j}{\partial \alpha_{ji}}, j = 1, \dots, J; i = 0, \dots, p$$

$$\frac{\partial Q}{\partial \beta_{kj}} = \frac{\partial Q}{\partial y_k} \frac{\partial y_k}{\partial \beta_{kj}}, k = 1, \dots, K; j = 0, \dots, J$$

Example:

Nonlinear regression with a single output

- $h(\cdot)$ the identity function
- $g(\cdot)$ the sigmoid function
- squared error loss

Let $\{\tilde{y}_{(l)}, \mathbf{x}_{(l)}\}$ correspond to the observed input/output pair for the l -th sample (in the minibatch)

$$y_{(l)} = \sum_{j=1}^J \beta_j z_{j(l)} \quad (14.1)$$

$$z_{j(l)} = g\left(\sum_{i=0}^p \alpha_{ji} x_{i(l)}\right) \quad (14.2)$$

Squared-error objective function

$$Q(\alpha, \beta, \{\mathbf{x}, \tilde{\mathbf{y}}\}) = \frac{1}{2} \sum_l (\tilde{y}_{(l)} - y_{(l)})^2$$

Then, the gradients are

$$\frac{\partial Q}{\partial \beta_{kj}} = \frac{\partial Q}{\partial y_k} \frac{\partial y_k}{\partial \beta_{kj}} = - \sum_l (\tilde{y}_{(l)} - y_{(l)}) z_{j(l)} \quad (14.3)$$

$$\begin{aligned} \frac{\partial Q}{\partial \alpha} &= \sum_l \frac{\partial Q}{\partial y_k} \frac{\partial y_k}{\partial z_j} \frac{\partial z_j}{\partial \alpha_{ji}} \\ &= \sum_l \{-(\tilde{y}_{(l)} - y_{(l)})\} \{\beta_j\} \{z_{j(l)}(1 - z_{j(l)})x_{i(l)}\} \end{aligned} \quad (14.4)$$

The $(\tilde{y}_{(l)} - y_{(l)})$ term in the equations (14.3) and (14.4) are residuals.

The other terms indicate what fraction of the residual gets associated with the parameters in the hidden and output layers

Note: $\partial g(ab)/\partial a = g(a)(1 - g(a))b$ when $g(\cdot)$ is a sigmoid function

Then, basic SGD algorithm updates parameters

$$\beta_j^{(r+1)} = \beta_j^{(r)} - \epsilon^{(r)} \frac{1}{L} \sum_l \frac{\partial Q}{\partial \beta_j} \quad (14.5)$$

$$\alpha_{ji}^{(r+1)} = \alpha_{ji}^{(r)} - \epsilon^{(r)} \frac{1}{L} \sum_l \frac{\partial Q}{\partial \alpha_{ji}} \quad (14.6)$$

where

- $\epsilon^{(r)}$ is the learning rate parameter
- L is the number of samples in the summation

Updates are done in a two-pass algorithm (forward and backward - hence the name backpropagation)

- In the forward pass, the current parameter values are fixed at the r -th values and $z_{j(l)}$ and $y_{(l)}$ are calculated using (14.1) and (14.2)
- In the backward pass, the gradients are then calculated from (14.3) and (14.4) using the r -th updates; then, the parameters are updated using (14.5) and (14.6)

Notes:

- Under backpropagation algorithms, each hidden unit passes and receives info only to and from units that share a connection (i.e., “local” algorithm)
 - which facilitates computation in a parallel computing environment, which is beneficial to process large datasets (this is why deep learning took off).
- We can also add multiple outputs, $y_{k(l)}$, $k = 1, \dots, K$
- Surprisingly, the chain rule gives the same general form for the backpropagation gradient terms for cross-entropy loss (both 2-class and multi-class cases) as for squared error loss.
- In real real, we never calculate derivatives by-hand because now we have automatic differentiation that can symbolically/algorithmically get the derivatives

14.3.2 Deep Feedforward Networks

- also known as Multilayer Perceptrons
- Problems like acoustic processing, image processing, and natural language processing, have complex structure that requires the use of “deep learning” algorithms
- These are neural networks with **many hidden layers**, with outputs from one layer becoming the input of the next.
- The number of units in each layer is called **width**
- The number of layers is the **depth** of the network
- The usual challenge with deep models is that they can be severely over-parameterized, unless we use regularization.

Example of 2-Layer

Model characteristics:

- Two hidden layers
- One output layer for nonlinear regression

$$z_{1j(l)} = g\left(\sum_{i=0}^p \alpha_{1ji} x_{i(l)}\right), j = 1, \dots, J_1$$

$$z_{2j'(l)} = g\left(\sum_{j=1}^{J_1} \alpha_{2j'j} z_{1j(l)}\right), j' = 1, \dots, J_2$$

$$y_{(l)} = \sum_{j'=0}^{J_2} \beta_{j'} z_{2j'(l)}$$

Similar to Backpropagation, as the depth increases the more number of parameters you have to estimate (this is the problem)

It helps to have multiple hidden layers in a deep learning model because the chain of transformations allow the network to learn complex transformation of the input

For example, when applied to images (e.g., in a convolutional neural network), different scales in images are identified at different levels).

14.3.3 Implementation

- Preprocessing of the Data: Two choices:
 - Feature normalization per sample: it can help to center each input $\mathbf{x}_m$ (demean)
 - Global feature standardization (more commonly used): subtract the average from all the elements of each feature element across the entire training set, and divides by the associated standard deviation (we also have to apply this to the test set, but using the means and standard deviation from the training set).
- Model Initialization
 - Because neural models are highly nonlinear, they can be very sensitive to the initial parameter values. But we can follow the following general heuristic “principles”:
 - * Choose parameters such that each hidden variable is operating in the linear range of its activation functions. Because this depends on weights and the inputs, that’s why we standardize the inputs
 - * Choose parameters randomly (since hidden units are interchangeable, we need them to find different features, so it’s important that they don’t have the same weights)

- Exemptions: **pretraining**
- Regularization
 - Because neural networks tend to overfit, we have to use regularization
 - **Weight Decay** (commonly used): which is L_2 (ridge) penalty applied to the objective function of the form (e.g., $\lambda(\sum_{ji} \alpha_{ji}^2 + \sum_{kj} \beta_{kj}^2)$) where $\lambda \geq 0$ is the tuning parameter. It will add terms $2\lambda\beta_{kj}$ and $2\lambda\alpha_{ji}$ to the gradient expressions
 - **Weight Elimination**: is the same thing as L_1 (Lasso) penalties, in which small parameters are set to 0.
 - **Dropout**: randomly omit a certain (relatively small) percentage (say, γ) of the hidden units in each hidden layer for each minibatch of samples during the backpropagation training. This can help break up the dependence in the hidden units, and prevents overfitting (improving generalization). To implement, set the activation to 0 (no information propagates through that unit). Then, we re-weight those units that remain in the model to account for those that were removed. Dropout can estimate uncertainty for model predictions).
 - **Early Stopping** (simplest): rarely does the best generalization of a deep model occur at the local minimum obtained by the gradient descent algorithm. People come up with early stopping, which means stopping the gradient descent algorithm before it has “converged.” The choice of the iteration to stop is a hyperparameter that must be chosen via a validation set or by trial and error. It can be used with other forms of regularization
 - **Weight Sharing**: In some deep model, it makes sense that weights are similar (e.g., imagine weights connected to components of an image or a group of words). Then, we can force these weights to be identical, or add a penalty term that favors them being similar (benefit is that we can estimate less parameters).
 - **Data Augmentation**: since deep models requires a lot of data to train, when there are too few samples, it’s hard for the model to generalize to new data. For models like CNNs, we can use data augmentation to generate more training data (from existing training samples by transforming inputs to produce new input samples that are similar, but different to the training data). We can’t do it in inference problem, but since we are doing prediction problem, this method is fine.
 - **Pretraining**: we use a pretrained network as starting values in backpropagation algorithm (or in cases that you have a huge sample has been used to train a network, just use the pretrained model directly). We typically use this method in image classification because many

architectures have been trained on ImageNet already (about 1k categories, and 1.4 mil images). In the early days of deep models, other form of unsupervised generative pre training were used (e.g, layered restricted Boltzmann machines, autoencoders).

- Batch Size Selection:
 - Minibatches in stochastic gradient descent is advantageous computationally. In deep learning training samples are divided by the **batch size** and then we cycle through all training data. Each iteration over the training data is called an **epoch**. At the beginning of the optimization, it's preferable to have large gradient variance so the solution can jump out of local minima, which suggests that the minibatch sample should be smaller initially. Then, we adjust it to have larger minibatches in alter stages of the optimization (when we don't want to ump around as much). Thus, we can vary minibatch size through the iterations.
- Metrics: to monitor the progress of the training (i.e., at the end of each epoch, we evaluate model performance). Examples
 - An estimate of the loss from the training and validation data
 - Classification accuracy (error rate)
- Momentum: we usually include a “momentum” term in the Stochastic Gradient Descent algorithm to take into account the previous gradients (i.e., we take a convex combination of the previous gradient and the current gradient estimates). The momentum value is typically chosen between $0.9 \rightarrow 0.99$
- Learning Rate: the actual learning rate is a function of both the “learning rate parameter” (ϵ) and the minibatch size M_b . Hence, we adjust the ratio $\epsilon^* = \epsilon/M_b$ to find a good learning rate. In cases that you vary minibatch size, the learning rate is changed as well.
- Network Architecture: The number of layers and the number of hidden units at each layer are parameters as well. We need enough hidden units to capture features (these are really just feature extraction models), and we don't want “too many” (i.e., overfitting). Hence, this is where the “art” comes into deep models. In general, wide/shallow models are easier to overfit and narrow/deep models are easier to underfit. It's recommended that the lower levels have more hidden units.

14.4 Convolutional Neural Networks

- A success story of multi-level neural networks

- The patterns they learn are translation invariant and they also learn spatial hierarchies of patterns (i.e., whereas densely connected network learn global patterns in their input space, convolution layers learn local patterns).

14.4.1 Convolution

$$(f \times g)(t) = \int_{R^d} f(\tau)g(t - \tau)d\tau = \int_{R^d} f(t - \tau)g(\tau)d\tau$$

Discrete convolution (e.g., $d = 2$)

$$f[x, y] \times g[x, y] = \sum_{i=-\infty}^{\infty} \sum_{j=-\infty}^{\infty} f[i, j]g[x - i, y - j]$$

where $f[\]$ is a “kernel” weight function that is applied to elements of the spatial image $g[\]$ (similar to Kernel Smoothing)

The difference is that Kernel Smoothing integrates to 1 (or sum to 1 with discrete data), while Convolution we don’t have to satisfy this constraint. Hence, sometimes people can refer to $f[\]$ as “kernels” or “filters”

Notes:

- Due to the edge effect (less data on edges), the convolved output is smaller. Hence, we can mitigate it by **padding** the original input space with 0 around edges (so that the convolution can be centered on every pixel value, thus giving a convolved feature map that is the same size as the original image).
- Ideally, we learn these kernels (filters) instead of specifying them (even though we have some standard specifications). Each of these filters convolved with the input space produces a new **feature map**. Typically, they are 3×3 or 5×5
- CNN reduces the parameter space by assuming there is **only one set of weights for each convolution**
 - Since weights are shared, we can benefit from dimension reduction in the parameter space
 - After learning the weights, we will use “pooling layer” (or “subsampling” or “down sampling”).
- **Tensors:** In practice, input (images) are often tensors. Hence, there is a **multivariate value** at each spatial location (pixel).
 - Example: in color images in RGB format, there are three values that define the pixel color. Then, CNNs consider a tensor convolution where the kernel averaging is over **location and the variables** (or, “channels” such as red, green, blue color value). Each layer of a

CNN has “height, width, and depth” (i.e., 3-d). For black and white images, each pixel has one value, so the “depth” is 1.

- Hidden layers in the CNN will always have depth >1 (in practice, this depth is way larger). The filter operates on a relatively small part of the image (in x and y), but across the entire depth; so the weights (in 3-d) form a tensor.
- **Stride:** When we skip pixels in convolution, it’s called strides (less weights to understand and computation)
- **Bach normalization:** Since there are multiple filters for each layer, it is beneficial to normalize the convolved pixels across these multiple filters (for a given layer), or a subset of those filters, which should be applied immediately after the ReLU step. This can also help with gradient propagation and allow for deeper networks in general
- **Data Augmentation:** augmenting the samples by a number of random transformations that yield believable images
- **Pretrained networks:** using a saved network that was previously trained on a large dataset (e.g., ImageNet): the spatial hierarchy of features learned by the pretrained network can effectively act as a generic model of the visual world. To implement, you can either use it as
 - feature extraction: use the convolution base layer (which is “frozen”) from the pre-trained model, and only train the flattened fully connected last layer(s)
 - fine-tuning: unfreezing the few top layers of the pre-trained model and jointly training both the newly added part of the model (the last fully connected layer) and the top layers. However, you can only fine-tune the top layers after the top level is already trained.

Convolution calculation

- very expensive
- Traditional approach used the fast Fourier transform (FFT)

Let $F(f), F(g)$ be the Fourier transforms of f and g . Then

$$F(f \times g) = F(f) \cdot F(g) \quad f \times g = F^{-1}(F(g) \cdot F(f))$$

Thus, we do element-wise multiplication in Fourier space, which is way faster.

14.4.2 Pooling

- The pooling layer considers a small rectangular block from the convolutional step and subsamples it to produce a single output.

- Ways to do pooling
 - Block maximum (“max pooling”) (favored): usually done with 2×2 windows and a stride of 2
 - Block average (mean)
 - Block weighted average
 - Downsampling (systemically subsampling one element per block)
- Pooling helps **make the network invariant to translations of the input**, especially when structure is more important than location
- **Rectification**: Convolved images typically go through a ReLU rectification ($ReLU(x) = \max(0, x)$) before pooling

14.4.3 Structure and training

- Convolution step and the activations for the flattened layers requires training, while activation and pooling layers are typically specified (don’t learn)
- The parameters for the convolutions for any given filter are the same for the entire feature map, which allows the presence of a shape in any part of the feature map to be processed the same way, regardless of its special location (i.e., translation invariant)
- Different from the convolution operator, the pooling operator applies to each feature map separately (i.e., don’t pool across the depth dimension).
- Each feature in the final spatial layer is connected to each hidden state in the first fully connected layer, which works like the layers in the deep FNN. There can be multiple fully connected layers (but remember it can be expensive). Most of the parameters in the CNN correspond to these layers.
- In practice, the backpropagation algorithm can be implemented using matrix and vector multiplications.

14.4.4 Interpretation

- To visualize what CNNs learn (e.g., explainable AI), we can visualize
 - intermediate outputs (activations)
 - the intermediate filters
 - “heatmaps” of class activation on an image (related to “layer-wise relevance propagation”, which operates by propagating the prediction backward in the neural network, using a set of purposely designed propagation rules),

14.5 Recurrent Neural Networks

- Traditional neural networks (CNNs) don't account for the sequential dependence of inputs and outputs, such as
 - document and time series classification
 - time series comparisons
 - sequence-to-sequence learning (translation)
 - sentiment analysis in text
 - time series forecasting
- Statisticians considered sequential dependence long before in time-series and spatio-temporal modeling, but these model can still be limited when it comes to complex nonlinear or non-Markovian dependence
- RNNs were developed to handle feedback models for sequence data. Now, they become a successful deep learning method, particularly for language processing applications.
- RNNs are closely related to multivariate state-space models for dynamical systems (e.g., time series, econometric, or spatio-temporal statistics).

Consider a classical dynamical system

$$\mathbf{s}_t = f(\mathbf{s}_{t-1}; \theta)$$

where

- $\mathbf{s}_t$ represents the state of the system at time t
- This is called “recurrent” because the state at the time t is a function of the state at time $t - 1$

In “unfolded” form

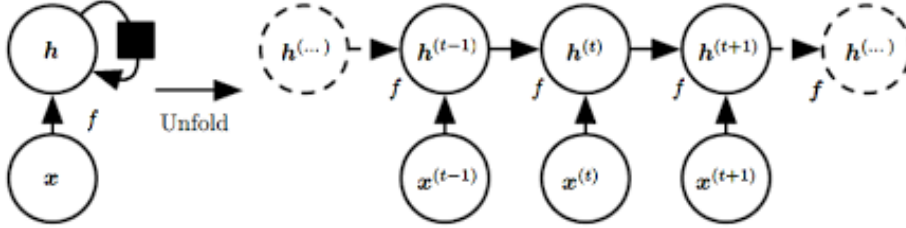
$$\begin{aligned}\mathbf{s}_t &= f(\mathbf{s}_{t-1}; \theta) \\ &= f(f(\mathbf{s}_{t-2}; \theta); \theta) \\ &= f(f(f(\mathbf{s}_{t-3}; \theta); \theta); \theta) \\ &= \dots\end{aligned}$$

Where parameters θ are shared across all states

Similar to traditional state space model, the states are considered “hidden” (denoted as $\mathbf{h}_t$) and add external inputs $\mathbf{x}_t$

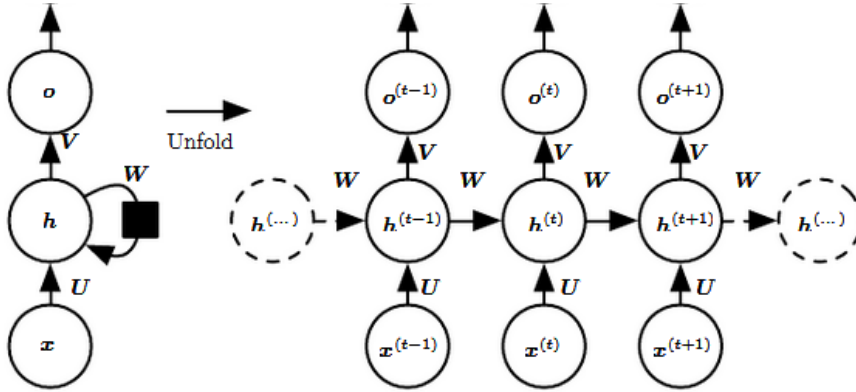
$$\mathbf{h}_t = f(\mathbf{h}_{t-1}, \mathbf{x}_t; \theta)$$

Hence, sequences are processed by iterating through the sequence elements and maintaining a state (containing info relative to what it has seen so far via an internal feedback loop)



(picture by Goodfellow et al. (2016) Ch. 10)

Incorporating RNN with output



(picture by Goodfellow et al. (2016) Ch. 10)

A basic RNN setup is

$$\mathbf{y}_t = g_o(\mathbf{o}_t)\mathbf{o}_t = \mathbf{W}_{hy}\mathbf{h}_t\mathbf{h}_t = g_h(\mathbf{W}_{hh}\mathbf{h}_{t-1} + \mathbf{W}_{xh}\mathbf{x}_t)$$

where

- $g_o(\cdot), g_h(\cdot)$ are activation functions
- $\mathbf{W}_{hy}, \mathbf{W}_{hh}, \mathbf{W}_{xh}$ are weight matrices (typically contain bias terms as well).

To estimate these parameters, we still define loss function and optimize by back-propagation. However, with common parameters across time, they introduce some complications. To solve this,

- Traditionally, the **backpropagation through time** (BPTT) algorithm via stochastic gradient descent can handle. But it introduces the problem of **vanishing gradient/exploding gradient**: when the gradients get

propagated over many times they tend to either (1) vanish (most of the time) or (2) explode

- Even though there are ways to mitigate exploding gradient (to ensure stability), the series of dependencies through time can still lead to exponentially smaller weights
- Or we add “gates” that selectively allow info from the past to influence the present. These gates serve to save info for later, breaking the gradient

14.5.1 Long Short-Term Memory (LSTM) RNNs

- This is the most common gated RNN.
- It creates time paths that have gradients that don’t vanish or explode

Setup

Name	Formula
Output	$\mathbf{y}_t = g_o(\mathbf{V}\mathbf{h}_t)$
Hidden State	$\mathbf{h}_t = \tanh(\mathbf{c}_t) \odot \mathbf{o}$
Internal Memory	$\mathbf{c} = \mathbf{c}_{t-1} \odot \mathbf{f} + \mathbf{g} \odot \mathbf{i}$ (first part = forgetting, second part = new info)
Candidate	$\mathbf{g} = \tanh(\mathbf{U}^g \mathbf{x}_t + \mathbf{W}^g \mathbf{h}_{t-1})$
Hidden State	
Output Gate	$\mathbf{o} = g(\mathbf{U}^o \mathbf{x}_t + \mathbf{W}^o \mathbf{h}_{t-1})$
Forget Gate	$\mathbf{f} = g(\mathbf{U}^f \mathbf{x}_t + \mathbf{W}^f \mathbf{h}_{t-1})$
Input Gate	$\mathbf{i} = g(\mathbf{U}^i \mathbf{x}_t + \mathbf{W}^i \mathbf{h}_{t-1})$

where $g(\cdot)$ is typically a sigmoid function and $\odot$ is Hadamard product (component-wise multiplication for matrices).

- The input gate selects hidden units that get input to time t
- The forget gate selects the hidden states at the previous time to reset to 0 at time t
- The output gate selects the states that will be related to the response
- The memory units indicate when to remember or forget previous hidden states, which will mitigate the vanishing/exploding gradient problem. And its assumption is also plausible because many processes that events in the distant past can influence presence irrespective of the intervening states.

Application:

- Classification /Segmentation
- Object Detection/Localization

- Similarity learning
- Generative models
- Video analysis

14.5.2 Gated Recurrent Units

Name	Formula
Output	$\mathbf{y}_t = g_o(\mathbf{V}\mathbf{h}_t)$
Hidden state	$\mathbf{h}_t = (1 - \mathbf{z}) \circ \mathbf{s} + \mathbf{z} \odot \mathbf{h}_{t-1}$
Candidate	$\mathbf{s} = \tanh(\mathbf{U}^s \mathbf{x}_t + \mathbf{W}^s (\mathbf{r} \odot \mathbf{h}_{t-1}))$
Hidden State	
Reset Gate	$\mathbf{r} = g(\mathbf{U}^r \mathbf{x}_t + \mathbf{W}^r \mathbf{h}_{t-1})$
Update Gate	$\mathbf{z} = g(\mathbf{U}^z \mathbf{x}_t + \mathbf{W}^z \mathbf{h}_{t-1})$

If $\mathbf{r} = 1, \mathbf{z} = 0$, then it is the vanilla RNN

Notes:

- Long Short-Term Memory (LSTM) RNNs and Gated Recurrent Units are computationally intensive to fit, and require parallelized implementation
- Similar to the standard Deep Neural Networks and Convolutional Neural Networks require a lot of data
- RNNs are more “black boxes” given their complexity, which has the benefit of making them modular and connectable, hence much deeper
- **Recurrent Dropout:** helps with overfitting. But it should not be applied before a recurrent layer as it measures information in the flow. Instead of allowing the dropout pattern to vary randomly with each time step, we use *the same pattern of dropout for every time step within the recurrent layer* (known as **recurrent dropout**). We can still use regular dropout for this input layers
- **Stacking Recurrent Layers:** Similar to Deep Neural Networks and Convolutional Neural Networks, we can also improve learning and build more powerful RNNs by stacking the recurrent layers, but it can increase dramatically the cost of implementation.
- **Recurrent Attention:** Recurrent Neural Networks can’t localize in certain regions of the sequence. Generally, a **neural attention mechanism** equips a neural network with the ability to focus on a subset of its inputs (more detail). Recurrent Attention Unit was described, which integrates an attention mechanisms into the interior of RNN by adding an attention gate. The attention gate enhances RNN’s ability to remember long-term memory and help memory cells quickly discard unimportant content Zhong and Yue (2020)

- We can also use one-dimensional (1D) CNNs for sequence data. This method can be cheaper to implement, but not as good when **global order matters** (e.g., time series where the recent past has more info). However, it's not necessarily true for text data, 1D CNNs can outperform RNNs. People also combine 1D CNN layers and RNN layers (usually the CNN layer is first to reduce dimensionality) (Chollet, 2018)

Chapter 15

Artificial Intelligence

15.1 Explainable AI

- denoted as XAI
- Historical criticisms for deep learning
 - lack of uncertainty quantification
 - inability to do inference (i.e., unknown important inputs)
- XAI tries to solve these problems (and fairness and transparency)
- Review papers
 - Samek et al. (2021)
 - Abdar et al. (2021)

Uncertainty Quantification

Four general topics:

1. Variational (Bayesian) inference
2. Monte Carlo Dropout: justifies dropout as a UQ procedure from a perspective of deep Gaussian Processes
3. Mixture Density Networks: this model could predict a mixture density for the output of a NN (instead of the mean response) (Pattern Recognition Bishop 2006's book)
4. Ensemble techniques: UQ by generating multiple realization (ensembles) of a deep model

Explainability

There are 4 primary approaches (Samek et al., 2021)

- Interpretable Local Surrogates (e.g., LIME)
- Occlusion Analysis (e.g., Shapley values, SHAP, Kernel SHAP, meaningful perturbation)
- Integrated Gradients (e.g., SmoothGrad)
- Layerwise Relevance Propagation

Additionally,

- Self-explainable model (e.g., attention mechanisms)
- Specialized models (e.g., graph neural networks)

Fail: People can modify the ML such that we can have the same predictions, but it matches up better with stories that they like to tell. (so be careful!!!).

15.1.1 Mixture Density Networks

It's theoretically possible/plausible to model any distributions as a mixture of Gaussians

We can model any distribution as a mixture of Gaussian

$$p(y|\mathbf{x}) = \sum_{k=1}^K \alpha_k(\mathbf{x}) \phi(\mu_k(\mathbf{x}), \sigma_k(\mathbf{x}))$$

where $\alpha_k(\mathbf{x})$ are mixture coefficients associated with the Gaussian distribution defined by mean and standard deviation $\phi(\mu_k(\mathbf{x}), \sigma_k(\mathbf{x}))$ and all of these parameters depend on the inputs $\mathbf{x}$

Instead of training the neural network to estimate the conditional mean, we estimate all the mixture parameters $\{\alpha_k(\mathbf{x}), \mu_k(\mathbf{x}), \sigma_k(\mathbf{x})\}$ (i.e., these parameters are the output of the NN)

This method can use mixtures of other distributions as well

15.1.2 Local Interpretable Model-Agnostic Explanations

- denoted as LIME (Ribeiro et al., 2016)
 - **Local:** explain individual predictions (local surrogate models - a model within model)
 - **Interpretable:** Understandable
 - **Model-agnostic:** explain prediction for any type of model
- It's hard to find the variable importance for predicting certain output.
- The attribution to a base feature (inputs, instead of parameters) can be interpreted as the importance of the feature to the prediction

- Treat all models as black boxes and understand why they make each individual prediction
- Undersigning why a model makes each individual prediction will help the user determine whether to trust the model
- **Local surrogate models:** approximate the predictions of black-box models locally
- Instead of training a global surrogate model, LIME trains many local surrogate models to explain individual prediction
- These local surrogate models must behave similarly to how the black-box model does in the neighborhood of each instance being explained (local faithfulness)

Steps to train local surrogate models

1. Select a prediction from the black-box model that you want to explain
2. Perturb the dataset and get the black-box predictions for these new points
3. Weight the new samples based on their proximity to the point of interest (i.e., your chosen prediction)
4. train a weighted, interpretable model (e.g., regression) on the perturbed dataset, using the black-box predictions as the response
5. Explain the prediction by interpreting the local model

LIME produces an explanation for a given instance of $(\mathbf{y}, \mathbf{x})$

We want to know which of the $\mathbf{x}$ features are most responsible for a given prediction $\hat{\mathbf{y}}$

$$\text{explanation}(\mathbf{x}) = \underset{g \in G}{\operatorname{argmax}} L(f, g, \pi_x) + \Omega(g)$$

where

- f : the original black-box model
- g the explanation model for the instance
- G the family of possible explanation models (e.g., linear regression models)
- π_x proximity measure that determines how large the neighborhood is around $\mathbf{x}$
- $\Omega(g)$: model complexity (want to be small - fewer features in explanation)

To define these neighborhoods

- π_x is an exponential smoothing kernel
- Small kernel width means that observation must be very close to influence the local model

- Large kernel width means that observations that are very different from the instance we are seeking to explain can have influence on the local model
- kernel widths are very subjective (have to experiment and see which ones result in meaningful explanations)

Notes:

- LIME is easy to use in R (`lime` or `iml` package)
- The difficult part is evaluating how good the local surrogate model is at estimating the complex model locally, which can affect the interpretability
- Heat map can show how all the variables have an influence on the predictions

References for LIME

- <https://www.oreilly.com/content/introduction-to-local-interpretable-model-agnostic-explanations-lime/>
- <https://homes.cs.washington.edu/~marcotcr/blog/lime/>
- Ribeiro et al. (2016)

15.1.3 Shapley Values

- Shapley values are a game theoretic approach to explain the output of any machine learning model (migrated from econ to stat)
- These values can also combined with other approaches such as LIME.
- Similar to LIME, the problem is distributing the prediction score of a model for a specific input to its base features
- Game theory inspiration:
 - Introduced in 1951 by Lloyd Shapley as a method to value individual play contributions in a cooperative game (which won him a Nobel Prize in Econ)
 - Example
 - * Determine value of players individually and in subsets
 - * Determine marginal values fro players from all possible orderings
 - * Find the Shapley value by taking average of marginal values

Formally, for a game with N players, the characteristics (value) function ν amps subsets $S \in \{0, 1, \dots, N\}$ to a real value $\nu(S)$, the value to be gained from player cooperation

The marginal contribution for each play is

$$\Delta_\nu(i, S) = \nu(S \cup i) - \nu(S)$$

The Shapley value (contribution to the “grand coalition”) for player i is

$$\phi_i(\nu) = \frac{1}{N} \sum_{S \subseteq N_{\{i\}}} \binom{n-1}{|S|}^{-1} \Delta_\nu(i, S)$$

which means

$$\varphi(\nu) = \frac{1}{\text{number of players}} \sum_{\text{coalitions excluding } i} \frac{\text{marginal contribution of } i \text{ to coalition}}{\text{number of coalitions excluding } i \text{ of this size}}$$

where x_S has missing values for the features not in set S

Desirable properties of Shapley Values

- Unique
- Symmetry: if two players have equal marginal contribution, they will have equal Shapley values
- Dummy: if the marginal contribution is 0, the Shapley value will be 0
- Additivity: for a single player, i , with two value functions, ν and w ,

$$\phi_i(\nu) + \phi_i(w) = \phi_i(\nu + w)$$

They are useful to ensure that the feature attribution correspondence to human understanding.

Application to predictive modeling

If the game is a single outcome, with players as features $x_1, \dots, x_p$ with a value function ν_f with respect to the model f , then the Shapley value for feature i can be viewed as the influence of that feature on the outcome

The key is then how one defines the value function, ν_f

Possible approaches:

- Conditional Expectation variants
 - Shapley regression values (Lipovetsky and Conklin, 2001b,a)
 - Shapley sampling values (Štrumbelj and Kononenko, 2013)
 - SHAP (Lundberg and Lee, 2017)

- Intervention variants
 - Quantitative input influence (Datta et al., 2016)
 - Formulate, Approximate, Explain framework (?)

SHapley Additive exPlanation (SHAP) values

The value function is defined as

$$\nu_{f,x}(S) = f(x_S) = E[f(x)|x_S] = E_{x_{\bar{S}}|x_S}[f(x)] \text{ assume feature independence} \approx E_{x_{\bar{S}}}[f(x)] \text{ assume model linearity}$$

where x_S has missing values for the features not in set S

Exact computation is challenging, but there are numerous model-agnostic approximation methods and model-type-specific approximation methods

these have varying assumptions, such as feature independence and model linearity (as in the equation above)

Approximation Methods

- Model-agnostic
 - If we assume feature independence, we can use sampling approximation methods from (Štrumbelj and Kononenko, 2013) and (Datta et al., 2016); reasonable to compute for small number of inputs only
 - Kernel SHAP (LIME + Shapley): reformulate as a weighted linear regression problem to jointly estimate all SHAP values
- Model-type-specific
 - Linear SHAP
 - Low-order SHAP
 - Max SHAP
 - Deep SHAP (DeepLift + Shapley)
 - Tree SHAP (Lundberg et al., 2018)

Implementation

- Python: `shap`: developed by Lundberg provides Kernel SHAP + model-type-specific SHAP algorithm; support for TensorFlow/Keras/PyTorch models
- R: `shapr` provides Kernel SHAP + Aas et al. (2021) method for accounting for feature dependence; support for `gam`, `lm`, `glm`, `xgb.Booster`, `ranger`, `shapper`: port for Python library `shap`
- Others: `fastshap`, `iml`, `shap-value`, `DALEX`, `shapleyR`

Critiques

- Mathematical Issues
 - Indirect influence: a feature which may have no effect on the response but is predictive of other features may attribute influence
 - How should statistically-related features be handled? Should they be considered a “single player”?
 - Intervention methods rely on out-of-distribution sampling and extrapolate outside of feature space
- Human Interpretability Issues
 - Shapley value answer question: “Why $f(x)$ instead of $E[f(x)]$?, while we may not actually observe data with the outcome $E[f(x)]$ ”
 - Does not provide guidance on changing outcomes
 - Prone to misinterpretation

References for Shapley

- Datta et al. (2016)
- Kumar et al. (2020)
- Lipovetsky and Conklin (2001a)
- Lundberg et al. (2018)
- Lundberg and Lee (2017)
- Merrick and Taly (2020)
- Štrumbelj and Kononenko (2013)
- Sundararajan and Najmi (2020)

Part VII

5. OTHERS

Chapter 16

Support Vector

16.1 Maximal Margin Classifier

We have n pairs of training data, $(x_1, y_1), \dots, (x_n, y_n)$, where $x_i \in R^p$ and $y_i \in \{-1, 1\}$, corresponding to two possible classes

Hyperplane in p -dimensional space is a “flat” affine subspace of dimension $p - 1$

If $p = 2$, the hyperplane is a line

If $p = 3$, the hyperplane is a plane

In two dimensions, a hyperplane is defined by the equation

$$\beta_0 + \beta_1 X_1 + \beta_2 X_2 = 0$$

so that any point X_1, X_2 that satisfies this equation is on the hyperplane

Then, in p -dimensions, the hyperplane is defined by

$$\beta_0 + \beta_1 X_1 + \dots + \beta_p X_p = 0$$

or equivalently

$$\beta_0 + \mathbf{X}^T \beta = 0$$

If the point $X = (X_1, \dots, X_p)^T$ does not lie on the hyperplane, it lies on one side or the other

$$\beta_0 + X^T \beta > 0 \quad \beta_0 + X^T \beta < 0$$

Hence, the hyperplane divides the p -dimensional space in two sides to determine which side the point is on, we can find the sign of $\beta_0 + X^T\beta$ (i.e., $G(x) = \text{sign}[\beta_0 + X^T\beta]$)

We want to classify, $x^* = (x_1^*, \dots, x_p^*)^T$ from the separating hyperplane

We start by assuming that it's possible to construct a hyperplane that separates the observations perfectly

Thus, our classification of (x_1, y_i) is

$$\begin{cases} \beta_0 + x_i^T\beta > 0 & \text{if } y_i = 1 \\ \beta_0 + x_i^T\beta < 0 & \text{if } y_i = -1 \end{cases}$$

This is equivalent to noting that the separating hyperplane has the property

$$y_i(\beta_0 + x_i^T\beta) > 0$$

for all $i = 1, \dots, n$

Thus, our classifier classifies x^* into the first group if $f(x^*) = \beta_0 + \beta_1 x_1^* + \dots + \beta_p x_p^*$ is positive and

into the second group if this is negative (i.e., this $G(x)$ defined above)

The magnitude of $f(x^*)$ tells us how far x^* is from the hyperplane

Clearly, **the separating hyperplane is just a linear decision boundary**

We can have an infinite number of such separating hyperplanes.

We can define an **optimal separating hyperplane** as **the maximal margin hyperplane**, which is just the hyperplane that is farthest from the training observations, which can be done by considering the perpendicular distance between each observation and hyperplane;

The smallest such distance is the minimal distance from the observations to the hyperplane, and is known as the **margin**

The **maximal margin hyperplane** is then the separating hyperplane that has the largest margin (i.e., the one that has the farthest minimum distance to the training observations).

The classifier built with this hyperplane is then known as **the maximal margin classifier**

The maximal margin hyperplane presents the mid-line between the two classes with the widest “slab” that we can insert between the classes

Points that are used to make the hyperplane are examples of **support vectors** (i.e., they are vectors in p -dimensional space that “support” the maximal margin hyperplane in that if they were located at different locations, the maximal margin hyperplane would move as well)

All of the other points do not affect the maximal margin hyperplane (i.e., if they move, it doesn’t change the location of the maximal margin hyperplane - so long as they don’t move across the existing margin boundaries).

This property of support vectors is important in the construction of support vector classifiers and support vector machines

We can find the maximal margin hyperplane by solving the following convex optimization problem

$$\max_{\beta_0, \dots, \beta_p} M$$

subject to

$$\sum_{j=1}^p \beta_j^2 = 1$$

i.e., $\|\beta\| = 1$

and

$$y_i(\beta_0 + x_i^T \beta) \geq M, \forall i = 1, \dots, n$$

The last constraint guarantees that each observation will be on the correct side of the hyperplane, provided M is positive, with a cushion M

The first constraint identifies the parameters such that the perpendicular distance between the i -th observation and the hyperplane is given by $y_i(\beta_0 + x_i^T \beta)$. This ensures that each observation is at least a distance M from the hyperplane - hence, M is the margin

However, this method would not work if no separating hyperplane exists (i.e., if no hyperplane can separate the two classes)

Two cases where a separating hyperplane may not be desirable even if possible

1. If it has a very small margin
2. it very sensitive to one or two observations, it will most likely overfit

16.2 Support Vector Classifiers

Due to the previous two problems, we seek a classifier that is based on a hyperplane that does not perfectly separate the classes to give

1. greater robustness to individual observation
2. better classification of most of the training observation

Such a **soft margin classifier** is the support vector classifier, which is constructed similarly to the maximal margin classifier but we allow some observations to be on the incorrect side of the margin or even the incorrect side of the hyperplane

The procedure for implementing a support vector classifier is also an optimization problem with a modification of one of the constraints and the addition of another

The support vector classifier is given by the solution to the following optimization problem

$$\max_{\beta_0, \dots, \beta_p, \epsilon_1, \dots, \epsilon_n} M$$

subject to

$$\sum_{j=1}^p \beta_j^2 = 1$$

(i.e., $\|\beta\| = 1$) and

$$y_i(\beta_0 + x_i^T \beta) \geq M(1 - \epsilon_i), \forall i = 1, \dots, n$$

$$\epsilon_i \geq 0, \sum_{i=1}^n \epsilon_i \leq C$$

where C is a non-negative tuning parameter

M is the margin

we have $\epsilon_i, i = 1, \dots, n$, which are called **slack variables** that provide the flexibility to be on the wrong side of the margin or hyperplane

Then, we classify a new observation x^* via the sign of $f(x^*)$ as before

The ϵ_i parameters give an indication of where the observation is located relative to the margin

- $\epsilon_i = 0$ means it's on the correct side of the margin

- $\epsilon_i > 0$ means it's on the wrong side of the margin
- $\epsilon_i > 1$ means it's on the wrong side of the hyperplane

Since C is the upper bound of the sum of the ϵ_i , it determines the number of severity of the violations to the margin (and hyperplane) that we can tolerate, which is like a budget for the amount that the margin can be violated by n observations (James et al., 2013, p. 376)

If $C = 0$, then there is no budget for violations and $\epsilon_i = 0, i = 1, \dots, n$ (which is just the [Maximal Marginal Classifier] if the classes are separable)

If $C > 0$, no more than C observations can be on the wrong side of the hyperplane (recall, $\epsilon_i > 1$ if the observation is misclassified)

As C increases, we become more tolerant to violations of the margin, so the margin widens

C acts as a tuning parameter that controls the variance-bias trade-off and can be chosen by cross-validation

- C small: narrow margins, few margin violations, low bias, high variance, fewer support vectors
- C large: wide margins, more violations, more bias, but less variance, many support vectors

Again, support vectors are those observations that are on or between the margins

The fact that the support vector decision rule is based on a relative few observations (support vectors) means that it's **relative robust to the behavior of observations far away from the hyperplane**, which is different from the linear discrimination rules

16.3 Nonlinear Decision Boundaries

Instead of fitting a support vector classifier using the p features, $X_1, \dots, X_p$, we consider $2p$ features (linear and quadratic)

$$X_1, X_1^2, \dots, X_p, X_p^2$$

Then, the optimization problem become

$$\max_{\beta_0, \dots, \beta_p, \epsilon_1, \dots, \epsilon_n}$$

subject to

$$y_i(\beta_0 + \sum_{j=1}^p \beta_{j1}x_{1j} + \sum_{j=1}^p \beta_{j2}x_{ij}^2) \geq M(1 - \epsilon_i), \forall i = 1, \dots, n$$

$$\epsilon_i \geq 0, \sum_{i=1}^n \epsilon_i \leq C, \sum_{j=1}^p \sum_{k=1}^2 \beta_{jk}^2 = 1$$

Although this is nonlinear (quadratic) in the original feature space, it's linear in the expanded (2p-dimensional) space

Then, we could extend the feature space to be beyond quadratic terms, such as polynomials and interactions, or other functions of the predictors (e.g., basis)

There are an infinite number of choices for such terms.

The support vector machine (svm) provides a principled way to enlarge the feature space used in a support vector classifier in a way that facilitates computation

There are two things that make this tractable

1. Only the support vectors are relevant for determining the classification boundary
2. A kernel representation that is easily generalizable

16.4 Support Vector Machines

- is an extension of the Support Vector Classifiers that results from enlarging the feature space to include transformations of the inputs using kernels

Consider the collection of m basis functions in terms of the features, x

$$h(x_i) = (h_1(x_i), h_2(x_i), \dots, h_m(x_i))^T, i = 1, \dots, n$$

Then, we could use the Support Vector Classifiers to find the parameters $\hat{\beta}$ that minimize $f(x) = \beta_0 + h(x)^T \beta$

Then, classification can be accomplished by $\hat{G}(x) = \text{sign}(\hat{f}(x))$

Then, we can we choose the basis functions?

We don't need to select basis function, but rather it's sufficient to choose a positive definite kernel function $K(x_i, x_{i'})$ which will imply a finite collection of basis functions

Reminder

Inner Products

The inner product of two r -dimensional vectors a and b is defined as $\langle a, b \rangle = \sum_{j=1}^r a_j b_j$

Then, the inner product of two inputs x_i and x'_i is

$$\langle x_i, x'_i \rangle = \sum_{j=1}^p x_{ij} x'_{ij}$$

Importantly, we can represent a positive definite kernel in terms of the inner product of some set of basis functions:

$$K(x_i, x_{i'}) = \langle h(x_i), h(x_{i'}) \rangle$$

Example with a second-degree polynomial kernel

$$K(x, \tilde{x}) = (1 + \langle x, \tilde{x} \rangle)^2$$

and $p = 2$; e.g., $x = (x_1, x_2)$, $\tilde{x} = (\tilde{x}_1, \tilde{x}_2)$

Then,

$$\begin{aligned} K(x, \tilde{x}) &= (1 + \langle x, \tilde{x} \rangle)^2 \\ &= (1 + x_1 \tilde{x}_1 + x_2 \tilde{x}_2)^2 \\ &= 1 + 2x_1 \tilde{x}_1 + 2x_2 \tilde{x}_2 + (x_1 \tilde{x}_1)^2 + (x_2 \tilde{x}_2)^2 + 2x_1 \tilde{x}_1 x_2 \tilde{x}_2 \end{aligned}$$

If we write $K(x, \tilde{x}) = \langle h(x), h(\tilde{x}) \rangle$, we have $m = 6$ basis function with

$$h_1(x) = 1, h_2(x) = \sqrt{2}x_1, h_3(x) = \sqrt{2}x_2, h_4(x) = x_1^2, h_5(x) = x_2^2, h_6(x) = \sqrt{2}x_1x_2$$

This is a powerful tool in regularized (penalized) modeling, known as **Reproducing Kernel Hilbert Spaces** (RKHS)

16.4.1 Regularization by RKHS

(Hastie et al., 2001, 5.8)

In general, the regularization (penalized loss) problem is

$$\min_{f \in H} \left[\sum_{i=1}^n L(y_i, f(x_i)) + \lambda J(f) \right] \quad (16.1)$$

where

- $L(y_i, f(x_i))$ is the loss function
- $J(f)$ is a penalty function
- H is a space of functions on which $J(f)$ is defined

Consider a **positive definite kernel** $K(x, \tilde{x})$ for $x, \tilde{x} \in R^p$, that has an eigen-expansion given by

$$K(x, \tilde{x}) = \sum_{j=1}^{\infty} \delta_j \phi_j(x) \phi_j(\tilde{x})$$

with eigenvalue $\delta_j > 0$, $\sum_{j=1}^{\infty} \delta_j^2 < \infty$ and eigenfunctions $\{\phi_j(\cdot)\}$

The idea is to use eigenfunctions as basis functions

We seek functions $f(x)$ in terms of these eigenfunctions,

$$f(x) = \sum_{j=1}^{\infty} c_j \phi_j(x)$$

that satisfy the constraint that $J(f) = \|f\|_{H_K}^2 \equiv \sum_{j=1}^{\infty} c_j^2 / \delta_j < \infty$

This function space, H_K , is called a reproducing kernel Hilbert space because the inner product of K with itself gives K (i.e., $\langle K(\cdot, x_i), K(x, x_{i'}) \rangle = K(x_i, x_{i'})$ it reproduces itself)

Then, we can rewrite the regularization problem as

$$\min_{\{c_j\}_1^{\infty}} \left[\sum_{i=1}^n L(y_i, \sum_{j=1}^{\infty} c_j \phi_j(x_i)) + \lambda \sum_{j=1}^{\infty} \frac{c_j^2}{\delta_j} \right] \quad (16.2)$$

Even though we might think that we have **infinite** number of choices for ϕ_j , the solution to the regularization problem here can be shown to be given by (**finite**)

$$f(x) = \sum_{i=1}^n \alpha_i K(x, x_i)$$

where (**finite**)

$$J(f) = \sum_{i=1}^n \sum_{j=1}^n K(x_i, x_j) \alpha_i \alpha_j$$

Critically, we note that **although the optimization function is infinite dimensional, this solution is finite-dimensional (this is the kernel trick in machine learning)**

In fact, the infinite-dimensional optimization problem reduces to the following finite dimensional optimization problem

$$\min_{\alpha} [L(\mathbf{y}, \mathbf{K}) + \lambda \mathbf{1}^T \mathbf{K}] \quad (16.3)$$

where $\mathbf{K}$ is a matrix with (i, j) th entry given by $K(x_i, x_j)$

This is powerful because it works for **any** (convex) loss function and **any** (positive definite) kernel function (from (16.1) to (16.2) to (16.3))

This idea relates back to Support Vector Machines because all we need is the loss function associated with SVMs and to choose a positive definite kernel.

It has been worked out that the loss function associated with the support vector classifier is given by $[1 - y_i f(x_i)]_+$ (where the “+” indicates the positive portion of the function in the brackets), which is called “hinge loss.”

Then the solution of the following constrained minimization problem gives the same estimates for β

$$\min_{\beta_0, \beta} \left[\sum_{i=1}^n [1 - y_i f(x_i)]_+ + \frac{\lambda}{2} \|\beta\|^2 \right] \quad (16.4)$$

where λ is proportional to C as before

Suppose a basis set $\{h_j(x)\}$ comes from the eigen-decomposition of $K(x, \tilde{x}) = \sum_{j=1}^{\infty} \delta_j \phi_j(x) \phi_j(\tilde{x})$; hence $h_j(x) = \sqrt{\delta_j} \phi_j(x)$. Let $\delta_j \equiv \sqrt{\delta_j} \beta_j$, (recall that the desired function is an expansion in these basis functions), the minimization above can be written

$$\min_{\beta_0, \theta} \left[\sum_{i=1}^n [1 - y_i (\beta_0 + \sum_{j=1}^{\infty} \theta_j \phi_j(x_i))]_+ + \frac{\lambda}{2} \sum_{j=1}^{\infty} \frac{\theta_j^2}{\delta_j} \right] \quad (16.5)$$

One of the strengths of the RKHS approach to the SVM problem is that the penalized loss approach suggests that it's fairly simple to consider any convex loss function. For example, using logistic regression as a classifier is similar to the SVM because the binomial deviance loss is similar to the hinge loss

Then, the RKHS theory that we discussed gives a finite-dimensional solution ((16.3)) $f(x) = \beta_0 + \sum_{i=1}^n \alpha_i K(x, x_i)$, that equivalently minimizes

$$\min_{\beta_0, \alpha} \left[[1 - y_i f(x_i)]_+ + \frac{\lambda}{2} \mathbf{1}^T \mathbf{K} \right] \quad (16.6)$$

For example, we chose a linear kernel, $K(x, x_i) = \langle x, x_i \rangle$ (which is essentially the Pearson correlation), then we obtain the solution $f(x) = \beta_0 + \sum_{i=1}^n \alpha_i \langle x, x_i \rangle$, which is just the Support Vector Classifiers

For SVM analysis, we typically select a more complicated kernel to allow flexible classification boundaries. Common choices are:

- d-th degree polynomial kernel: $K(x, \tilde{x}) = (1 + \langle x, \tilde{x} \rangle)^d$
- radial basis kernel (effectively is the Gaussian): $K(x, \tilde{x}) = \exp(-\gamma \|x - \tilde{x}\|^2)$
- neural network kernel: $K(x, \tilde{x}) = \tanh(\kappa_1 \langle x, \tilde{x} \rangle + \kappa_2)$

The drawback is that you don't know which one to pick.

An advantage of being able to use finite-dimensional kernel representation is that we only need to consider the $\binom{n}{2}$ distinct pairs of x_i and $x_{i'}$ in the kernel matrix. Even though it can still be a large number of kernel, but it's certainly better than infinite.

Moreover, $\alpha_i = 0$ if the observed feature vector x_i is not a support vector, because the hinge loss function is 0 for observations for which $y_i f(x_i) \geq 1$, which are those on the correct side of the margin. This simplifies the calculation in the procedure and we write

$$f(x) = \beta + \sum_{i \in S} \alpha_i K(x, x_i)$$

where S is the set of support vectors.

It can be shown that for a given λ that $\beta = (1/\lambda) \sum_{i \in S} \alpha_i y_i x_i$

Alternative Loss Functions

- **Huberized loss**, which has some of the attractive features of the hinge loss (support points) and logistic regression (smooth loss function for which one can estimate probabilities) - good with outliers

Extensions to Multiple Classes

It isn't clear how we should extend the SVM machinery from the binary case to the multiple case. The notion of a separating hyperplane is hard to generalize to more than two classes. There are few possible approaches: (1) one-versus-one or (2) one-versus-all approaches

The **one-versus-on approach** for $K > 2$ classes finds $\binom{K}{2}$ SVMs for each pair of classes. Then, given a new observation, each SVM classifier is used and one keeps track of the number of times the test observation was assigned to each class. The one that is assigned most frequently is chosen

The **one-versus-all** (preferable) approach fits K different SVMs comparing each class to all of the others, thus getting K functions, $f_k(x), k = 1, \dots, K$. Given test observation x^* , one selects the class, k for which $f_k(x^*)$ is largest

Given that logistic regression gives similar results to SVMs for binary classes, a multinomial-logistic classifier (i.e., multinomial loss function) might be a better way to do classification for $K > 2$

16.4.2 SVM Regression

We can extend the SVM method to the case of continuous responses (regression), known as support vector machine regression. Given the “loss plus penalty” paradigm, we can do this by looking for the β coefficients in the function $f(x) = \beta_0 + x^T \beta$ relative to some loss function. To make it analogous to the SVM case, we need a loss function that accommodates something like the margin in the SVM case (i.e., we seek a loss function where only residuals larger in absolute value than some positive constant contribute to the loss):

$$\min_{\beta_0, \beta} [V_\epsilon(y_i - f(x_i)) + \frac{\lambda}{2} \|\beta\|^2]$$

where

$$V_\epsilon(r) \equiv \begin{cases} 0, & \text{if } |r| < \epsilon \\ |r| - \epsilon, & \text{otherwise} \end{cases}$$

where errors $< \epsilon$ are ignored (Hastie et al., 2001, 12.3.6)

The SVM regression loss function is similar to other robust regression loss functions (e.g., the Huber loss)

Chapter 17

Discriminant Analysis

Assumption:

- Multivariate normality
- Homoskedasticity: assumption for LDA only, QDA can still be used with unequal covariances
- No Multicollinearity: because it can decrease your predictive power
- Independent predictors

LDA with p inputs x for $k = 1, \dots, K$ classes proceeds by finding the minimum value of the discriminant function for a given x

$$\delta_k(x) = x^T \Sigma^{-1} \mu_k - \frac{1}{2} \mu_k^T \Sigma^{-1} \mu_k + \log \pi_k$$

for each class, where

- Σ is the pooled covariance matrix for all classes
- μ_k is the class-specific mean (centroid)
- π_k is the prior probability of being in class k

In practice, we don't know the parameters Σ, μ_k, π_k and either estimate them or in the case of π_k specify them.

The quadratic discriminant function (QDA) follows similarly, except we accommodate a different covariance matrix for each class, Σ_k , which gives

$$\delta_k(x) = -\frac{1}{2} \log |\Sigma_k| - \frac{1}{2} (x - \mu_k)^T \Sigma_k^{-1} (x - \mu_k) + \log \pi_k$$

Virtues of LDA

- Simple and typically good classifier
- It's the estimated Bayes classifier if the observations are really from a multivariate normal distribution with a common covariance matrix (unlikely in practice)
- the decision boundaries are linear, making the decision rule simple to describe and implement
- Provides a natural low-dimensional view of the data
- It often works well because it's simple and has low variance.

Problems with LDA

- Linear boundaries are often not adequate to separate classes. when the sample size is larger, we can efficiently estimate nonlinear boundaries
- A single centroid for each class is not sufficient many applications
- In some problems, there may be too many predictors (p is large relative to n) and LDA has too many parameters that are estimated with high variance

Extensions of LDA

- Recast the LDA problem as a linear regression problem, and then generalize it to include functions of the x inputs - so called "flexible discriminant analysis" (FDA)
 - FDA also stands for "Functional Data analysis" in the broader statistical jargon
- Penalize the coefficients in the LDA when there are too many coefficients
- Model each class as a mixture of two or more Gaussians with different centroids, but with all of the Gaussians sharing the same covariance matrix

Background

- Related to finding lower-dimensional linear classifiers (R.A. Fisher), his idea was to obtain a reasonable representation of the class population that involved only a few linear combination of the observation (e.g., $a'_1x, a'_2x, \dots$ (which can be useful for visual inspection or graphical description of the class centroids). This is also described as "reduced-rank linear discriminant analysis" (Hastie et al., 2001, 4.3.3)
- The idea is we seek a reduced set of **canonical** or **discriminant variables** to represent the classification. To implement this, we need
 - the between-class covariance matrix $\mathbf{B}$
 - the within-class covariance matrix $\mathbf{W}$

17.1 Reduced-Rank LDA

We define the between and within-class covariance matrices as:

$$\mathbf{B} = \sum_{k=1}^K (\mu_k - \mu)(\mu_k - \mu)'$$

$\mathbf{B}$ looks at how far away the centroids are from the overall mean for each group

$$\mathbf{W} = \sum_{k=1}^K \sum_{j=1}^{n_k} (x_{kj} - \mu_k)(x_{kj} - \mu_k)'$$

$\mathbf{W}$ looks at how far away from the group centroids.

where

- μ_k is the p -dimensional k -th class mean
- μ is the p -dimensional overall mean
- x_{kj} corresponds to the p -dimensional input vector for the j -th element in the k -th group

We then seek the linear combination $Z = a^T X$ such that the **between-class variance is maximized relative to the within-class variance** (i.e., the between-class variance of Z is $a^T \mathbf{B} a$ and the within-class variance is $a^T \mathbf{W} a$)

We seek the a that maximizes the quotient of $a^T \mathbf{B} a$ divided by $a^T \mathbf{W} a$

The solution to the maximization problem is the “ a ” given by the eigenvector associated with the largest eigen value of $\mathbf{W}^{-1} \mathbf{B}$, which we call a_1

One can show that the next direction a_2 that maximizes this quotient and is orthogonal to a_1 is the eigenvector associated with the next largest eigenvalue, etc.

- The a_l are referred to as **discriminant coordinates** (not discriminant functions) or canonical **variates**

This sequence of discriminant then provides the low-dimensional presentation for the data that separates the populations as much as possible.

In addition to separating the populations, these discriminants can be used to classify.

Let $z_l(x_i) = a_l^T x_i, l = 1, \dots, K - 1$ be the discriminants for the i -th input x_i , which can be shown to have variance of one and are uncorrelated with each other

Then, it can be shown that for an observation, x , the squared distance from $z_l(x)$ to the k -th class centroid (μ_k) is given by $\sum_{l=1}^{K-1} (z_l(x) - \mu_{z_l^k})^2$, where $\mu_{z_l^k}$ is the mean of $\{z_l(x_j)\}$ for observations x_j in the k -th class.

A reasonable classification rule classifies x to class k with the smallest value of this squared distance.

17.2 Flexible Discriminant Analysis

FDA could be confused with functional data analysis

If we have $L = K - 1$ score functions $\theta_l(i)$ (a function in R^1 that assigns scores to the classes so that they can be modeled by “regression” functions) then we seek these scores and the coefficients β_l that minimize the average squared residual

$$\frac{1}{n} \sum_{l=1}^L \left[\sum_{i=1}^n (\theta_l(i) - x_i^T \beta_l)^2 \right]$$

Importantly, **the sequence of β_l 's can be shown to be the same as the discriminant (canonical) vectors (a_l)** in the Reduced-Rank LDA methodology, up to a constant, and the scores (θ_s) correspond to the Z 's

The β coefficients act like linear regression coefficients, which mean we can generalize this procedure as we did in nonparametric regression and replace the regression fits by some more flexible class of functions, $\eta_l(x)$ (e.g., Spline, MARS, etc.)

Hence, a more general classifier is defined by

$$\frac{1}{n} \sum_{l=1}^L \left[\sum_{i=1}^n (\theta_l(i) - \eta_l(x_i))^2 + \lambda J(\eta_l) \right]$$

where we include the regularization term from some forms of nonparametric regression (e.g., Smoothing splines, etc.)

Additive spline is a more flexible approach (called “bruto” can be found in `mda` package in R). More details can be found (Hastie et al., 2001, 12.5.1)

17.3 Penalized Discriminant Analysis

Flexible Discriminant Analysis is a form of regularized discriminant analysis. For example, consider an FDA with a linear regression onto a basis expansion $h(x)$, with a quadratic penalty on the coefficients,

$$\frac{1}{n} \sum_{l=1}^L \left[\sum_{i=1}^n (\theta_l(i) - h^T(x_i) \beta_l)^2 + \lambda \beta_l^T \beta_l \right]$$

where the choice of λ depends on the specific problem,

For example, if $\eta_l(x) = h(x)^T \beta_l$ is an expansion on the spline basis functions, Ω might constrain η_l to be smooth

The idea is that we seek to **give more weight to “smooth” coefficients and less weight to “rough” ones, which can be seen if we consider highly correlated predictors** (such as with images).

This procedure is known as Penalized Discriminant Analysis

17.4 Mixture Discriminant Analysis

We sometimes refer to LDA as a prototype analysis in that there is just a single centroid for each class.

More generally, such a single prototype classifier might be sufficient if the classes are inhomogeneous.

In that case, we might consider a mixture of Gaussian models to present each class.

Recall, a Gaussian mixture model for the k -th class has density

$$P(X|G = k) = \sum_{r=1}^{R_k} q_{kr} \phi(X; \mu_{kr}, \Sigma)$$

where the mixing proportions q_{kr} sum to 1.

In this case, there are R_k mixtures (prototypes) for the k -th class

But, notice that we are assuming the **same covariance matrix** Σ for each class and each prototype (estimating different covariance matrices can be expensive).

Estimating the parameters corresponding to this distribution can be tricky as the optimization can be difficult.

Typically, we use the expectation-maximization (E-M) algorithm (Hastie et al., 2001, 12.7) or a fully-Bayesian implementation.

17.5 Large p, small n Approaches

Consider some procedures that are especially suited for the large p, small n case

- Diagonal Linear Discriminant Analysis

- Nearest Shrunk Centroids

17.5.1 Diagonal Linear Discriminant Analysis

When $p \gg n$, it's not possible to fit a full LDA without some sort of regularization.

A simple type of regularization assumes **independence of the features within class**

Thus, the within-class covariance matrix is just a diagonal matrix

But in reality, this typically isn't true, but we don't have enough data to estimate the covariances anyway

the benefit is that this greatly reduces the number of parameters that must be estimated.

Thus, the diagonal-covariance LDA rule is given by the discriminant score (we classify into the class with the maximum of this score):

$$\delta_k(x^*) = - \sum_{j=1}^p \frac{(x_j^* - \bar{x}_{kj})^2}{s_j^2} + 2 \log \pi_k$$

where

- x^* is the p -dimensional test observation
- s_j is the pooled within-class standard deviation of the j -th feature
- $\bar{x}_{kj}$ is the mean of the n_k values for the j -th feature in class k
- π_k is the prior probability for class k

We say $\bar{x} = (\bar{x}_{k1}, \dots, \bar{x}_{kp})^T$ is the centroid of class K

The first part of the diagonal LDA discriminant score is just the (negative) standardized squared distance of x^* to the k -th centroid, and

The second term corresponds to the prior classification probability for class k

Thus, we classify x^* into the k -th class that has the highest discriminant score. This classifier is often pretty good in high-dimensional settings.

This procedure is equivalent to the nearest centroid classifier

Unfortunately, this classifier has a drawback in that it uses all p of the features, and thus, it prohibits interpretation.

It's possible with additional regularization, to gain interpretability and improve test error, which gives us the Nearest Shrunk Centroids

17.5.2 Nearest Shrunk Centroids

To regularize, we remove features that are not contributing to successful classification.

Hence, we shrink the class means toward the overall mean, for each feature separately.

This gives a regularized version of the Diagonal Linear Discriminant Analysis or, equivalently, a regularized version of the nearest centroid classifier

The procedure is known as the Nearest Shrunk Centroids approach

First, let

$$d_{kj} = \frac{\bar{x}_{kj} - \bar{x}_j}{m_k(s_j + s_0)}$$

where $\bar{x}_j$ is the overall mean for the j -th feature

$m_k^2 = \frac{1}{n_k} - \frac{1}{n}$ and s_0 is a small positive constant (usually the median of the s_j values)

We then shrink the d_{kj} toward 0 using the soft thresholding formula

$$d'_{kj} = \text{sign}(d_{kj})(|d_{kj}| - \Delta)_+$$

where Δ is usually determined by cross-validation

Note:

- $d'_{kj} = 0$ if the argument in $(\cdot)_+$ is less than 0

Then, replacing d_{kj} with d'_{kj} and rearranging gives

$$\bar{x}'_{kj} = \bar{x}_j + m_k(s_j + s_0)d'_{kj}$$

which are the **shrunk centroids**.

These are then used in place of $\bar{x}_{kj}$ in the discriminant score function for the Diagonal Linear Discriminant Analysis

Now, only those **features that have a nonzero d'_{kj} for at least one of the classes contribute to the classification rule**, and thus many features are discarded

Example from (Hastie et al., 2001, 18.4)

17.5.3 Regularized Discriminant Analysis

(Hastie et al., 2001, 4.3.1) describes a hybrid approach between quadratic discriminant analysis and LDA where one shrinks the class-specific covariance matrices to a common pooled covariance matrix, which is known as **regularized discriminant analysis**

$$\hat{\Sigma}_k(\alpha) = \hat{\Sigma}_k + (1 - \alpha)\hat{\Sigma}$$

In the large p , small n situation, a simple variant of this is sometimes used to address the fact that the sample covariance matrix, $\hat{\Sigma}_k$ (or $\hat{\Sigma}$) is singular when $n < p$

In particular, we can consider

$$\hat{\Sigma}_k(\gamma) = \gamma\hat{\Sigma}_k + (1 - \gamma)\text{diag}(\hat{\Sigma}_k)$$

for $\gamma \in [0, 1]$ (i.e., when $\gamma \neq 1$ there is shrinkage of the sample covariance matrix toward its diagonal)

This has a similar flavor to ridge regression 2.3.2.1, but shrinks to a diagonal matrix.

Typically, γ is chosen by cross-validation

Sometimes this is known as L_2 -penalized discriminant analysis

17.5.4 Logistic Regression with Quadratic Regularization

We can also regularize multi-class logistic regression.

Consider a multi-class extension to logistic regression known as the **symmetric multiclass logical model** (multinomial):

$$Pr(G = k | X = x) = \frac{\exp(\beta_{k0} + x^T \beta_k)}{\sum_{l=1}^K \exp(\beta_{l0} + x^T \beta_l)}$$

To deal with the $p \gg n$ problem, we can regularize by maximizing the L_2 -penalized log-likelihood

$$\max_{\{\beta_{k0}, \beta_k\}_1^K} \left[\sum_{i=1}^n \log Pr(g_i | x_i) - \frac{\lambda}{2} \sum_{k=1}^K \|\beta_k\|^2 \right]$$

This automatically forces $\sum_{k=1}^K \hat{\beta}_{kj} = 0$ for $j = 1, \dots, p$

A nice feature of the L_2 penalty is that it tends to shrink **correlated parameters toward each other**.

However, it doesn't shrink the parameters to exactly zero.

17.5.5 Other Regularization Penalties

We could consider an L_1 (lasso 2.3.2.2) penalty for the logistic classification, which would have the advantage in that a subset of the β parameters will be shrunk to exactly 0.

Thus, we want to maximize the following penalized log-likelihood:

$$\max_{\{\beta_{k0}, \beta_k\}_1^K} \left[\sum_{i=1}^n \log \Pr(g_i | x_i) - \lambda \sum_{k=1}^K \sum_{j=1}^p |\beta_{kj}| \right]$$

Another useful penalty is the **elastic net penalty**, which is a mixture of the L_1 and L_2 penalties and takes the form:

$$\sum_{j=1}^p (\alpha |\beta_j| + (1 - \alpha) \beta_j^2)$$

This is a useful penalty because

- the L_2 portion encourages common shrinkage for correlated parameter and
- the L_1 portion encourages sparsity (e.g., many coefficients shrunk to exactly 0)

One can also build penalty function that performs regularization in the context when “nearby” parameters are likely to be dependent e.g., as in spatial, time series, and functional data analysis)

The **fused lasso** is one such example.

The penalty function for equally spaced parameters is then

$$\lambda_1 \sum_{j=1}^p |\beta_j| + \lambda_2 \sum_{j=1}^{p-1} |\beta_{j+1} - \beta_j|$$

the second component can be modified as described in (Hastie et al., 2001, 18.4.2) if the parameters are not equally spaced but are still “neighbors.”

This can be extended to two-dimensional distances in the case of image data. (Hastie et al., 2001, 18.4.2) show an example for a model of genome order of a gene in a tumor sample using what they call the fused lasso signal approximator which solves

$$\min_{\beta \in R^n} \left\{ \sum_{i=1}^n (y_i - \beta_0 - \beta_i)^2 + \lambda_1 \sum_{i=1}^n |\beta_i| + \lambda_2 \sum_{i=1}^{n-1} |\beta_{i+1} - \beta_i| \right\}$$

The interesting thing about this model is that basically each data point has its own mean, so there are $n + 1$ parameters and n samples. (Hastie et al., 2001, 18.8) (basically, we force all points to have the same mean unless the data suggest otherwise)

As it turns out, the support vector classifier 16.2 works well in the large p , small n problem.

Although we can add regularization to the SVM, typically when $p \gg n$ the unregularized support vector classifier works about well or better than the regularized version

Overfitting is not as much of a problem with the SVM because the inherent dependence on a relatively few support vectors

Remember in the multiclass problem, we can consider the “one versus one” or “one versus all” approaches.

Part VIII

UNSUPERVISED LEARNING

Chapter 18

Introduction

Up until now, we covered **supervised learning methods**

That is, we have a set of n observations of p features, $X^T = (X_1, \dots, X_p)$ and we seek to predict the response Y on those n observations

In this sense, the response “supervises” (or teaches) the model to find the correct answer, usually relative to some loss function and tested with cross-validation or a validation sample

Now, we focus on **unsupervised learning** where we still have a set of n observations of p features X , but instead of prediction, our goal is to **find structure in these features**.

We will focus on two general types of unsupervised learning

1. Cluster Analysis (i.e., data segmentation)
2. Dimension Reduction

Chapter 19

Cluster Analysis

Cluster analysis consists of a relatively large set of techniques that are used for finding subgroups, cluster, or segments in a set of features

The idea is that we decide on some way to measure similarity or dissimilarity and try to form clusters that are homogeneous so that observations **in a subgroup are more similar to each other than to other observations not in the subgroup.**

There are a large number of clustering algorithm.

two of the best-known approaches are **K-means clustering** and **hierarchical clustering**

- In the case of K-means clustering, we attempt to partition observations into a pre-determined number of clusters
- In hierarchical, clustering, we either
 - start with each observation as its own cluster and aggregate them, or
 - start with one cluster and successively disaggregate them,
 - in both cases giving a tree or dendrogram representation of the clustering
- **We can choose to perform clustering on the basis of the features to get subgroups or observations, or in terms of the observations to get subgroups of features**
- Critical all clustering approaches is the choice of **distance or proximity or dissimilarity measure between two objects.**

Both Discriminant Analysis and Cluster Analysis are used to classify.

	Cluster Analysis	Discriminant Analysis
Under framework	Unsupervised Learning	Supervised Learning
Dependent variable (category)	No	Yes
Classification rule based on	Closeness	Difference

19.1 Proximity Matrices

It's convenient to represent the data in terms of **proximity** (alikehood of affinity)

This can be thought of in terms of similarity or dissimilarity

A similarity measure increases as two objects are more alike, while a dissimilarity measure decreases as two objects become more alike (e.g., Euclidean distance).

Formally, given two p -dimensional objects x_i and x'_i , a dissimilarity **measure** $d_{ii'}$ satisfies:

1. $d_{ii'} \geq 0 \forall x_i, x'_i$
2. $d_{ii'} = 0$ iff $x_i = x'_i$
3. $d_{ii'} = d_{i'i}$

A measure that satisfies all of these conditions is said to be a **semi-metric**

If it further satisfies the triangle inequality, $d_{ii'} < d_{ik} + d_{i'k}$, then it is said to form a **metric**.

We typically form the $n \times n$ **proximity matrix**, $\mathbf{D}$, which in the case of a dissimilarity measure, has (i, j) th element given by d_{ij}

19.1.1 Dissimilarities based on Attributes

In general, we can define the dissimilarity by summing across the p attributes (i.e., equal weighting):

$$d_{ii'} = D(x_i, x'_i) = \sum_{j=1}^p d_j(x_{ij}, x_{i'j})$$

A very common choice for the dissimilarity metric is squared distance

$$d_j(x_{ij}, x_{i'j}) = (x_{ij} - x_{i'j})^2$$

Issues:

1. We may wish more general distance measures
2. We may have categorical variables
3. We may wish to weight the attributes differently

Fixes

1. A general class of dissimilarities is the **Minkowski metric**

$$d_{ii'} = \left(\sum_{j=1}^p |x_{ij} - x_{i'j}|^\lambda \right)^{1/\lambda}$$

Consider the situation when we have ordinal variables (e.g, categorical variables that have order)

Typically, one converts these to a numerical value then uses a quantitative metric.

For example, say we have M ordinal values, we can then replace these with $(i - 1/2)/M$ for $i = 1, \dots, M$

For unordered categorical (nominal) variables, the degree-of-difference between pairs of values must be specified directly. These are selected in a way to maintain the symmetric nature of the dissimilarity matrix

And it's common to weight the elements of the discrepancy function differently depending on attribute

$$D(x_i, x_{i'}) = \sum_{j=1}^p w_j d_j(x_{ij}, x_{i'j}); \sum_{j=1}^p w_j = 1$$

where w_j is a weight assigned to the j -th attribute

As discussed in (Hastie et al., 2001, 14.3.3), setting the w_j to the same value for each variable does not necessarily give all of the attributes equal weight.

They show that setting w_j proportional to $1/\bar{d}_j$ where

$$\bar{d}_j = \frac{1}{n^2} \sum_{i=1}^n \sum_{i'=1}^n d_j(x_{ij}, x_{i'j})$$

gives all attributes equal influence, which is not always a good thing in terms of finding subgroups

19.2 Combinatorial Cluster Algorithms

Let each observation be uniquely labeled by an integer $i \in \{1, \dots, n\}$. A pre-specified number of clusters $K < n$ is then assumed, and each cluster is labeled by an integer $k \in \{1, \dots, K\}$

Each observation is assigned to one unique cluster.

The assignment is characterized by an **encoder**, $k = C(i)$ which assigns the i -th observation to the k -th cluster

We seek an encoder based on the dissimilarities $d(x_i, x_{i'})$ between every pair of observations.

We have to specify some “**loss**” **function** that we would like to minimize for clustering. One approach is to define

$$W(C) = \frac{1}{2} \sum_{k=1}^K \sum_{C(i)=k} \sum_{C(i')=k} d(x_i, x_{i'})$$

$$B(C) = \frac{1}{2} \sum_{k=1}^K \sum_{C(i)=k} \sum_{C(i') \neq k} d(x_i, x_{i'})$$

$$T = \frac{1}{2} \sum_{i=1}^n \sum_{i'=1}^n d(x_i, x_{i'}) = W(C) + B(C)$$

K-Means Clustering

1. randomly assign a number ($1 \rightarrow K$), to each of the observations, which will be initial cluster assignments.
2. Iterate until the cluster assignments stop changing:
 1. For each of the K clusters, compute the cluster **centroid**. The k -th cluster centroid is the vector of the p feature means for the observations in the k -th cluster
 2. Assign each observation to the cluster whose centroid is close (where closest is defined by Euclidean distance).

Hierarchical Clustering

1. Begin with n observations and a measure (e.g., Euclidean distance) of all the $\binom{n}{2} = n(n-1)/2$ pairwise dissimilarities. Treat each observation as its own cluster.
2. For $i = n, n-1, \dots, 2$:
 1. Examine all pairwise inter-cluster dissimilarities among the i clusters and identify the pair of clusters that are least dissimilar (i.e., most similar). Fuse these two clusters. The dissimilarity between these two clusters indicates the height in the dendrogram at which the fusion should be placed
 2. Compute the new pairwise inter-cluster dissimilarities among the $i-1$ remaining clusters.

Chapter 20

Interpretable Machine Learning

<https://christophm.github.io/interpretable-ml-book/> https://en.wikipedia.org/wiki/Explainable_artificial_intelligence <https://arxiv.org/pdf/1905.08883.pdf>
<https://arxiv.org/abs/2004.14545> <https://www.kaggle.com/learn/machine-learning-explainability> <https://towardsdatascience.com/interperable-vs-explainable-machine-learning-1fa525e12f48>

TCAV:

- <https://github.com/tensorflow/tcav>
- <http://proceedings.mlr.press/v80/kim18d.html>
- <https://arxiv.org/abs/1711.11279>

PAIR:

- <https://pair-code.github.io/facets/>
- <https://research.google/teams/brain/pair/>

Embedding Project:

- <https://projector.tensorflow.org/>

What-If tools:

- <https://pair-code.github.io/what-if-tool/>
- <https://cloud.google.com/ai-platform/prediction/docs/using-what-if-tool>

20.1 SHAP

<https://github.com/slundberg/shap> <https://meichenlu.com/2018-11-10-SHAP-explainable-machine-learning/> <https://bjlkeng.github.io/posts/model-explanability-with-shapley-additive-explanations-shap/>

20.2 Video Classification

<https://hackernoon.com/continuous-video-classification-with-tensorflow-inception-and-recurrent-nets-250ba9ff6b85> <https://cloud.google.com/video-intelligence> <https://arxiv.org/pdf/1609.06782.pdf> <https://www.tubefilter.com/2019/02/12/taxonomy-youtube-videos-develop-original-content-that-works/> <https://medium.com/swlh/converting-news-into-video-stories-5d8d4fc5a32b> <https://towardsdatascience.com/introduction-to-video-classification-6c6acbc57356> <https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0228579>

20.3 Image Memorability

(AMNET)

https://www.researchgate.net/publication/335603628_Image_Memorability_Using_Diverse_Visual_Features_and_Soft_Attention https://www.researchgate.net/publication/308812354_Deep_Learning_for_Image_Memorability_Prediction_the_Emotional_Bias <https://arxiv.org/pdf/2005.02969.pdf> <https://github.com/dazcona/memorability> <https://arxiv.org/abs/1804.03115> https://www.researchgate.net/publication/324387176_AMNet_Memorability_Estimation_with_Attention http://ceur-ws.org/Vol-2283/MediaEval_18_paper_24.pdf <https://jov.arvojournals.org/article.aspx?articleid=2771173> https://openaccess.thecvf.com/content_cvpr_2018/papers/Fajtl_AMNet_Memorability_Estimation_CVPR_2018_paper.pdf <https://ieeexplore.ieee.org/document/8578764>

<https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=7382237&tag=1> <https://people.umass.edu/~yangzhou/>

<http://people.csail.mit.edu/khosla/projects/regionmem/> <https://github.com/adikhosla/feature-extraction> <https://expertprogrammanagement.com/2019/12/elaboration-likelihood-model/>

Part IX

DEEP LEARNING

Chapter 21

Introduction

Note: Even though deep learning can be a powerful tool when it comes to fitting a function to data. Unless you have a causal diagram/model, its knowledge is still not transferable. For example, when you get a new population, you will have to train your deep learning model again.

(LeCun et al., 2015) for a review. (LeCun et al., 2015) defines a deep-learning architecture as “a multilayer stack of simple modules, all (or most) of which are subject to learning, and many of which compute non-linear input-output mappings”.

Chapter 22

Neural Network

Neuron

$$z = a_1 w_1 + \dots + a_k w_k + b$$

Sigmoid function

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

Different connections lead to different network structures

The neurons have different values of weights and biases.

Fully connected Feedforward Network

Framework

Step 1: A set of function

Step 2: Goodness of function f

Step 3: Pick the best function

22.1 Step 1: A set of function

Deep = many hidden layers

Universality Theorem

Any continuous, function f

$$f : R^N \rightarrow R^M$$

can be realized by a network with one hidden layer (given enough hidden neurons).

Output layer

- Softmax layer as the output layer

$$\sigma(\vec{z})_i = \frac{e^{z_i}}{\sum_{j=1}^k e^{z_j}}$$

– where

- * σ = softmax
- * $\vec{z}$ = input vector
- * e^{z_i} = standard exponential function for input vector
- * K = number of classes in the multi-class classifier
- * e^{z_j} = standard exponential function for output vector

– probability:

- * $1 > y_i > 0$
- * $\sum_i y_i = 1$

– Monotonicity

– Non-locality

22.2 Step 2: Goodness of function

Loss can be square error or cross entropy between the network output and target.

With softmax, the summation of all the outputs would be one.

A good function should make the loss of all examples as small as possible

Total Loss:

$$L = \sum_{r=1}^R l_r$$

Find a function in function set (the network parameters θ^*) that minimizes total loss.

22.3 Step 3: Pick the best function

Find network parameters θ^* that minimize total loss L .

Gradient Descent

Network parameters $\theta = (w_1, w_2, \dots, b_1, b_2, \dots)$

- Pick an initial value for w
- Compute $\frac{\partial L}{\partial w}$,
 - if negative, increase w
 - if positive, decrease w
- $w_{new} = w_{old} - \eta \frac{\partial L}{\partial w}$ where η is the learning rate.
- You have problems with:
 - plateau: $\frac{\partial L}{\partial w} \approx 0$
 - saddle point: $\frac{\partial L}{\partial w} = 0$
 - Stuck at local minima: $\frac{\partial L}{\partial w} = 0$
- solution:
 - Different initial point to reach different minima, so different results.

Backpropagation

- is an efficient way to compute $\partial L / \partial w$

22.4 Recipe of Deep learning

1. Not good result on training data
 - Choosing proper loss
 - Mini-batch
 - New activation function
 - Adaptive Learning Rate
 - Momentum
2. Not good result on testing data
 - Early Stopping
 - Regularization
 - Dropout
 - Network Structure

22.4.1 Choosing proper loss

When using softmax output layer, choose cross entropy. Problem

22.4.2 Mini-batch

```
model.fit(x_train,y_train,batch_size = 100,nb_epoch = 20) # 100 examples in a mini-batch
# repeat 20 times
```

- Randomly initialize network parameters

- One epoch
 - Pick the 1st batch $L' = l^1 + l^{31} + ..$ Update parameters once.
 - Pick the 2nd batch $L'' = l^2 + l^{16} + ...$ Update parameters once
 - Until all mini-batches have been picked
- Repeat the above process
- Mini-batch is faster (with mini-batch. if there are 20 batches, update 20 times in one epoch).

22.4.3 New activation function

- Rectified Linear Unit (ReLU)

Reason:

1. Fast to compute
2. Biological reason
3. Infinite sigmoid with different biases
4. Vanishing gradient problem.

Parametric ReLU (PReLU) (Glorot et al., 2011)

ReLU - Variant

- Leaky ReLU
- Parametric ReLU

22.4.4 Adaptive Learning Rate

- Set the learning rate η carefully
- If learning rate is too large
- Total loss may not decrease after each update.
- If learning rate is too small
- Training would be too slow
- Popular and simple idea: reduce the learning rate by some factor every few epochs.
 - At the beginning, we are far from the destination, so we use larger learning rate
 - After several epochs, we are close to the destination, so we reduce the learning rate
 - E.g., 1/t decay $\eta^t = \eta\sqrt{t+1}$
- Learning rate cannot be one-size-fits-all
 - Giving different parameters different learning rates.

- η is unit-dependent (from 0 to 1)

Adagrad

- Original: $w \leftarrow w - \eta \partial L / \partial w$
- Adagrad: $w \leftarrow -\eta_w \partial L / \partial w$
 - where η_w is parameter dependent learning rate.

$$\eta_w = \frac{\eta}{\sqrt{\sum_{i=1}^t (g^i)^2}}$$

- η is constant
- g^i is $\partial L / \partial w$ obtained at the i-th update.
- Learning rate is smaller and smaller for all parameters
- Smaller derivatives, larger learning rate, and vice versa.

link

22.4.5 Momentum

Hard to find optimal network parameters

Movement = negative of $\partial L / \partial w$ + Momentum

```
# RMSProp (Advanced Adagrad) + Momentum
model.compile(loss='categorical_crossentropy',optimizer = SGD(lr=0.1),metrics = ['accuracy'])
model.compile(loss='categorical_crossentropy',optimizer=Adam(),metrics = ['accuracy'])
```

22.4.6 Early Stopping

- training data and testing data can be different.
- Learning target is defined by the training data
- The parameters achieving the learning target do not necessary have good results on the testing data.

22.4.7 Regularization

22.4.8 Dropout

Training

- each time before updating the parameters
 - each neuron has p% to dropout.
 - * The structure of the network is changed.
 - Using the new network for training
 - * For each min-batch, we resample the dropout neurons.

Testing

- No dropout
 - If the dropout rate training is $p\%$, all the weights times $1-p\%$
 - Assume that the dropout rate is 50%. If a weight $w = 1$ by training, set $w = 0.5$ for testing.

Dropout is a kind of ensemble.

- training of dropout: M neurons $\rightarrow 2^M$ possible networks.
- Using one mini-batch to train one network
- Some parameters in the network are shared.
- Training of dropout: assume dropout rate is 50%
- Testing of dropout: weight

```
model = Sequential()
model.add(Dense(input_dim = 28*28,output_dim = 500))
model.add(Activation('sigmoid'))
model.add(dropout(0.8))
model.add(Dense (output_dim=500))
model.add(Activation('sigmoid'))
model.add(dropout(0.8))
model.add(Dense(output_dim = 10))
model.add(Activation('softmax'))
```

22.4.9 Network Structure

Convolutional Neural Network (CNN)

- Widely used in image processing
- use in the first step for deep learning

The Whole CNN

- Property 1: Some patterns are much smaller than the whole image
- Property 2: The same patterns appear in different regions.
- Property 3: Subsampling the pixels will not change the object.

CNN in Keras: only modified the network structure and input format (vector to 3-D tensor)

```
model2.add(Convolution2D(25,3,3),input_shape= (1,28,28))
# 25 3x3 filters; 1:black/weight, 3: RFB, 28x28 pixels
model2.add(MaxPooling2D(2,2))
model2.add(Convolution2D(50,3,3))
```

```
model2.add(MaxPooling2D(2,2))
model2.add(Flatten())
model2.add(Dense(output_dim = 100))
model2.add(Activation('relu'))
model2.add(Dense(output_dim=10))
model2.add(Activation('softmax'))
```

will we always have to know our classes? Yes, and you have to retrain your model all the time. Or you could use
Transfer learning in deep learning

Neural topic modeling

graph representation learning

GNN

DGL

recurrent network

BERT model

Chapter 23

Overview

- What is deep learning? Input to output via layers of representation
- What are layers? A layer is a geometric transformation function on the data that goes through it Weights determine the data transformation behavior of a layer

Learning Representation

- Transforming input data into useful representation

The “deep” in deep learning

- multiple layers
- Other possibly more appropriate names for the field:
 - Layered representations learning
 - Hierarchical representations learning
 - Chained geometric transformation learning

New problem domains for R:

- Computer vision
- Computer speech recognition
- Reinforcement learning applications

How do we train deep learning models?

- Basic of machine learning algorithms
- Machine learning vs. statistical modeling
- MNIST example
 - Model definition in R
 - Layers of representation

- The training loop

Machine learning algorithms

Learning model parameters via exposure to many example data points

Statistics: Often focused on inferring the process by which data is generated

Machine Learning: Principally focused on predicting future data

Deep learning frontiers

- Computer vision
- Natural language processing
- Time series
- Biomedical

Chapter 24

Tensor Flow and R

TensorFlow APIs

- Keras API
- Estimator API
- Core API

24.1 R packages

TensorFlow APIs

- `keras`: Interface for neural networks, with a focus on enabling fast experimentation.
- `tfestimators`: Implementations of common model types such as regressors and classifiers.
- `tensorflow`: *Low level interface to the TensorFlow computational graph.*
- `tfdatasets`: Scalable input pipelines for TensorFlow models.

Supporting Tools

- `tfruns` : Track , visualize, and manage TensorFlow training runs and experiments
- `tfdeploy` : Tools designed to make exporting and serving TensorFlow models straightforward.
- `cloudml` : R interface to Google Cloud Machine Learning Engine.

R interface to Keras

- Step by step example
- Keras layers

- Compiling models
- Losses, Optimizers, and Metrics
- More examples

24.2 Keras layers

65 layers available

- Dense layers: classic “fully connected” neural network layers
- Convolutional layers: “Filters for learning local patterns in data
- Recurrent layers: Layers that maintain state based on on previously seen data
- Embedding layers: Vectorization of text that reflects semantic relationships between words

Chapter 25

Compiling models

Model compilation prepares the model for training by:

1. Converting the layers into a TensorFlow graph
2. Applying the specified loss function and optimizer
3. Arranging for the collection of metrics during training

Chapter 26

Losses, Optimizers, and Metrics

All available at Keras for R cheatsheet

Example of Image classificaiton

Chapter 27

NYU

Materials in this chapter is based on NYU Deep Learning

Deep in Deep Learning means multi-layers. “A deep network has several layers and uses them to build a hierarchy of features of increasing complexity”

Supervised Learning (= Function Optimization): With inputs and outputs, you train the machine to tweak its parameter to have the correct outputs from inputs. In “traditional machine learning”, you have to engineer **feature extractor**, but in deep learning you can multi-layers feature extractor can be trained. And all layers are non-linear, because if they are linear, then all layers would be collapse into one layer.

Natural Data is compositional. Hence, it is efficiently representable hierarchically.

Revolutions in Deep Learning:

- Speech Recognition: 2010
- Image Recognition: 2013
- Natural Language Processing (NLP): 2015

Deep Learning = Learning Representations/ Features Representation means you turn raw data into something useful

Image Recognition:

Pixel, edge, texture, motif, part, object

Text

Character, group, word group, clause, sentence, story

Speech

Sample, spectral band, sound, phone, phoneme, word.

Difference between Deep Learning Support-Vector Machines (SVM)

SVM is a 2-layer neural net, where

- layer 1 = templates
- layer 2 = linear classifier.

where Deep Learning is “A deep network has several layers and uses them to build a hierarchy of features of increasing complexity”

Deep Machines are more efficient than SVM in representing certain classes of functions.

The Manifold Hypothesis “states that real-world high-dimensional data lie on low-dimensional manifolds embedded within the high-dimensional space”

Difference between Deep Learning and PCA (Principal Components Analysis) is that Deep learning uses non-linear function, while PCA uses linear function.

Chapter 28

Dimension Reduction

Chapter 29

Applications

```
# Step 1: define a set of function
model = Sequential()
model.add(Dense(input_dim = 28*28,
output_dim = 500))

model.add(Activation('sigmoid'))
model.add(Dense(output_dim = 500))
model.add(Activation('sigmoid'))
model.add(Dense(output_dim=10))
model.add(Activation('softmax'))

# Step 2: goodness of function
model.compile(loss = 'mse',optimizer = SGD(lr=0.1),metrics = ['accuracy'])

# Step 3: Pick the best function
# Step 3.1: Configuration
model.compile(loss = 'mse',optimizer=SGD(lr=0.1),metrics = ['accuracy'])

# Step 3.2: Find the optimal network parameters
model.fit(x_train,y_train,batch_size = 100,nb_epoch=20) # x_train = training data (images)
# y_train: labels (digits)

# Save and load models
# http://keras.io/getting-started/faq/#how-can-i-save-a-keras-model

# How to use the neural network (testing):
# case 1
score = model.evaluate(x_test,y_test)
```

```
print('Total loss on Testing Set:',score[0])
print('Accuracy of Testing set:',score[1])

# case 2
result = model.predict(x_test)
```

Part X

**REINFORCEMENT
LEARNING**

Chapter 30

Contextual Bandits

https://vowpalwabbit.org/tutorials/contextual_bandits.html

Other terms:

- Multi-world testing
- Associative bandits
- Learning with partial feedback
- learning with bandit feedback
- bandits with side information
- Multi-class classification with bandit feedback
- associative reinforcement learning
- one-step reinforcement learning

Applications:

- A/B testing
- Tiktok algorithm
- Clinical trials

Chapter 31

Research Methods

31.1 But-for World

(Mahajan et al., 1993) (Landsman and Stremersch, 2020)

Chapter 32

Model Card

Originate from the (Mitchell et al., 2019) Google's Model Card. This is scientists' attempt to bring transparency, fairness and equity to other stakeholders that are affected by AI models. Two specific examples that serve as a template are:

- Face Detection
- Object Detection

Although we have pre-defined template in Python and example, and deploy, there doesn't seem to be one in R

Part XI

MISCELLANEOUS

Chapter 33

Model Tuning

```
library("finetune")  
knitr::opts_chunk$set(eval = FALSE)
```

(Kuhn, 2014) for detail

- Grid Search
- Annealing Search

Chapter 34

Model Stacking

Codes are from stack vignettes basic and

Source Code

Steps in doing model staking:

1. Define your candidate (ensemble) models
2. Initialize a `data_stack` object with `stacks()`
3. Add candidate candidate models using `add_candidates()`
4. Evaluate the combined predictions `blend_predictions()`
5. `'fit_members()'` allows you to fit models with non-zero staking coefficients
6. `predict()` new data

```
library(tidymodels)
library(stacks)
library(dplyr)
library(purrr)
library(tune)
library(ggplot2)
```

```
data("tree_frogs")
# subset the data
# drops <- c("clutch", "hatched")
# tree_frogs[ , !(names(tree_frogs) %in% drops)]
```

```
# subset the data
tree_frogs <- tree_frogs %>%
  filter(!is.na(latency)) %>%
  select(-c(clutch, hatched))
```

```
ggplot(tree_frogs) +
  aes(x = age, y = latency, color = treatment) +
  geom_point() +
  labs(x = "Embryo Age (s)", y = "Time to Hatch (s)", col = "Treatment")
ggplot(tree_frogs) +
  aes(x = treatment, y = age, color = reflex) +
  geom_jitter() +
  labs(y = "Embryo Age (s)",
       x = "treatment",
       color = "Response")
```

```
# some setup: resampling and a basic recipe
set.seed(1)

tree_frogs_split <- initial_split(tree_frogs)
tree_frogs_train <- training(tree_frogs_split)
tree_frogs_test  <- testing(tree_frogs_split)

folds <- rsample::vfold_cv(tree_frogs_train, v = 5)

tree_frogs_rec <-
  recipe(reflex ~ ., data = tree_frogs_train) %>%
  step_dummy(all_nominal(), -reflex) %>%
  step_zv(all_predictors())

tree_frogs_wflow <-
  workflow() %>%
  add_recipe(tree_frogs_rec)
ctrl_grid <- control_stack_grid() # same control settings
ctrl_res <- control_stack_resamples()
```

Add random forest model

```
rand_forest_spec <-
  rand_forest(mtry = tune(),
              min_n = tune(),
              trees = 500) %>%
  set_mode("classification") %>%
  set_engine("ranger")

rand_forest_wflow <-
  tree_frogs_wflow %>%
  add_model(rand_forest_spec)
```



```

rand_forest_res <-
  tune_grid(
    object = rand_forest_wflow,
    resamples = folds,
    grid = 10,
    control = ctrl_grid
  )

```

Add neural network model

```

nnet_spec <-
  mlp(hidden_units = tune(),
       penalty = tune(),
       epochs = tune()) %>%
  set_mode("classification") %>%
  set_engine("nnet")

nnet_rec <-
  tree_frogs_rec %>%
  step_normalize(all_predictors())

nnet_wflow <-
  tree_frogs_wflow %>%
  add_model(nnet_spec)

nnet_res <-
  tune_grid(
    object = nnet_wflow,
    resamples = folds,
    grid = 10,
    control = ctrl_grid
  )

```

Build stacks

```

tree_frogs_model_st <-
  # initialize the stack
  stacks() %>%
  # add candidate members
  add_candidates(rand_forest_res) %>%
  add_candidates(nnet_res) %>%
  # determine how to combine their predictions
  blend_predictions() %>%
  # fit the candidates with nonzero stacking coefficients
  fit_members()

tree_frogs_model_st

```

Check performance

```
theme_set(theme_bw())
autoplot(tree_frogs_model_st)
autoplot(tree_frogs_model_st, type = "members")
autoplot(tree_frogs_model_st, type = "weights") # to see penalty
```

See parameters

```
collect_parameters(tree_frogs_model_st, "rand_forest_res")
```

Predict new data

```
tree_frogs_pred <-
  tree_frogs_test %>%
  bind_cols(predict(tree_frogs_model_st, ., type = "prob"))
```

ROC AUC

```
yardstick::roc_auc(tree_frogs_pred,
  truth = reflex,
  contains(".pred_"))
```

Compare predictions with actual observations

```
# ggplot(tree_frogs_pred) +
#   aes(x = latency,
#       y = .pred) +
#   geom_point() +
#   coord_obs_pred()
```

Comparison between stacks and member model

```
member_preds <-
  tree_frogs_test %>%
  dplyr::select(latency) %>%
  bind_cols(predict(tree_frogs_model_st, tree_frogs_test, members = TRUE))
```

Compare different model predictions

```
map_dfr(setNames(colnames(tree_frogs_pred), colnames(tree_frogs_pred)),
  ~ mean(tree_frogs_pred$reflex == pull(tree_frogs_pred, .x))) %>%
  pivot_longer(c(everything(), -reflex))
```

Chapter 35

Data Storage

	Structured Databases	Unstructured Databases
Structure	every element has the same number of attributes (i.e., column)	Different elements can have different number of attributes (more efficient)
Latency	Slower	Faster
Ease of learning	Easy	Harder, more steep
Storage Volume	Not appropriate for strong Big Data	handle Big Data well
Data Types	numerical, texture	any (e.g., audio, video)
Examples	MySQL, PostgreSQL	MongoDB, Neo4j

35.1 Structured Databases

35.2 Unstructured Databases

35.2.1 MongoDB

Setup Enviroment

Code to access virtual machine: Cd m103 Cd m103-vagrant-env

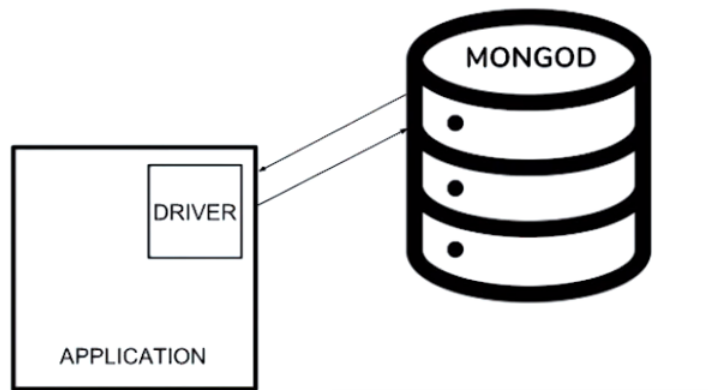
Open virtual machine: vagrant up Bring up environment: vagrant provision
Connect to ssh: vagrant ssh Version: mongod --version Make sure box is running properly: validate_box Exit the box: exit Stop the box virtual machine: vagrant halt

Mongod is the main daemon processing of mongDB Run mongod use driver to communicate with application.

Can access mongod or mongo: In Mongo: - show dbs: show database - exit: use admin → db.shutdownServer() → exit

In Mongod: – port : tell MongoDB which port to listen to (default: 27017) – dbpath the location to store database files – logpath the location of a logfile for mongod to log information to. – fork a flag to tell mongod to run as a background process, rather than as an active process which blocks the shell

What's a daemon?



If set fork path, must set log path Set up mongod on port 30000 with a dbpath to local directory called first_mongod and fork the process so it would not block the shell. `mkdir first_mongod mongod -port 30000 -dbpath first_mongod -logpath first_mongod/mongod.log -fork`

then connect to mongo: `mongo -port 30000`

```

import pymongo

### Create a Database
myclient = pymongo.MongoClient("mongodb://localhost:27017/")

mydb = myclient["mydatabase"]

### Check if database exists

#check if a database exist by listing all databases in you system
print(myclient.list_database_names())
  
```

```

#check a specific database by name
dblist = myclient.list_database_names()
if "mydatabase" in dblist:
    print("The database exists.")

### Insert Into Collection
#The first parameter of the insert_one() method is a dictionary containing the name(s) and value(s)

myclient = pymongo.MongoClient("mongodb://localhost:27017/")
mydb = myclient["mydatabase"]
mycol = mydb["customers"]

mydict = { "name": "John", "address": "Highway 37" }

x = mycol.insert_one(mydict)

### Return the _id field
mydict = { "name": "Peter", "address": "Lowstreet 27" }

x = mycol.insert_one(mydict)

print(x.inserted_id)

### Insert Multiple documents
import pymongo

myclient = pymongo.MongoClient("mongodb://localhost:27017/")
mydb = myclient["mydatabase"]
mycol = mydb["customers"]

mylist = [
    { "name": "Amy", "address": "Apple st 652"},
    { "name": "Hannah", "address": "Mountain 21"},
    { "name": "Michael", "address": "Valley 345"},
    { "name": "Sandy", "address": "Ocean blvd 2"},
    { "name": "Betty", "address": "Green Grass 1"},
    { "name": "Richard", "address": "Sky st 331"},
    { "name": "Susan", "address": "One way 98"},
    { "name": "Vicky", "address": "Yellow Garden 2"},
    { "name": "Ben", "address": "Park Lane 38"},
    { "name": "William", "address": "Central st 954"},
    { "name": "Chuck", "address": "Main Road 989"},
    { "name": "Viola", "address": "Sideway 1633"}
]

```

```

x = mycol.insert_many(mylist)

#print list of the _id values of the inserted documents:
print(x.inserted_ids)

### Insert Multiple Documents, with Specified IDs
import pymongo

myclient = pymongo.MongoClient("mongodb://localhost:27017/")
mydb = myclient["mydatabase"]
mycol = mydb["customers"]

mylist = [
    { "_id": 1, "name": "John", "address": "Highway 37"},
    { "_id": 2, "name": "Peter", "address": "Lowstreet 27"},
    { "_id": 3, "name": "Amy", "address": "Apple st 652"},
    { "_id": 4, "name": "Hannah", "address": "Mountain 21"},
    { "_id": 5, "name": "Michael", "address": "Valley 345"},
    { "_id": 6, "name": "Sandy", "address": "Ocean blvd 2"},
    { "_id": 7, "name": "Betty", "address": "Green Grass 1"},
    { "_id": 8, "name": "Richard", "address": "Sky st 331"},
    { "_id": 9, "name": "Susan", "address": "One way 98"},
    { "_id": 10, "name": "Vicky", "address": "Yellow Garden 2"},
    { "_id": 11, "name": "Ben", "address": "Park Lane 38"},
    { "_id": 12, "name": "William", "address": "Central st 954"},
    { "_id": 13, "name": "Chuck", "address": "Main Road 989"},
    { "_id": 14, "name": "Viola", "address": "Sideway 1633"}
]

x = mycol.insert_many(mylist)

#print list of the _id values of the inserted documents:
print(x.inserted_ids)

### Find One

import pymongo

myclient = pymongo.MongoClient("mongodb://localhost:27017/")
mydb = myclient["mydatabase"]
mycol = mydb["customers"]

x = mycol.find_one()

print(x)

```

Find All

```
import pymongo

myclient = pymongo.MongoClient("mongodb://localhost:27017/")
mydb = myclient["mydatabase"]
mycol = mydb["customers"]

for x in mycol.find():
    print(x)
```

Return Only Some Fields

```
import pymongo

myclient = pymongo.MongoClient("mongodb://localhost:27017/")
mydb = myclient["mydatabase"]
mycol = mydb["customers"]

for x in mycol.find({}, { "_id": 0, "name": 1, "address": 1 }):
    print(x)
```

Example exclude "address"

```
import pymongo

myclient = pymongo.MongoClient("mongodb://localhost:27017/")
mydb = myclient["mydatabase"]
mycol = mydb["customers"]

for x in mycol.find({}, { "address": 0 }):
    print(x)
```

get error if you specify both

```
import pymongo

myclient = pymongo.MongoClient("mongodb://localhost:27017/")
mydb = myclient["mydatabase"]
mycol = mydb["customers"]

for x in mycol.find({}, { "name": 1, "address": 0 }):
    print(x)
```

Filter the Result

```
import pymongo

myclient = pymongo.MongoClient("mongodb://localhost:27017/")
mydb = myclient["mydatabase"]
mycol = mydb["customers"]

myquery = { "address": "Park Lane 38" }

mydoc = mycol.find(myquery)

for x in mydoc:
    print(x)
```

Advanced Query

```
import pymongo

myclient = pymongo.MongoClient("mongodb://localhost:27017/")
mydb = myclient["mydatabase"]
mycol = mydb["customers"]

myquery = { "address": { "$gt": "S" } } # o find the documents where the "address" fie

mydoc = mycol.find(myquery)

for x in mydoc:
    print(x)
```

Filter With Regular Expressions

```
import pymongo

myclient = pymongo.MongoClient("mongodb://localhost:27017/")
mydb = myclient["mydatabase"]
mycol = mydb["customers"]

myquery = { "address": { "$regex": "^S" } } #Find documents where the address starts w

mydoc = mycol.find(myquery)

for x in mydoc:
    print(x)
```


Sort the Result

```
import pymongo
```

```
myclient = pymongo.MongoClient("mongodb://localhost:27017/")
mydb = myclient["mydatabase"]
mycol = mydb["customers"]
```

```
mydoc = mycol.find().sort("name") #method takes one parameter for "fieldname" and one parameter j
```

```
for x in mydoc:
    print(x)
```

Sort Descending

```
import pymongo
```

```
myclient = pymongo.MongoClient("mongodb://localhost:27017/")
mydb = myclient["mydatabase"]
mycol = mydb["customers"]
```

```
mydoc = mycol.find().sort("name", -1) #Sort the result reverse alphabetically by name:
```

```
for x in mydoc:
    print(x)
```

Delete Document

```
import pymongo
```

```
myclient = pymongo.MongoClient("mongodb://localhost:27017/")
mydb = myclient["mydatabase"]
mycol = mydb["customers"]
```

```
myquery = { "address": "Mountain 21" } #delete document with address "Mountain 21"
```

```
mycol.delete_one(myquery)
```

Delete Many Documents

```
import pymongo
```

```
myclient = pymongo.MongoClient("mongodb://localhost:27017/")
mydb = myclient["mydatabase"]
mycol = mydb["customers"]
```

```

myquery = { "address": {"$regex": "^S"} } #Delete all documents where the address starts with S

x = mycol.delete_many(myquery)

print(x.deleted_count, " documents deleted.")

### Delete All Documents in a Collection
import pymongo

myclient = pymongo.MongoClient("mongodb://localhost:27017/")
mydb = myclient["mydatabase"]
mycol = mydb["customers"]

x = mycol.delete_many({}) # Delete all documents in the "customers" collection:

print(x.deleted_count, " documents deleted.")

### Delete Collection

import pymongo

myclient = pymongo.MongoClient("mongodb://localhost:27017/")
mydb = myclient["mydatabase"]
mycol = mydb["customers"] #Delete the "customers" collection:

mycol.drop()
# The drop() method returns true if the collection was dropped successfully, and false otherwise

### Update Collection
# If the query finds more than one record, only the first occurrence is updated.
import pymongo

myclient = pymongo.MongoClient("mongodb://localhost:27017/")
mydb = myclient["mydatabase"]
mycol = mydb["customers"]

myquery = { "address": "Valley 345" } #Change the address from "Valley 345" to "Canyon 123"
newvalues = { "$set": { "address": "Canyon 123" } }

mycol.update_one(myquery, newvalues)

#print "customers" after the update:
for x in mycol.find():
    print(x)

```

```
### Update Many
#o update all documents that meets the criteria of the query, use the update_many() method.

myclient = pymongo.MongoClient("mongodb://localhost:27017/")
mydb = myclient["mydatabase"]
mycol = mydb["customers"]

myquery = { "address": { "$regex": "^S" } } #Update all documents where the address starts with S
newvalues = { "$set": { "name": "Minnie" } }

x = mycol.update_many(myquery, newvalues)

print(x.modified_count, "documents updated.")

### Limit the Result
# Limit the result to only return 5 documents:
import pymongo

myclient = pymongo.MongoClient("mongodb://localhost:27017/")
mydb = myclient["mydatabase"]
mycol = mydb["customers"]

myresult = mycol.find().limit(5)

#print the result:
for x in myresult:
    print(x)
```


Chapter 36

API

36.1 R

Example

check Blair's talk

<https://www.rplumber.io/> <https://community.rstudio.com/tag/plumber>
<https://github.com/rstudio/webinars> <https://github.com/meztez/rapidoc>
<https://github.com/meztez/plumberDeploy> <https://github.com/rstudio/plumber> <https://github.com/sol-eng/plumberExamples>

Chapter 37

BERT

<https://github.com/google-research/bert>

<https://www.freecodecamp.org/news/google-bert-nlp-machine-learning-tutorial/>

<https://jalammar.github.io/a-visual-guide-to-using-bert-for-the-first-time/>

<https://towardsml.com/2019/09/17/bert-explained-a-complete-guide-with-theory-and-tutorial/>

Chapter 38

Parallel Computing

According to *Matt Jones*, computation is bound by

- CPU (take much CPU time)
- Memory: take memory
- I/O: take time to read/write from disk
- Network: transfer time

Parallel computing can help with CPU-bound

By default, computer programming software (e.g., Python and R) usually use one core processor for serial computation. However, with big data, the iterative process can take a lot of time. If we don't have to data analysis in a sequential step, a in an analysis or can break the big data set into smaller chunks and process them simultaneously, it will increase the speed of the analysis process.

Hence, we have parallel computing that utilizes the philosophy of “divide and conquer.” Hence, parallel computing

There are two actors in a parallel backend

1. Director: who tells workers what to do, and manages computing resources (processors, RAM memory, network bandwidth, etc.)
2. Workers: who execute what the director told them to do

A parallel backend is the way workers and director communicate

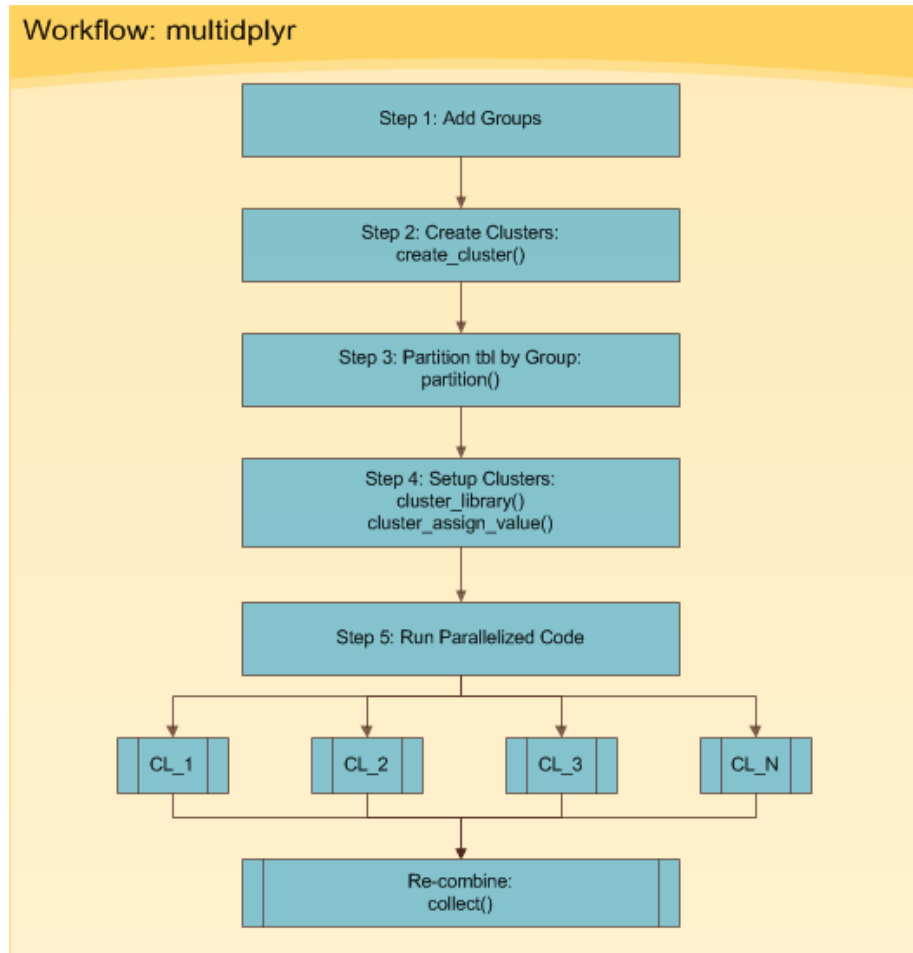
There are 2 types of parallel backends

Type Available	Work in clusters	Col4	Col5
FORK Unix machines (Linux, Mac, etc)	Nope	Workers share the same environment (data, loaded packages, functions) as the director	
PSOCK Unix and Windows (default for <code>foreach</code>)	Yep	environment is copied to the environment of each worker	

FORK is more efficient because there is only one environment and only outputs are sent back to the director (40% faster than PSOCK)

1. FORK
2. PSOCK

38.1 multidplyr



Typically, there are two situations that we can utilize this process:

1. A big dataset that needs to be broken up into small ones and perform analysis on each
2. A data set that needs to perform multiple analyses (e.g., models).

Old example provided by Business Science (no longer available)

Example provided by the package's author

```
library(multidplyr)
```

```
cluster <- new_cluster(parallel::detectCores() - 2) # use all cores except 2  
cluster_library(cluster, "dplyr") # you have to load your libraries to be used across workers (c
```

First way (more efficient)

read different files on each worker (cluster)

```
# Create a filename vector containing different values on each worker
cluster_assign_each(cluster, filename = c("a.csv", "b.csv", "c.csv", "d.csv"))

# Use vroom to quickly load the csvs
cluster_send(cluster, my_data <- vroom::vroom(filename))

# Create a party_df using the my_data variable on each worker
my_data <- party_df(cluster, "my_data")
```

If you put your files separately in a folder already, you can use

```
files <- dir(path, full.names = TRUE)
cluster_assign_partition(cluster, files = files)
```

Second way

If you load the dataset already, then you have to **partition** them across workers (clusters).

You can also **group_by** before **partition** if your analysis requires observations to be in the same group

```
library(nycflights13)

flight_dest <- flights %>% group_by(dest) %>% partition(cluster)
flight_dest

flight_dest %>%
  summarise(delay = mean(dep_delay, na.rm = TRUE), n = n()) %>%
  collect()
```

For finished computation across cores, use `collect()` to get the data back to one dataframe or one place

```
rm(cluster)
detach(package:multidplyr)
rm(list = ls())
```

38.2 parallel

Example by Blas M. Benito

```
#automatic install of packages if they are not installed already
list.of.packages <- c(
  "foreach",
  "doParallel",
```

```

    "ranger",
    "palmerpenguins",
    "tidyverse",
    "kableExtra"
  )
new.packages <- list.of.packages[!(list.of.packages %in% installed.packages()[,"Package"])]

if(length(new.packages) > 0){
  install.packages(new.packages, dep=TRUE)
}

#loading packages
for(package.i in list.of.packages){
  suppressPackageStartupMessages(
    library(
      package.i,
      character.only = TRUE
    )
  )
}

#loading example data
data("penguins")

```

Set up for local computer

```

n.cores <- parallel::detectCores() - 2 # use all cores except 2

#create the cluster
my.cluster <- parallel::makeCluster(
  n.cores,
  type = "PSOCK" # alternatively, we can use FORK
)

#check cluster definition (optional)
print(my.cluster)

#register it to be used by %dopar%
doParallel::registerDoParallel(cl = my.cluster)

#check if it is registered (optional)
foreach::getDoParRegistered()

#how many workers are available? (optional)
foreach::getDoParWorkers()

```

```
x <- foreach(
  i = 1:10,
  .combine = 'c'
) %dopar% {
  sqrt(i)
}

# when done, stop cluster

parallel::stopCluster(cl = my.cluster)
```

Example by Jens Moll-Elsborg

parLapply equivalent to lapply

```
# Generate data
data <- 1:1e9
data_list <- list("1" = data,
                  "2" = data,
                  "3" = data,
                  "4" = data)

library(parallel)
cl <- parallel::makeCluster(detectCores() - 2)
parallel::parLapply(cl,
                    data_list,
                    mean)

# Close cluster
parallel::stopCluster(cl)
```

Another example by Blas Benito regarding tuning random forest hyperparameters using parallel computing

38.3 doParallel

Iterations in a loop

```
library(doParallel)
library(parallel)
library(foreach)

# Detect the number of available cores and create cluster
cl <- parallel::makeCluster(detectCores() - 2) # use all cores except 2
```

```
# Activate cluster for foreach library
doParallel::registerDoParallel(cl)

r <- foreach::foreach(i = 1:length(data_list),
                      .combine = rbind) %dopar% {
  mean(data_list[[i]])
}

# Stop cluster to free up resources
parallel::stopCluster(cl)

pacman::p_unload(pacman::p_loaded(), character.only = TRUE)

# detach(package:parallel)
```


Chapter 39

Generalized Method of Moments

Chapter 40

Interpretable Machine Learning

<https://christophm.github.io/interpretable-ml-book/> https://en.wikipedia.org/wiki/Explainable_artificial_intelligence <https://arxiv.org/pdf/1905.08883.pdf>
<https://arxiv.org/abs/2004.14545> <https://www.kaggle.com/learn/machine-learning-explainability> <https://towardsdatascience.com/interperable-vs-explainable-machine-learning-1fa525e12f48>

TCAV:

- <https://github.com/tensorflow/tcav>
- <http://proceedings.mlr.press/v80/kim18d.html>
- <https://arxiv.org/abs/1711.11279>

PAIR:

- <https://pair-code.github.io/facets/>
- <https://research.google/teams/brain/pair/>

Embedding Project:

- <https://projector.tensorflow.org/>

What-If tools:

- <https://pair-code.github.io/what-if-tool/>
- <https://cloud.google.com/ai-platform/prediction/docs/using-what-if-tool>

40.1 SHAP

<https://github.com/slundberg/shap> <https://meichenlu.com/2018-11-10-SHAP-explainable-machine-learning/> <https://bjlkeng.github.io/posts/model-explanability-with-shapley-additive-explanations-shap/>

40.2 Video Classification

<https://hackernoon.com/continuous-video-classification-with-tensorflow-inception-and-recurrent-nets-250ba9ff6b85> <https://cloud.google.com/video-intelligence> <https://arxiv.org/pdf/1609.06782.pdf> <https://www.tubefilter.com/2019/02/12/taxonomy-youtube-videos-develop-original-content-that-works/> <https://medium.com/swlh/converting-news-into-video-stories-5d8d4fc5a32b> <https://towardsdatascience.com/introduction-to-video-classification-6c6acbc57356> <https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0228579>

40.3 Image Memorability

(AMNET)

https://www.researchgate.net/publication/335603628_Image_Memorability_Using_Diverse_Visual_Features_and_Soft_Attention https://www.researchgate.net/publication/308812354_Deep_Learning_for_Image_Memorability_Prediction_the_Emotional_Bias <https://arxiv.org/pdf/2005.02969.pdf> <https://github.com/dazcona/memorability> <https://arxiv.org/abs/1804.03115> https://www.researchgate.net/publication/324387176_AMNet_Memorability_Estimation_with_Attention http://ceur-ws.org/Vol-2283/MediaEval_18_paper_24.pdf <https://jov.arvojournals.org/article.aspx?articleid=2771173> https://openaccess.thecvf.com/content_cvpr_2018/papers/Fajtl_AMNet_Memorability_Estimation_CVPR_2018_paper.pdf <https://ieeexplore.ieee.org/document/8578764>

<https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=7382237&tag=1> <https://people.umass.edu/~yangzhou/>

<http://people.csail.mit.edu/khosla/projects/regionmem/> <https://github.com/adikhosla/feature-extraction> <https://expertprogrammanagement.com/2019/12/elaboration-likelihood-model/>

Chapter 41

Supervised Learning

From the data, find a function f of the inputs that best approximates the outputs.

$$\hat{Y} = f(X_1, \dots, X_n)$$

- Predictive: Make predictions for a new object described by its attributes
- Informative: help understand the relationship between inputs and outputs

Divide data into

- Training set: train model
- Validation set: tune hyper parameters
- Test set: evaluation of the model for generalization error.

Learning algorithm

- defined by
 - a family of candidate models (= **hypothesis space H**)
 - a quality measure for a model
 - an optimization strategy
- takes as input a learning sample and outputs a function h in H of maximum quality

Model selection

- Why not choose the model that minimizes the error rate on the learning sample? (**re-substitution error**)
- How well are you going to predict future data drawn from the same distribution? (**generalization error**)
- Evaluation on test set

1. Randomly choose a percentage (e.g., 30%) of the data to be in a test set
 2. The remainder is training set
 3. Learn the model from the training set
 4. Estimate its future performance on the test set.
- Problems:
 - overfits: occurs when the learning algorithm starts fitting noise.
 - underfits
 - Evaluation on test set:
 - Good:
 - * Very simple
 - * Computationally efficient
 - Bad:
 - * Waste data
 - * very unstable when database is small
 - Another evaluation: **Leave-one-out cross validation**
 - For $k = 1$ to N
 - * remove the k -th object from the learning sample
 - * learn the model on the remaining objects
 - * apply the model to get a prediction for the k -th object
 - report the proportion of mis-classified objects
 - Good:
 - Do not waste the data (you get an estimate of the best method to apply to $N-1$ data)
 - Bad:
 - Computationally intensive (train N models)
 - high variance.
 - Another evaluation: **K-fold cross validation**:
 - Break up data into k folds
 - For each fold
 - * Choose the fold as a temporary test set
 - * Train on $k-1$ folds, compute performance on the test fold.
 - Report average performance of the 10 runs.
 - randomly partition the dataset into k subsets (e.g., 10)

- For each subset:
 - * learn the model on the objects that are not in the subset
 - * compute the error rate on the points in the subset. Report the mean error rate over the k subsets.
- When k = the number of objects -> leave-one-out-cross validation.

Rule-of-thumb for evaluation

- Lots of data (>1000) test set validation
- small data (100-1000): 10-fold CV
- very small data (<100) leave one out CV.

Use the testing set to estimate the performance of the final model

Performance measures

- The error rate is no the only way to assess a predictive model.
- In binary classification, results can be summarized in a contingency table (i.e., confusion matrix)

Various criteria

$$\text{Error rate} = \frac{FP + FN}{N + P} \quad \text{Accuracy} = \frac{TP + TN}{N + P} = 1 - \text{Error rate} \quad \text{Sensitivity or recall} = \frac{TP}{P} \quad \text{Specificity} = \frac{TN}{TN + FP} \quad \text{Precision} = \frac{TP}{TP + FP}$$

ROC and precision/recall curves

- Each point corresponds to a particular choice of the decision threshold.

Also known as robustness check

You want large area under the curve

Comparison of learning models

- Three main criteria:
 - Accuracy: Measures by the generalization error (estimated by CV)
 - Efficiency: Computing times and scalability for learning and testing
 - Interpretability: Comprehension brought by the model about the input-output relationship.

- Depends on the application, you have to make trade-off.

Learning Techniques

- Supervised learning categories and techniques
 - Linear classifier (numerical functions):

- * Perceptron
 - * Logistic regression
 - * Support vector machine (SVM): linear to nonlinear: feature transform and kernel function
 - * Ada-line
 - * Multi-layer perceptron (MLP)
- Parametric (Probabilistic functions)
 - * Naive Bayes, Gaussian discriminant analysis (GDA), Hidden Markov models (HMM), probabilistic graphical model
- Non-parametric (Instance-based functions)
 - * K-nearest neighbors, Kernel regression, Kernel density estimation, local regression.
- Non-metric (symbolic functions)
 - * Classification and regression tree (CART), decision tree
- Aggregation
 - * Bagging (bootstrap + aggregation), Adaboost, Random forest.
- Unsupervised learning categories and techniques:
 - Clustering
 - * K-means clustering
 - * Spectral clustering
 - Density Estimation
 - * Gaussian mixture model (GMM)
 - * Graphical models
 - Dimensionality reduction
 - * Principal component analysis (PCA)
 - * Factor analysis
- Semi-supervised learning
 - goal: exploit both labeled and unlabeled data to build better models than using each one alone
 - Self-training:
 - * Iteratively label some unlabeled examples with a model learned from the previously labeled examples
 - * Active learning.
- Reinforcement learning
 - Learning from interactions

- Goal: learn to choose sequence of actions (=policy) that maximizes $r_0 + \gamma r_1 + \gamma^2 r_2 + \dots$ where $0 \leq \gamma < 1$

41.1 Decision Tree

- Is a tree-structured plan of a set of attributes to test to predict the output.
- Give one attribute, try to predict the value of new people's attribute by means of some other available attribute
- Input attributes:
 - **d** features/ attributes $x^{(1)}, x^{(2)}, \dots, x^{(d)}$
 - Each $x^{(j)}$ has domain O_j
 - * Categorical: $O_j = (read, blue)$
 - * Numerical: $H_j = (0, 10)$
 - Y is output variable with domain O_Y :
 - * Categorical: Classification
 - * Numerical: regression
- Data D:
 - n examples (x_i, y_i) where
 - * x_i is a d-dim feature vector
 - * $y_i \in O_Y$
- Decision trees:
 - Split the data at each internal node
 - Each leaf node makes a prediction

How to construct a tree

- Find best split
- Stopping Criteria
- Find Prediction

41.1.1 Find best split

Pick an attribute and value that optimizes some criterion

+ Regression: Purity + Classification: Information Gain: Measures how much a given attribute X tells us about the class Y.

Information Gain? Entropy

The entropy of X: $H(X) = -\sum_{j=1}^m p_j \log p_j$

- “High entropy”: X is from a uniform distribution
 - A histogram of the frequency distribution of values of X is flat
- “Low Entropy”: X is from varied distribution
 - A histogram of the frequency distribution of values of X would have many lows and one or two highs.
- Def: Information Gain:
 - $IG(Y|X) = H(Y) - H(Y|X)$ I must transmit Y. How many bits on average would it save me if both ends of the line knew X?

41.1.2 Stopping Criteria

- When the leaf is “pure”: the target variable does not vary too much: $\text{Var}(Y) < \text{threshold}$.
- When number of examples in the leaf is too small.
- Could also use information as as a stopping criteria.

41.1.3 Find Prediction

- Regression: Predict average y of the examples
- Classification: Predict most common y of the examples the leaf.

41.2 Random Forest

Ensemble methods

- A single decision tree does not perform well.
- But, it is super fast.
- What if we learn multiple trees?

Bagging

- If we split the data in random different ways, decision trees give different results, high variance.
- Bagging: Bootstrap aggregating is a method that result in low variance.

- If we had multiple realizations of the data (or multiple samples) we could calculate the prediction multiple times and take the average of the fact that averaging multiple onerous estimations produce less uncertain results
- say for each sample b , we calculate $f^b(x)$, then $\hat{f}_{avg}(x) = \frac{1}{B} \sum_{b=1}^B \hat{f}^b(x)$
- Bootstrap: Construct B (hundreds) of trees. Learn a classifier for each bootstrap sample and average them. Very effective.
- Reduces overfitting (variance)
- Could use one classifier, or multiple classifiers.
- Easy to parallelize
- Variable Importance Measures
 - Bagging results in improved accuracy over prediction using a single tree
 - Unfortunately, difficult to interpret the resulting model. Bagging improves prediction accuracy at the expense of interpretability.
- Issues:
 - Each tree is identically distributed (i.d)
 - * The expectation of the average of B such trees
 - * The bias of bagged trees is the same as that of individual trees
 - An average of B i.i.d random variables, each with variance σ^2 , has variance σ^2/B . If i.d. (identical but not independent) and pair correlation ρ is present, then the variance is : $\rho\sigma^2 + \frac{1-\rho}{B}\sigma^2$. As B increases, the second term disappears but the first term remains.
- Suppose that there is one very strong predictor in the data set, along with a number of other moderately strong predictors. Then all bagged trees will select the strong predictor at the top tree and therefore all trees will look similar.
 - We can penalize the splitting with a penalty term that depends on the number of times a predictor is selected at a given length.
 - restrict how many items a predictor can be used.
 - allow a certain number of predictors.
- Since we want i.i.d so that the bias to be the same and variance to be less.
 - in bagging, we build a number of decision trees on bootstrapped training samples each time a split in a tree is considered, a random sample of m predictors is chosen as split candidates from the full set of p predictors (if $m = p$, then this is bagging).

Random Forest software

Random Forests Algorithm

For $b = 1$ to B :

- a. draw a bootstrap sample Z^* of size N from the training data.
- b. Grow a random-forest tree to the bootstrapped data, by recursively repeating the following steps for each terminal node of the tree, until the minimum node size n_{min} is reached.
 - Select m variables at random from the p variables.
 - Pick the best variable/split-point among the m .
 - Split the node into two daughter nodes.

Output the ensemble trees.

	Regression	Classification
To make a prediction at a new point x	average the results	majority vote
Tuning	The default value for m is $p/3$ and the minimum node size is 5	The default value for m is $\sqrt{p}$ and the minimum node size is one

Another issue:

When the number of variables is large, but the fraction of relevant variables is small, random forests are likely to perform poorly when m is small. Because: At each split the chance can be small that the relevant variables will be selected.

Example: with 3 relevant and 100 not so relevant variables the probability of any of the relevant variables being selected at any split is approx 0.25.

random forests “cannot overfit” the data because the number of trees (B) does not mean increase in the flexibility of the model.

Always include Random Forest (ensembles) and XGBoost (like bagging)

Alternative

- Support vector machine (SVM) dont usually use this because of computational reasons and because it is only 1 model.

Chapter 42

Unsupervised Learning

Unsupervised learning tries to find regularities in the data without guidance about inputs and outputs

Unsupervised learning categories and techniques:

- Clustering
 - K-means Clustering
 - Spectral Clustering
- Density Estimation
 - Gaussian mixture model (GMM)
 - Graphical models
- Dimensionality reduction
 - Principal component analysis (PCA)
 - Factor Analysis

42.1 Clustering

Clustering is used to find similar groups in data (i.e., clusters)

No class value denoting an a priori grouping (hence, Unsupervised Learning)

Clustering algorithm

- Partitional clustering
- Hierarchical clustering

A distance (similarity, or dissimilarity) function

Clustering quality

- Inter-clusters distance $\rightarrow$ maximized
- Intra-clusters distance $\rightarrow$ minimized

which can be use to judge good clustering

- Internal criterion: A good clustering will produce high quality clusters in which:
 - The intra-class (intra-cluster) similarity is high
 - The inter-class similarity is low.
 - The measured quality of a clustering depends on both the document representation and the similarity measure used.
- External criteria:
 - Quality measured by its ability to discover some or all of the hidden patterns or latent classes in gold standard data.
 - Assesses a clustering with respect to **ground truth**
 - Assume documents with C gold standard classes, while our clustering algorithm produce cluster $w_1, w_2, \dots, w_k$ with $n_1, \dots, n_k$ members.
 - Simple measure: **purity**, the ratio between the dominant class in the cluster π_i and the size of cluster w_i

$$Purity(w_i) = \frac{1}{n_i} \max_j (n_{ij}) j \in C$$

- others are entropy of classes in clusters (or mutual inforamtion between classes and clusters).

The quality of a clustering result depends on

- The algorithm
- the distance function
- the application

42.1.1 Similarity Measures

It's your choice of how to define similarity

(Dis)similarity measure

- Equivalently, we can use dissimilarity measures
- Jagota defines a dissimilarity measure as a function $f(x,y)$ such that $f(x,y) > f(w,z)$ if and only if x is less similar to y than w is to z .
Also known as **pair-wise** measure

Distance functions

Different distance functions for

- Different types of data
 - Numeric data
 - Nominal data
- Different applications

Numeric data

- $dist(x_i, x_j)$, where
 - $dist(x_j, x_j) = ((x_{i1} - x_{j1})^h + (x_{i2} - x_{j2})^h + \dots + (x_{ir} - x_{jr})^h)^{\frac{1}{h}}$
 - x_i and x_j are data points (vectors)
- Euclidean distance
 - If $h = 2$, it is the Euclidean distance: $dist(x_j, x_j) = ((x_{i1} - x_{j1})^2 + (x_{i2} - x_{j2})^2 + \dots + (x_{ir} - x_{jr})^2)^{\frac{1}{2}}$
 - Weighted Euclidean distance: $dist(x_j, x_j) = (w_1(x_{i1} - x_{j1})^2 + w_2(x_{i2} - x_{j2})^2 + \dots + w_r(x_{ir} - x_{jr})^2)^{\frac{1}{2}}$
 - Squared Euclidean distance: to place progressively greater weight on data points that are further apart: $dist(x_j, x_j) = (x_{i1} - x_{j1})^2 + (x_{i2} - x_{j2})^2 + \dots + (x_{ir} - x_{jr})^2$
- Manhattan distance
 - If $h = 1$, it is the Manhattan distance: $dist(x_j, x_j) = |x_{i1} - x_{j1}| + |x_{i2} - x_{j2}| + \dots + |x_{ir} - x_{jr}|$
- Minkowski distance (h is positive integer)
 - $dist(x_j, x_j) = ((x_{i1} - x_{j1})^h + (x_{i2} - x_{j2})^h + \dots + (x_{ir} - x_{jr})^h)^{\frac{1}{h}}$
 - 1st dimension, 2nd dimension, etc.

- Chebychev distance: define 2 data points as “different” if they are different on any one of the attributes

$$- \text{dist}(x_j, x_j) = \max(|x_{i1} - x_{j1}| + |x_{i2} - x_{j2}| + \dots + |x_{ir} - x_{jr}|)$$

Binary and nominal Attributes

- Binary attribute: has 2 values or states but no ordering relationship (e.g., gender)
- We use a confusion matrix to introduce the distance function/measures
- Let the i-th and j-th data points be X_i and x_j (vectors)

Confusion matrix

		Data point j		
		1	0	
Data point i	1	a	b	a + b
	0	c	d	c + d
		a + c	b + d	a + b + c + d

- a: the number of attributes with the value of 1 for both data points
- b: The number of attributes with which $x_{if} = 1$ and $x_{jf} = 0$ where x_{if} is the value of f attribute of the data point x_i
- c: the number of attributes of which $x_{if} = 0$ and $x_{jf} = 1$
- d: the number of attributes with the value of 0 for both data points.

Symmetric binary attributes

- A binary attribute is symmetric if both of its states (0 and 1) have equal importance, and carry the same weights, e.g., Gender.
- distance function: **Simple Matching Coefficient**, proportion of mismatches of their values

$$\text{dist}(x_i, x_j) = \frac{b + c}{a + b + c + d}$$

Asymmetric binary attributes

- if one of the states is more important or more valuable than the other
 - By convention, state 1 represents the more important state.

- Jaccard coefficient is popular measure

$$\text{dist}(x_i, x_j) = \frac{b + c}{a + b + c}$$

- We can have some variations, adding weights

Nominal attributes: with more than 2 states or values.

* the common used distance measure is also based on the simple matching method.

* Given two data points x_i, x_j let the number of attributes be r , and the number of values that match in x_i and x_j be q . $\text{dist}(x_i, x_j) = \frac{r-q}{r}$

Data standardization

- In the Euclidean space, standardization of attributes is recommended so that all attributes can have equal impact on the computation of distances.
- standardize attributes: to force the attributes to have a common value range.
- Real data have different types:
 - Interval-scaled
 - symmetric binary
 - asymmetric binary
 - ratio-scaled
 - ordinal
 - nominal
- Convert to a single type:
 - To deal with mixed attributes:
 - * Decide the dominant attribute type
 - * Convert the other types to this type

42.1.2 K-mean Algorithm

K-means is a partitioning clustering algorithm.

Let the set of data points (or instances) D to be $(x_1, \dots, x_n)$, where $x_i = (c_{i1}, \dots, c_{ir})$ is a vector in a real-valued space $X \subseteq R^r$ and r is the number of attributes (dimensions) in the data

The k-means algorithm partitions the given data into k clusters

- Each cluster has a cluster center (centroid)
- k is specified by user.

Steps

1. Randomly choose k data points (seeds) to be the initial centroids (cluster centers)
2. Assign each data point to the closest centroid
3. Re-compute the centroids using the current cluster memberships
4. If a convergence criterion is not met, go back to 2.

Convergence Criterion

1. No(or minimum) re-assignments of data points to different clusters,
2. No (or minimum) change of centroids, or
3. minimum decreases in the sum of squared error (SSE)

$$SSE = \sum_{j=1}^k \sum_{x \in C_j} \text{dist}(x, m_j)^2$$

- C is the j -th cluster, m is the centroid of cluster C_j (the mean vector of all the data points in C). and $\text{dist}(x, m)$ is the distance between data point x and centroid m_j .

Pros

- Simple: easy to understand and implement
- Efficient: time complexity: $O(tkn)$, where n is the number of data points, k is the number of clusters, and t is the number of iterations.
- Since both k and t are small, k -means is considered a linear algorithm
- Note: it terminates at a local optimum if SSE is used. The global optimum is hard to find due to complexity.

Cons

- The algorithm is only applicable if the mean is defined.
 - For categorical data, k -mode
- The user needs to specify k
- The algorithm is sensitive to outliers.
 - outliers could be errors in the data recording or some special data points with very different values.
- is not suitable for discovering clusters that are not hyper-ellipsoids (or hyper-spheres)

Solution

- To deal with outliers:
 - remove some data points in the clustering process that are much further away from the centroids than other data points.
 - Or perform random sampling. Since in sampling we only choose a small subset of the data points, the chance of selecting an outlier is very small.
- To deal with sensitive algorithm
 - Use different seeds

Chapter 43

Semi-supervised learning

- Goal: exploit both labeled and unlabeled data to build better models than using each one alone
- Self-training
 - Iteratively label some unlabeled examples with a model learned from the previously labeled examples
 - Active learning

Chapter 44

Reinforcement learning

- Learning from interactions
- Goal: learn to choose sequence of actions (= policy) that maximizes $r_0 + \gamma r_1 + \gamma^2 r_2 + \dots$

Chapter 45

Minimum Distance

Chapter 46

Quantile Regression

For academic review on quantile regression, check (Yu et al., 2003)

Linear Regression is based on the conditional mean function $E(y|x)$

In Quantile regression, we can view each points in the conditional distribution of y . Quantile regression estimates the conditional median or any other quantile of Y .

In the case that we're interested in the 50th percentile, quantile regression is median regression, also known as least-absolute-deviations (LAD) regression, minimizes $\sum_i |e_i|$

Properties of estimators β

- Asymptotically normally distributed

Advantages

- More robust to outliers compared to OLS
- In the case the dependent variable has a bimodal or multimodal (multiple humps with multiple modes) distribution, quantile regression can be extremely useful.
- Avoids parametric distribution assumption of the error process. In another word, no assumptions regarding the distribution of the error term.
- Better characterization of the data (not just its conditional mean)
- is invariant to monotonic transformations (such as log) while OLS is not. In another word, $E(g(y)) = g(E(y))$

Disadvantages

- The dependent variable needs to be continuous with no zeroes or too many repeated values.

$$y_i = x_i' \beta_q + e_i$$

Let $e(x) = y - \hat{y}(x)$, then $L(e(x)) = L(y - \hat{y}(x))$ is the loss function of the error term.

If $L(e) = |e|$ (called absolute-error loss function) then $\hat{\beta}$ can be estimated by minimizing $\sum_i |y_i - x_i' \beta|$

More specifically, the objective function is

$$Q(\beta_q) = \sum_{i: y_i \geq x_i' \beta}^N q |y_i - x_i' \beta_q| + \sum_{i: y_i < x_i' \beta}^N (1 - q) |y_i - x_i' \beta_q|$$

where $0 < q < 1$

The sum penalizes $q|e_i|$ for under-prediction and $(1 - q)|e_i|$ for over-prediction

We use simplex method to minimize this function (cannot use analytical solution since it's non-differentiable). Standard errors can be estimated by bootstrap.

The absolute-error loss function is symmetric.

Interpretation For the j th regressor (x_j), the marginal effect is the coefficient for the q th quantile

$$\frac{\partial Q_q(y|x)}{\partial x_j} = \beta_{qj}$$

At the quantile q of the dependent variable y , β_q represents a one unit change in the independent variable x_j on the dependent variable y .

In other words, at the q th percentile, a one unit change in x results in β_q unit change in y .

46.1 Application

```
# generate data with non-constant variance

x <- seq(0,100,length.out = 100)      # independent variable
sig <- 0.1 + 0.05*x                    # non-constant variance
b_0 <- 3                               # true intercept
b_1 <- 0.05                             # true slope
set.seed(1)                             # reproducibility
e <- rnorm(100,mean = 0, sd = sig)      # normal random error with non-constant variance
y <- b_0 + b_1*x + e                    # dependent variable
dat <- data.frame(x,y)
```

```
hist(y)
library(ggplot2)
ggplot(dat, aes(x,y)) + geom_point()
ggplot(dat, aes(x,y)) + geom_point() + geom_smooth(method="lm")
```

We follow (Koenker, 1996) to estimate quantile regression

```
library(quantreg)
qr <- rq(y ~ x, data=dat, tau = 0.5) # tau: quantile of interest. Here we have it at 50th percent
summary(qr)
```

adding the regression line

```
ggplot(dat, aes(x,y)) + geom_point() +
  geom_abline(intercept=coef(qr)[1], slope=coef(qr)[2])
```

To have R estimate multiple quantile at once

```
qs <- 1:9/10
qr1 <- rq(y ~ x, data=dat, tau = qs)
#check for its coefficients
coef(qr1)

# plot
ggplot(dat, aes(x,y)) + geom_point() + geom_quantile(quantiles = qs)
```

To examine if the quantile regression is appropriate, we can see its plot compared to least squares regression

```
plot(summary(qr1), parm="x")
```

where red line is the least squares estimates, and its confidence interval. x-axis is the quantile y-axis is the value of the quantile regression coefficients at different quantile

If the error term is normally distributed, the quantile regression line will fall inside the coefficient interval of least squares regression.

```
# generate data with constant variance

x <- seq(0, 100, length.out = 100) # independent variable
b_0 <- 3 # true intercept
b_1 <- 0.05 # true slope
set.seed(1) # reproducibility
e <- rnorm(100, mean = 0, sd = 1) # normal random error with constant variance
y <- b_0 + b_1 * x + e # dependent variable
dat2 <- data.frame(x, y)
qr2 = rq(y ~ x, data = dat2, tau = qs)
plot(summary(qr2), parm = "x")
```


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